BIO 004 · HUMAN ANATOMY · SOLANO COLLEGE

Module 5 Note Sheets.

Urinary · Male and female reproductive · Brain and brainstem · Meninges and CSF · Nervous tissue · Spinal cord · Peripheral, cranial and autonomic nerves

PART A	**The brain dump prompts** All 61 Module 5 competencies, numbers 133 to 193, with two drawing prompts each. Read this part once, then pick.
PART B	**The note sheets** One sheet per competency. The competency and its two prompts on the left, one big drawing box on the right. Sixty one sheets.
PART C	**Check your drawing** For every prompt, what should be on your drawing, and the most common mistake to look for.

How this packet works

Every Module 5 competency is on one sheet with two prompts beside one open box. The number on each sheet is the same number it has in your competency packet. You build one prompt in the box. The other is there if you want more practice, and it is a good place to go the week before the exam.

Before you start

1. Pick two pens you can tell apart. Pass one is the course notes, before any video. Pass two is a second color, after the concept video, into the same box.
2. Stop pass one while part of the box is still empty. That empty part is the point. It is what the video fills in.
3. Read your prompt first, then divide the box to fit it. Some prompts want one labeled drawing. Some want a table. Some want two views side by side. The division is yours to make.
4. Words are allowed inside the drawing, as labels and inside little boxes with arrows between them. A sentence running the width of the box is not. If a box reads like a paragraph, redo it as a picture.

Why two prompts

A and B are never the same question twice. One usually asks for a labeled drawing and the other for a table or a sequence, so a student who thinks in pictures and a student who thinks in categories both get a way in. Neither is the easy option.

About the paper

You do not have to print this. Rule one box per competency onto your own paper, or set the same thing up on a tablet, and copy the prompt letter you chose at the top. A hand ruled sheet counts exactly the same as a printed one.

Lecture and lab

Each competency is tagged Lecture, Lab, or both, the same as your competency packet. Lab tagged competencies are on the lab practical, so practice those on the models and the cadaver too.

PART A

The brain dump prompts.

Sixty one competencies, grouped by topic in the order of your competency packet. Each one gives you two prompts. Build one of them in that competency's box on your note sheet in Part B.

The Urinary System

133 Kidney gross anatomy and layers

MODULE 5 · LECTURE · LAB

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

A LABELED LOCATION VIEW

Draw the back wall of the abdomen with the vertebrae from T12 to L3, and place both kidneys with the right one slightly lower. On one kidney, label the renal hilum with the ureter, renal artery, renal vein, and nerves. Around the other, draw and label the renal capsule, adipose capsule, and renal fascia.

Then: Write beside the right kidney why it sits lower, and write beside the left what retroperitoneal tells you about where it sits.

B LAYERS TABLE

Draw a table with the renal capsule, adipose capsule, renal fascia, and renal hilum down the side. Across the top: where it sits (inner, middle, outer, or medial border), what it is made of or what passes through it, and its job. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch both kidneys between T12 and L3 with the right one lower, and mark the hilum on each.

134 Internal kidney regions

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

A LABELED FRONTAL SECTION

Draw a kidney cut in a frontal section. Label the renal capsule, renal cortex, renal medulla, the renal pyramids with their bases toward the cortex, the renal columns between them, the renal papillae, minor calyces, major calyces, renal pelvis, and renal sinus. Show the ureter leaving at the hilum.

Then: Shade the parenchyma in one color and write beside it the two regions it includes and what it holds.

B SORTING BOX

Draw two boxes, one labeled parenchyma and one labeled spaces that collect urine. Sort the cortex, renal columns, renal pyramids, renal papillae, minor calyces, major calyces, and renal pelvis into the two boxes. Beside the boxes, draw one pyramid with its papilla draining into a minor calyx, then a major calyx and the renal pelvis.

Then: Label a renal column beside your pyramid and write which region of the kidney it is an extension of.

135 Nephron structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

A LABELED NEPHRON

Draw one nephron with its collecting duct. Label the renal corpuscle, the glomerulus, the glomerular capsule with its parietal and visceral layers, the podocytes, the capsular space, the proximal convoluted tubule, the descending and ascending limbs of the nephron loop, the distal convoluted tubule, and the collecting duct.

Then: Draw arrows for the filtrate from the capsular space to the collecting duct, and mark the segment lined with a brush border.

B PARTS TABLE

Draw a table with the glomerulus, glomerular capsule, podocytes, capsular space, proximal convoluted tubule, nephron loop, distal convoluted tubule, and collecting duct down the side, with the renal corpuscle parts first and then the tubule parts in the order the filtrate meets them. Across the top: part of the renal corpuscle or the tubule, and one feature you would use to spot it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a podocyte wrapped around a glomerular capillary and label its foot processes and the filtration slits.

136 Cortical vs juxtamedullary nephrons

MODULE 5 · LECTURE · LAB

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

A SIDE BY SIDE NEPHRONS

Draw a strip of kidney with the cortex on top, the medulla below, and a line between them. Draw a cortical nephron and a juxtamedullary nephron side by side, showing how deep each nephron loop reaches. Wrap the cortical nephron in peritubular capillaries and the long juxtamedullary loop in the vasa recta.

Then: Write the percent of all nephrons each type makes up beside it, and star the type that concentrates the urine.

B COMPARE TABLE

Draw a table with cortical nephrons and juxtamedullary nephrons down the side. Across the top: share of all nephrons, length of the loop, how far the loop reaches into the medulla, and the capillary bed that serves it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch one renal pyramid with a short loop and a long loop drawn into it, and label each loop's nephron type.

137 JGA and filtration membrane

MODULE 5 · LECTURE · LAB

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

A LABELED CORPUSCLE CLOSE-UP

Draw a renal corpuscle with its afferent and efferent arterioles and the distal convoluted tubule touching them. Label the juxtaglomerular apparatus, the macula densa cells in the tubule, and the juxtaglomerular (granular) cells in the afferent arteriole. Add a zoomed box of the filtration membrane and label the fenestrated endothelium, basement membrane, and slit membranes.

Then: Draw a blood cell, a large protein, and a medium protein, each stopped at the layer that holds it back.

B WHERE AND WHAT TABLE

Draw a table with the macula densa cells, the juxtaglomerular cells, and the three layers of the filtration membrane down the side, the layers in order from the blood to the capsular space. Across the top: where it sits, and what it senses, releases, or holds back. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the distal convoluted tubule touching the arterioles of its own corpuscle and circle where the juxtaglomerular apparatus forms.

138 Kidney blood supply pathway

MODULE 5 · LECTURE · LAB

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

A LABELED VESSEL CUTAWAY

Draw a frontal section of a kidney showing two pyramids and the cortex. Draw and label the renal artery at the hilum, the segmental arteries branching in the renal sinus, the interlobar arteries running between the pyramids, the arcuate arteries along the pyramid bases, and the interlobular (cortical radiate) arteries up into the cortex. Add one nephron with its afferent arteriole, glomerulus, efferent arteriole, and peritubular capillaries.

Then: Draw the veins in a second shade beside the arteries and label the cortical radiate vein, arcuate vein, and renal vein.

B FLOW CHAIN

Draw a chain of boxes with arrows: renal artery, segmental and interlobar arteries, arcuate artery, interlobular (cortical radiate) artery, afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries and vasa recta, cortical radiate vein, arcuate vein, interlobar vein, renal vein. Under each artery box, write a word or two saying where it runs.

Then: Circle the two capillary beds in your chain and box the one vessel that sits between them.

139 Ureters, bladder, urethra and urine path

MODULE 5 · LECTURE · LAB

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

A LABELED URINARY TRACT

Draw a kidney, ureter, and bladder, with the kidney and bladder cut open. Label the minor and major calyces, renal pelvis, ureter, detrusor muscle in the bladder wall, the trigone with the two ureteral openings and the internal urethral orifice, the internal urethral sphincter, and the external urethral sphincter. Add arrows for urine from the collecting duct to the outside.

Then: Beside the bladder, sketch a male and a female urethra, and label the prostatic, membranous, and spongy parts of the male one.

B FLOW CHAIN

Draw a chain of boxes with arrows for urine: collecting duct, renal papilla, minor calyx, major calyx, renal pelvis, ureter, urinary bladder, urethra. Under the bladder box, list the trigone, detrusor muscle, and both urethral sphincters. Split the urethra box into male (prostatic, membranous, spongy) and female, and write which one is longer.

Then: Write beside each sphincter whether it is smooth or skeletal muscle and whether you control it.

140 Clinical anatomy of urinary disorders

MODULE 5 · LECTURE

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

A DISORDER MAP

Draw the urinary tract from the kidneys to the urethra, with the adipose capsule around one kidney. Write each disorder on the structure it affects: nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, renal failure, and polycystic kidney disease. Draw one stone in the renal pelvis and another lodged in the ureter.

Then: Draw a dropped kidney beside your map with a kinked ureter, and write what loss of fat has to do with it.

B DISORDER TABLE

Draw a table with nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, and polycystic kidney disease down the side. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a short female urethra beside a long male urethra and write why urinary tract infections are more common in women.

MODULE 5

The Male Reproductive System

141 Scrotum and testis structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.

A LABELED TESTIS SECTION

Draw the scrotum with one testis cut open inside it. Label the scrotal septum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, lobules, and seminiferous tubules. Add a zoomed circle of one tubule cut across, and label the sustentacular cells in its wall and the interstitial cells between the tubules.

Then: Write beside the dartos and the cremaster which kind of muscle each one is and what it moves.

B STRUCTURE TABLE

Draw a table with the scrotum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, seminiferous tubules, sustentacular cells, and interstitial cells down the side. Across the top: where it sits (scrotum, covering the testis, or inside the testis) and its job. Use a word or a tiny sketch in each cell.

Then: Circle the row whose cells produce testosterone, and star the two rows that set the temperature of the testes.

142 Sperm duct system and pathway

MODULE 5 · LECTURE · LAB

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

A LABELED DUCT VIEW

Draw a side view of the male pelvis with the testis, the epididymis on it, and the ductus deferens running up out of the scrotum in the spermatic cord. Label the seminal vesicle, the ejaculatory duct, the prostate, and the prostatic, membranous, and spongy urethra. Add arrows for sperm from the seminiferous tubules to the outside.

Then: Draw the spermatic cord cut across as a circle, and label the ductus deferens, blood vessels, nerves, and cremaster muscle inside it.

B FLOW CHAIN

Draw a chain of boxes with arrows: seminiferous tubules, epididymis, ductus deferens, ejaculatory duct, prostatic urethra, membranous urethra, spongy urethra. Under each box, write in a word or two what happens to the sperm there. Box the one structure that produces sperm.

Then: Draw a bracket over the duct that runs inside the spermatic cord, and list the cord's other three contents beside it.

143 Sperm cell structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

A LABELED SPERM CELL

Draw one sperm cell large enough to fill the box. Label the head with its nucleus, the acrosome capping the head, the neck, the middle piece, the principal piece, and the end piece. Draw the middle piece packed with small mitochondria and the end piece tapering to a point.

Then: Write beside each part what it holds or does, in three words or fewer.

B PARTS TABLE

Draw a table with the head, acrosome, neck, middle piece, principal piece, and end piece down the side, in order from front to back. Across the top: how you spot it and what it holds or does. Use a word or a tiny sketch in each cell.

Then: Draw a bracket beside the rows that make up the tail, and star the longest part of the tail.

144 Accessory glands and penis

MODULE 5 · LECTURE · LAB

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

A LABELED PENIS VIEW

Draw a side view of the penis and a cross section of its body. On the side view, label the root, body, glans penis, and prepuce. On the cross section, label the two corpora cavernosa on the dorsal side and the corpus spongiosum on the ventral side surrounding the spongy urethra.

Then: Add the seminal vesicles, prostate, and bulbourethral glands as labeled boxes beside your drawing, and write what fluid each adds to semen.

B TWO PART TABLE

Draw a table in two parts. First, put the seminal vesicles, prostate, and bulbourethral glands down the side, with what each adds to semen across the top. Second, put the root, body, glans penis, prepuce, corpora cavernosa, and corpus spongiosum down the side, with what each is. Use a word or a tiny sketch in each cell.

Then: Star the gland that adds the largest share of semen, and circle the erectile column that forms the glans.

145 Clinical anatomy of male disorders

MODULE 5 · LECTURE

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

A DISORDER MAP

Draw a side view of the male pelvis with the testis in the scrotum, the inguinal canal, and the urethra passing through the prostate. Write each disorder on the structure it affects: cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, prostate cancer, testicular cancer, and inguinal hernia.

Then: Draw an undescended testis with a dotted line to where it should sit, and write why staying at body temperature harms it.

B DISORDER TABLE

Draw a table with cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, and inguinal hernia down the side. Across the top: the structure it affects, where that structure sits, and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an enlarged prostate squeezing the urethra and write the urinary symptom it causes.

MODULE 5

The Female Reproductive System

146 Ovary structure and follicle stages

MODULE 5 · LECTURE · LAB

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

A LABELED OVARY VIEW

Draw the uterus with one ovary beside it. Label the ovarian ligament, suspensory ligament, broad ligament, and mesovarium. Draw the ovary cut open and label the germinal epithelium, tunica albuginea, cortex, and medulla. In the cortex, draw ovarian follicles, a mature follicle, a corpus luteum, and a corpus albicans.

Then: Number the follicle structures in the order they appear, and mark where the oocyte leaves at ovulation.

B SEQUENCE STRIP

Draw a row of circles showing a follicle growing into a mature follicle, releasing its oocyte, then becoming a corpus luteum and finally a corpus albicans. Beneath the row, list the ovary layers from the outside in, then the three ligaments with what each one attaches the ovary to.

Then: Circle the layer of the ovary where every follicle in your strip is found.

147 Uterine tubes and ovum pathway

MODULE 5 · LECTURE · LAB

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

A LABELED TUBE VIEW

Draw the uterus with one uterine tube and the ovary beside it. Label the fimbriae, infundibulum, ampulla, and isthmus, and show the infundibulum open to the pelvic cavity near the ovary. Draw an oocyte leaving a mature follicle, with arrows following it through the tube into the uterine cavity.

Then: Star where fertilization usually happens, and draw cilia on the tube lining with an arrow showing which way they move the oocyte.

B FLOW CHAIN

Draw a chain of boxes with arrows: ovary, fimbriae and infundibulum, ampulla, isthmus, uterus. Under each box, write what happens to the oocyte there and one feature of that region, such as its shape or width. Use a word or a tiny sketch under each box.

Then: Draw a dotted arrow from the open infundibulum into the pelvic cavity, and write one disorder this open route explains.

148 Uterus regions, ligaments, and wall

MODULE 5 · LECTURE · LAB

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

A LABELED UTERUS VIEW

Draw the uterus cut open from the front, with the uterine tubes and the top of the vagina. Label the fundus, body, cervix, uterine cavity, and cervical canal with the internal and external os. Label the perimetrium, myometrium, and endometrium in the wall. Add the broad, round, uterosacral, and cardinal ligaments as lines running to what each attaches to.

Then: Zoom in on the endometrium and split it into the layer that sheds each month and the layer that rebuilds it.

B REGION STACK AND TABLE

Draw a stack of three boxes from top to bottom for the fundus, body, and cervix, with the uterine cavity and cervical canal drawn inside. Beside it, draw a table with the four ligaments and the three wall layers down the side. Across the top: what it connects to or what it is made of. Use a word or a tiny sketch in each cell.

Then: Star the wall layer that contracts during childbirth, and circle the ligament that runs to the labia majora.

149 Vagina and external genitalia

MODULE 5 · LECTURE · LAB

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

A LABELED VULVA VIEW

Draw the vulva in an inferior view, with the mons pubis at the top. Label the labia majora, labia minora, clitoris, vestibule, urethral orifice, vaginal orifice, hymen, and the greater vestibular glands beside the vaginal opening. Outline the perineum around the genitalia and the anus.

Then: Sketch a side view of the vagina meeting the cervix, and label the fornix around the cervix.

B STRUCTURE TABLE

Draw a table with the vagina, fornix, hymen, mons pubis, labia majora, labia minora, clitoris, vestibule, and greater vestibular glands down the side. Across the top: where it sits and what it is or does. Use a word or a tiny sketch in each cell.

Then: Mark the three structures that are homologous to male structures, and write the male match beside each.

150 Mammary gland structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

A LABELED BREAST SECTION

Draw a side view of a breast cut open, sitting over the chest muscles. Label two or three lobes, the lobules inside one lobe, the alveoli inside one lobule, the lactiferous ducts running to the nipple, the areola, and the suspensory ligaments.

Then: Draw arrows for the path of milk from the alveoli to the nipple, and label each structure the arrows pass.

B NESTING DIAGRAM

Draw nested shapes: alveoli inside a lobule, lobules inside a lobe, and the lobe draining by a lactiferous duct to the nipple. Beside it, draw a small table with the nipple, areola, and suspensory ligaments down the side and what each is or does across the top. Use a word or a tiny sketch in each cell.

Then: Write beside your lobe how many lobes one mammary gland has.

The Brain

151 Brain gross organization

MODULE 5 · LECTURE · LAB

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

A LABELED MIDSAGITTAL VIEW

Draw the brain cut down the middle and seen from the side. Label the cerebrum, diencephalon, brainstem, and cerebellum, and outline each one so the four divisions are easy to tell apart. Along the edge of the cerebrum, draw a thin outer band of gray matter over a thick core of white matter, and label both.

Then: Write beside the gray matter band what it is made of, and beside the white matter what it is made of.

B DIVISION TABLE

Draw a table with the cerebrum, diencephalon, brainstem, and cerebellum down the side. Across the top: where it sits and its main job. Use a word or a tiny sketch in each cell. Below the table, draw a small block of brain tissue with gray matter on the outside and white matter on the inside, and label both.

Then: Star the largest division and the second largest division, and write the cranial fossa where the cerebellum sits.

152 Neural tube brain development

MODULE 5 · LECTURE

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

A BRANCHING FLOW CHAIN

Draw the neural tube at the top of the box. Branch it into the three primary vesicles, writing each one's formal name and common name (forebrain, midbrain, hindbrain). Branch those into the five secondary vesicles. Under each secondary vesicle, write the adult structures it becomes and its ventricle or fluid space.

Then: Write the germ layer the neural tube forms from, and circle the one primary vesicle that carries through without splitting.

B VESICLE TABLE

Write neural tube at the top of the box. Below it, draw a table with the five secondary vesicles down the side. Across the top: the primary vesicle it came from, the adult structures it becomes, and its ventricle or fluid space. Use a word or a tiny sketch in each cell.

Then: Draw a simple side view of the adult brain and number each region 1 to 5 to match the secondary vesicle it grew from.

153 Cerebral surface and lobes

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

A SIDE AND TOP VIEWS

Draw the left side of the brain. Label the frontal, parietal, temporal, and occipital lobes, a gyrus, a sulcus, the precentral and postcentral gyri, and the central, lateral, and parieto-occipital sulci. Draw the insula as a dashed shape inside the lateral sulcus. Show the transverse fissure above the cerebellum. Beside it, draw a small top view and label both hemispheres and the longitudinal fissure.

Then: Write beside each of the four outer lobes the cranial bone that lies over it and gives it its name.

B BOUNDARY TABLE

Sketch three folds of cortex and label a gyrus, a sulcus, and a fissure. Below, draw a table with the central sulcus, lateral sulcus, parieto-occipital sulcus, longitudinal fissure, and transverse fissure down the side. Across the top: the region on one side of it and the region on the other. Use a word or a tiny sketch in each cell.

Then: Draw the brain seen from above, label the longitudinal fissure, central sulcus, and each lobe you can see, then name the lobe no outside view shows.

154 Functional cortical areas

MODULE 5 · LECTURE · LAB

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

A LABELED CORTEX MAP

Draw the side view of the hemisphere where Broca's and Wernicke's areas are usually found. Draw the central and lateral sulci. Label the primary motor cortex, primary somatosensory cortex, Broca's area, Wernicke's area, prefrontal cortex, primary auditory cortex, and primary visual cortex. Draw the insula dashed for the gustatory cortex, and point an arrow to the medial temporal lobe for the olfactory cortex.

Then: Write M, S, or A on each labeled area to sort it as a motor, sensory, or association area.

B AREA LOCATION TABLE

Draw a table with these areas down the side: primary motor, primary somatosensory, primary visual, primary auditory, primary gustatory, and primary olfactory cortex, Broca's area, Wernicke's area, and the prefrontal cortex. Across the top: lobe, gyrus or landmark, motor, sensory, or association, and job. Use a word or a tiny sketch in each cell.

Then: Add a row for each association area the notes place next to a primary area, and name the primary area it sits beside.

155 Cerebral white matter tracts

MODULE 5 · LECTURE · LAB

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

A LABELED FRONTAL SECTION

Draw a frontal section through both hemispheres with gray cortex on the outside and white matter inside. Draw and label an association fiber arching between two gyri on the same side, the corpus callosum crossing between the hemispheres, and the projection fibers of the internal capsule running down beside the basal ganglia toward the brainstem.

Then: Shade the internal capsule and write beside it why a small stroke there can weaken a whole side of the body.

B FIBER CLASS TABLE

Draw a table with association, commissural, and projection fibers down the side. Across the top: what it connects, a named example, and a tiny arrow sketch of its path. Use a word or a tiny sketch in each cell. Below, sketch a midline view with the corpus callosum and a frontal section with the internal capsule, and label both.

Then: Circle the largest commissural tract, and write which tract links Broca's and Wernicke's areas and which fiber class it belongs to.

156 Basal ganglia

MODULE 5 · LECTURE · LAB

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

A LABELED FRONTAL SECTION

Draw a frontal section of one hemisphere with cortex on the outside, white matter inside, and deep masses of gray matter buried in the white matter. Label the caudate nucleus, the putamen, and the globus pallidus medial to the putamen. Below the hemisphere, sketch the diencephalon and midbrain, and mark the subthalamic nucleus and the substantia nigra.

Then: Draw dotted lines linking all five nuclei, and write the three roles the basal ganglia share.

B LOCATION SORT

Draw three boxes stacked from top to bottom labeled cerebrum, diencephalon, and midbrain. Write each nucleus in the box where it sits: caudate nucleus, putamen, globus pallidus, subthalamic nucleus, and substantia nigra. Circle the three core nuclei and underline the two grouped with them only by function. Add an arrow showing the putamen lateral to the globus pallidus.

Then: Write beside the substantia nigra what it looks like and the disease its loss causes.

157 Diencephalon

MODULE 5 · LECTURE · LAB

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

A LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the thalamus with the massa intermedia, the hypothalamus below it, the epithalamus at the back with the pineal gland, and the third ventricle. Add the corpus callosum above, the infundibulum and optic chiasm below the hypothalamus, and the midbrain below as landmarks.

Then: Shade the third ventricle, and write the percent of the diencephalon the thalamus makes up.

B THREE PART TABLE

Draw a table with the thalamus, hypothalamus, and epithalamus down the side. Across the top: where it sits, what it contains, and its main job. Use a word or a tiny sketch in each cell. Below the table, draw the third ventricle as a midline cavity and label it.

Then: Draw the right and left thalami on each side of your third ventricle, and join them across it with the massa intermedia.

158 Cerebellum

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

A BACK AND CUT VIEWS

Draw the cerebellum from behind and label the two hemispheres, the vermis, and the folia. Beside it, draw the cerebellum cut down the midline and attached to the brainstem. Label the cerebellar cortex, arbor vitae, a deep cerebellar nucleus, and the superior, middle, and inferior cerebellar peduncles as stalks running to the midbrain, pons, and medulla.

Then: Shade the gray matter on your cut view and leave the white matter blank.

B PARTS TABLE

Draw a table with these down the side: cerebellar hemispheres, vermis, folia, arbor vitae, deep cerebellar nuclei, and the superior, middle, and inferior cerebellar peduncles. Across the top: what it is, gray or white matter, and where it sits or what it joins. Use a word or a tiny sketch in each cell.

Then: Draw an arrow on each peduncle row showing whether it mostly carries signals into the cerebellum or out of it.

159 Limbic system

MODULE 5 · LECTURE · LAB

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

A LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the corpus callosum, the cingulate gyrus arching above it, and the diencephalon beneath it. Draw the hippocampus and amygdala as dashed shapes curving down and forward below the diencephalon, so that the three limbic structures form a ring around it.

Then: Write beside each limbic structure its main role from the notes.

B LIMBIC COMPARE TABLE

Draw a table with the cingulate gyrus, amygdala, and hippocampus down the side. Across the top: where it sits, its role, and a tiny sketch. Use a word or a tiny sketch in each cell. Below the table, draw a small oval for the diencephalon and arrange the three structures around it as a ring.

Then: Write in the center of your ring the two jobs the limbic system does as a whole.

160 Cerebral arterial supply

MODULE 5 · LECTURE · LAB

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

A LABELED INFERIOR VIEW

Draw the base of the brain seen from below with the brainstem in the middle. Draw the vertebral arteries joining into the basilar artery, then the ring: posterior cerebral, posterior communicating, internal carotid, anterior cerebral, and anterior communicating arteries. Add the middle cerebral arteries and the cerebellar branches: posterior inferior, anterior inferior, and superior cerebellar arteries.

Then: Shade the anterior circulation one way and the posterior circulation another, and write what each one feeds.

B TWO CIRCULATION CHAINS

Draw two flow chains side by side. Anterior: internal carotid artery to the anterior and middle cerebral arteries. Posterior: vertebral arteries to basilar artery to posterior cerebral arteries, with the posterior inferior cerebellar branch off each vertebral and the anterior inferior and superior cerebellar branches off the basilar. Join the two chains with the posterior communicating artery, and add the anterior communicating artery joining the right and left anterior cerebral arteries.

Then: Write beside each cerebral artery the part of the brain it supplies.

MODULE 5

The Brainstem

161 Brainstem regions

MODULE 5 · LECTURE · LAB

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

A LABELED SIDE VIEW

Draw the brainstem from the side with the diencephalon above it, the cerebellum behind it, and the spinal cord below. Label the midbrain, pons, and medulla oblongata from top to bottom, the basilar pons, and the pontomedullary junction. Mark the foramen magnum where the medulla becomes the spinal cord.

Then: Draw one arrow up and one arrow down through all three regions, and label them ascending sensory tracts and descending motor tracts.

B REGION ORDER CHAIN

Draw a vertical chain of boxes from top to bottom: cerebrum and diencephalon, midbrain, pons, medulla oblongata, and spinal cord, with the cerebellum to the side. In each brainstem box, write its position (most superior, middle, most inferior) and one structure you can find on it on the model.

Then: Write beside the chain what passes through all three regions and how many cranial nerves have nuclei in the brainstem.

162 Midbrain structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

A TWO LABELED VIEWS

Draw the midbrain from behind and in cross section. On the back view, label the tectum and its four colliculi, with the superior pair above the inferior pair. On the cross section, label the tectum, the cerebral aqueduct, the darkly pigmented substantia nigra, and the cerebral peduncles on the front.

Then: Draw an arrow into and out of the cerebral aqueduct, and name the ventricle at each end.

B MIDBRAIN STRUCTURE TABLE

Draw a table with the tectum, superior colliculi, inferior colliculi, cerebral peduncles, substantia nigra, and cerebral aqueduct down the side. Across the top: how to spot it on the model, and its job. Use a word or a tiny sketch in each cell.

Then: Circle the two reflex centers, and write beside them whether you become conscious of the sight or sound there.

163 Pons structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

A LABELED SIDE VIEW

Draw the brainstem from the left side with the cerebellum behind it. Label the midbrain, pons, and medulla, the basilar pons, and the thick middle cerebellar peduncle running back into the cerebellum. Dot the pontine nuclei inside the pons. Draw the fourth ventricle as a dashed space behind the pons and upper medulla.

Then: Draw an arrow from the cerebral cortex through the pontine nuclei and the middle cerebellar peduncle into the cerebellum.

B PONS FLOW CHAIN

Draw a flow chain for motor information: cerebral cortex, pontine nuclei, middle cerebellar peduncles, cerebellum. Write what each step does in a few words. Under the chain, sketch the pons with the midbrain above, the medulla below, and the fourth ventricle behind it, and label all four.

Then: Write what the word pons means, and name the three regions it bridges.

164 Medulla oblongata structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

A LABELED ANTERIOR VIEW

Draw the brainstem from the front with the spinal cord below. Label the pons, the pontomedullary junction, the medulla oblongata, the two pyramids, the medullary olive on each side of the pyramids, and the decussation of the pyramids where the medulla meets the cord. Inside the medulla, draw and label boxes for the cardiac, respiratory, and vasomotor centers.

Then: Draw crossing arrows at the decussation, and write why a stroke in the right hemisphere weakens the left side of the body.

B MEDULLA TABLE

Draw a table with the pyramids, medullary olive, decussation of the pyramids, cardiac center, respiratory center, and vasomotor center down the side. Across the top: seen on the surface or inside the medulla, and what it looks like or what it does. Use a word or a tiny sketch in each cell.

Then: Star the three vital centers, and write why injury to the medulla is immediately life threatening.

165 Cranial nerve nuclei distribution

MODULE 5 · LECTURE · LAB

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

A LABELED BRAINSTEM VIEW

Draw the brainstem from the front, divided into the midbrain, pons, and medulla oblongata, and label the pontomedullary junction. Beside each region, write the Roman numerals and names of the cranial nerves whose nuclei it holds, from CN III down to CN XII.

Then: Write which two cranial nerves have no nuclei in the brainstem, and circle the one nerve you wrote beside two regions.

B NERVE SORT TABLE

Draw a table with CN III through CN XII down the side. Across the top: the nerve's name, midbrain, pons, and medulla oblongata. Write each name, then put a check in the region that houses that nerve's nuclei. Use a word or a check in each cell.

Then: Count how many of the twelve cranial nerves have nuclei in the brainstem, and circle the nerve with checks in two regions.

166 Reticular formation

MODULE 5 · LECTURE

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

A LABELED CUTAWAY

Draw the brainstem from the side, cut open to show its core. Draw the reticular formation as a net of dots and lines woven through the midbrain, pons, and medulla. Draw arrows up from the net to the cerebral cortex labeled reticular activating system, and arrows down toward the spinal cord labeled descending component.

Then: Write beside each set of arrows two of its jobs from the notes.

B COMPARE TABLE

Draw a table with the reticular activating system and the descending component down the side. Across the top: which way it sends signals, what it acts on, and its jobs. Use a word or a tiny sketch in each cell. Above the table, sketch the brainstem core with a gray matter net running through its white matter.

Then: Write under the table what happens when the reticular activating system shuts down and what severe damage to it can cause.

MODULE 5

Meninges and Cerebrospinal Fluid

167 Cranial meninges and dural reflections

MODULE 5 · LECTURE · LAB

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

A LABELED MIDSAGITTAL VIEW

Draw the inside of the skull in midsagittal view with the brain removed. Label the falx cerebri, falx cerebelli, and tentorium cerebelli. Draw the superior sagittal sinus on the falx cerebri, then the confluence of sinuses, a transverse sinus, a sigmoid sinus, and the internal jugular vein. In a corner, draw the dura, arachnoid, and pia from skull to brain.

Then: Add arrows along the sinuses for the direction blood drains, and label the periosteal and meningeal layers on your dura strip.

B TABLE AND FLOW CHAIN

Draw a table with the dura mater, arachnoid mater, pia mater, falx cerebri, falx cerebelli, and tentorium cerebelli down the side. Across the top: where it sits and what it touches or separates. Use a word or a tiny sketch in each cell. Below the table, draw a chain of boxes from the superior sagittal sinus to the internal jugular vein.

Then: Mark the box in your chain where the arachnoid granulations return CSF to the blood.

168 Spinal meninges and meningeal spaces

MODULE 5 · LECTURE · LAB

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

A LABELED CROSS SECTION

Draw a cross section of a vertebra with the spinal cord inside. Label the vertebral bone, the epidural space with fat and veins, the dura mater, subdural space, arachnoid mater, subarachnoid space with CSF, and pia mater. Draw a denticulate ligament from the pia to the dura. Beside it, sketch the lower cord with the filum terminale running to the coccyx.

Then: Star the space that holds CSF, and write potential space beside the space that is only a potential space.

B SPACE COMPARE TABLE

Draw a table with the epidural, subdural, and subarachnoid spaces down the side. Across the top: the two layers it sits between, what fills it, and whether it is also found in the cranium. Use a word or a tiny sketch in each cell. Below the table, sketch the cord with two denticulate ligaments and the filum terminale.

Then: Write beside each anchor which meningeal layer it is made of and what it attaches the cord to.

169 Cord termination and lumbar puncture

MODULE 5 · LECTURE · LAB

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

A LABELED SIDE VIEW

Draw a side view of the lower vertebral column with the spinal cord inside. Label the conus medullaris at the L1 to L2 disc, the cauda equina, the filum terminale, and the lumbar cistern below the cord. Draw a needle entering in the midline between the L3 and L4 spines, and label each layer it crosses, in order, from the skin to the subarachnoid space.

Then: Write beside the conus medullaris why the needle is placed between the L3 and L4 or L4 and L5 spines.

B NEEDLE PATH SEQUENCE

Draw a vertical stack of boxes, one for each layer a lumbar puncture needle crosses, in order from the skin at the top to the subarachnoid space at the bottom. Name the layer and sketch what it looks like in each box. Beside the bottom box, sketch the lumbar cistern with the floating cauda equina, the conus medullaris above it, and the filum terminale.

Then: Mark the box where a clinician feels a firm ligament give way, and write the two spine levels where the needle can enter.

170 Ventricular system

MODULE 5 · LECTURE · LAB

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

A LABELED VENTRICLE VIEW

Draw a side view of the brain outline with the ventricles inside it, as on the brain model. Label a lateral ventricle, an interventricular foramen of Monro, the third ventricle, the cerebral aqueduct, the fourth ventricle, and the central canal of the spinal cord. Draw the median aperture and the two lateral apertures opening from the fourth ventricle into the subarachnoid space.

Then: Write beside each ventricle the region of the brain it sits in, and write what lines every chamber.

B CONNECTION TABLE

Draw a table with the lateral ventricles, third ventricle, fourth ventricle, and central canal down the side. Across the top: where it is, what it connects to next, and the opening or channel that makes the connection. Use a word or a tiny sketch in each cell.

Then: Circle the three narrow connection points in your table, and write beside the table what would happen if one were blocked.

171 CSF production and circulation

MODULE 5 · LECTURE · LAB

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

A LABELED FLOW DRAWING

Draw a midsagittal view of the brain and upper spinal cord showing the ventricles, the subarachnoid space, and the superior sagittal sinus. Draw choroid plexus in the lateral, third, and fourth ventricles. Add numbered arrows for CSF from the lateral ventricles through the interventricular foramina, third ventricle, cerebral aqueduct, fourth ventricle, apertures, and subarachnoid space to the arachnoid granulations.

Then: Add a side arrow into the central canal, and write which cells of the choroid plexus produce the CSF.

B FLOW CHAIN

Draw a chain of boxes with arrows for the one-way loop of CSF, from production at the choroid plexus of the lateral ventricles to reabsorption at the arachnoid granulations. Give each ventricle, opening, and space its own box. Add a tiny sketch in each box of what the structure looks like.

Then: Mark each box that holds choroid plexus, and add a final arrow into the venous blood of the superior sagittal sinus.

172 Blood-brain barrier structure

MODULE 5 · LECTURE · LAB

You should be able to: Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

A LABELED CAPILLARY SECTION

Draw a cross section of a brain capillary with blood in its center. Label the capillary endothelial cells joined by tight junctions, the basement membrane wrapped around them, and the astrocyte end-feet covering the outside. Draw one whole star-shaped astrocyte with its end-feet reaching the capillary and a neuron nearby.

Then: Draw an arrow from the blood toward the neuron, and number each barrier component the arrow has to cross.

B COMPARE TABLE

Draw a table with the three components of the blood-brain barrier down the side. Across the top: what it is, its position counting outward from the blood, and one feature that makes it part of the barrier. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an ordinary capillary beside a brain capillary and label the pores on the ordinary capillary that the brain capillary lacks.

MODULE 5

Functional Organization and Nervous Tissue

173 Nervous system divisions

MODULE 5 · LECTURE

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

A BRANCHING MAP

Draw a branching map that starts with the nervous system and splits into the CNS and PNS. Under the CNS, write the brain and spinal cord. Under the PNS, list nerves, ganglia, and sensory receptors, then branch into the somatic, autonomic, and enteric motor divisions. Split the autonomic division into sympathetic and parasympathetic.

Then: Write beside each motor division the effector it controls, and add the number of pairs of cranial and spinal nerves.

B LABELED BODY OUTLINE

Draw a body outline with the brain and spinal cord shaded as the CNS. Draw a few nerves leaving the cord, one ganglion, and a sensory receptor in the skin, and label them as PNS. Draw a skeletal muscle, the heart, a gland, and a loop of gut wall with its own neurons.

Then: Label each effector with the motor division that controls it: somatic, autonomic, or enteric.

174 Neuron structure

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

A LABELED NEURON

Draw a multipolar neuron as it looks on the neuron model. Label the cell body with its nucleus and nucleolus, the Nissl bodies, the dendrites, the axon hillock, the axon, and the axon terminals with their synaptic bulbs.

Then: Add arrows along the neuron for the direction a signal travels, from the dendrites through to the axon terminals.

B SIGNAL FLOW CHAIN

Draw a chain of boxes in the order a signal travels through a neuron: dendrites, cell body, axon hillock, axon, and axon terminals. In each box, sketch the part and write its job in a word or two. Draw Nissl bodies inside the cell body box and write what they are made of and what they do.

Then: Add neurofibrils to the cell body box, and write why an axon dies when it is cut away from its cell body.

175 Synapse structures

MODULE 5 · LECTURE · LAB

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

A LABELED SYNAPSE CLOSE-UP

Draw the end of an axon meeting a second neuron. Label the axon terminal, a synaptic end bulb, the synaptic vesicles inside it, the synaptic cleft, and the next cell. Draw neurotransmitter molecules leaving the vesicles and crossing the cleft. Beside it, draw three tiny sketches of an axon ending on a dendrite, on a cell body, and on another axon.

Then: Name each of your three tiny sketches by the site it contacts: axodendritic, axosomatic, or axoaxonic.

B SYNAPSE TYPE MAP

Draw one neuron with three axons from other neurons ending on it: one on a dendrite, one on the cell body, and one on its axon. Name each synapse by the site it contacts. Draw a zoom circle on one synapse showing the synaptic end bulb, synaptic vesicles, and synaptic cleft.

Then: Write beside the zoom circle what the vesicles store and what carries the signal across the cleft.

176 Neuron classification

MODULE 5 · LECTURE · LAB

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

A THREE SHAPE SKETCHES

Draw a multipolar, a bipolar, and a unipolar neuron side by side. Label the dendrites, cell body, and axon on each one, and write how many processes leave the cell body. Under the multipolar and bipolar shapes, write where each is found, and add the Purkinje cell and pyramidal cell under the shape they belong to.

Then: Label which shape most sensory neurons have, and which shape motor neurons and interneurons have.

B FUNCTIONAL PATHWAY MAP

Draw a skin receptor, a cross section of the spinal cord, and a skeletal muscle. Connect them with a sensory neuron carrying a signal toward the CNS, an interneuron inside the cord, and a motor neuron carrying a command out to the muscle. Draw each neuron in its structural shape and label the shape.

Then: In the margin, add a bipolar neuron and name the three places it is found.

177 Neuroglia identification

MODULE 5 · LECTURE · LAB

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

A LABELED GLIA SCENE

Draw a patch of CNS tissue with a neuron, a capillary, and the edge of a ventricle. Draw and label an astrocyte touching the capillary, an oligodendrocyte wrapping several axons, a microglial cell eating debris, and ependymal cells lining the ventricle. In a separate PNS box, draw Schwann cells on an axon and satellite cells around cell bodies in a ganglion.

Then: Write the support role of each glial cell beside its label in a few words.

B SORTING BOX

Draw a box split into a CNS side and a PNS side. Sort the six glial cells into the correct side. For each cell, add a tiny sketch of its shape and a few words for the support role it performs.

Then: Circle the two cells that make myelin, and star the cell that helps form the blood-brain barrier.

178 Myelination and regeneration

MODULE 5 · LECTURE · LAB

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

A LABELED AXON COMPARISON

Draw a PNS axon wrapped by a row of Schwann cells. Label the myelin sheath, the nodes of Ranvier, and the neurolemma. Beside it, draw one CNS oligodendrocyte with arms wrapping segments of several different axons, and label its myelin sheath and nodes of Ranvier. Show that the CNS axons have no neurolemma.

Then: Draw a cut across each axon, then write which one can regrow and the reason beside it.

B COMPARE TABLE

Draw a table with PNS and CNS across the top. Down the side: the cell that makes the myelin, how many axons one cell wraps, whether a neurolemma is present, and whether a cut axon can regrow. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a myelinated axon, write what myelin is made of, and draw the impulse jumping from node to node.

MODULE 5

Gross Anatomy and Neuronal Integration

179 CNS and PNS tissue terms

MODULE 5 · LECTURE

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

A TWO BY TWO GRID

Draw a grid with CNS and PNS across the top. Down the side: a cluster of cell bodies and a bundle of axons. Write nucleus, ganglion, tract, and nerve in the correct boxes. Add a tiny sketch in each box, with dots for cell bodies and lines for axons.

Then: Write a one-line definition under each term that uses the words cluster or bundle and CNS or PNS.

B LABELED NERVOUS SYSTEM MAP

Draw the brain and spinal cord with one nerve leaving the cord and a ganglion along that nerve outside the CNS. Draw a nucleus as a cluster of dots inside the brain and a tract as a bundle of lines running up the cord. Label all four structures.

Then: Draw a dashed line where the CNS ends and the PNS begins, and label each structure cell bodies or axons.

180 Gray and white matter

MODULE 5 · LECTURE · LAB

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

A TWO LABELED SECTIONS

Draw a spinal cord cross section and a slice through the cerebrum side by side, as on the models. Shade gray matter one way and white matter another. On the brain, label the cortex and the white matter beneath it. On the cord, label the gray matter with its dorsal and ventral horns, the outer white matter, and the central canal.

Then: Add a small cerebellum with its gray cortex, and write what gray and white matter are made of in your shading key.

B COMPARE TABLE

Draw a table with gray matter and white matter down the side. Across the top: what it is made of, where it sits in the brain, and where it sits in the spinal cord. Use a word or a tiny sketch in each cell.

Then: Below the table, draw two circles labeled brain and cord, and shade each to show which matter is on the outside.

181 Reflex arc components

MODULE 5 · LECTURE

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

A LABELED REFLEX ARC

Draw a receptor, a cross section of the spinal cord, and a muscle. Connect them with a sensory neuron running into the cord, a synapse in the cord, and a motor neuron running out to the muscle. Label the five components of the reflex arc and add arrows for the direction of the signal.

Then: Put an X at one spot on the arc and write what a missing reflex tells a clinician about that pathway.

B FLOW CHAIN

Draw a chain of five boxes for the reflex arc in order, from receptor to effector. In each box, name the component, sketch it, and write its job in a few words. Draw an arrow between each pair of boxes for the direction of the signal.

Then: Circle the box where the synapses are, and write beside the chain why a reflex is fast.

182 Types of reflexes

MODULE 5 · LECTURE

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

A COMPARE TABLE

Draw a table with six reflex types down the side: somatic, autonomic, monosynaptic, polysynaptic, innate, and learned. Across the top: what the name is based on and one example. Use a word or a tiny sketch in each cell.

Then: Draw a bracket joining each pair of types, and write the one feature that splits that pair.

B PAIRED SKETCHES

Draw three pairs of small sketches. For somatic and autonomic, show the effector each one uses. For monosynaptic and polysynaptic, show one synapse in the cord beside a chain with interneurons. For innate and learned, sketch one example of each. Label every sketch with its reflex type and its example.

Then: Write under each pair what the names are based on: the effector, the number of synapses, or whether it is inborn.

183 Neuronal pools and pathways

MODULE 5 · LECTURE

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

A LABELED CIRCUIT DRAWING

Draw a convergence circuit with many neurons feeding into one, and a divergence circuit with one neuron spreading to many, with arrows. Below them, draw a front view of the brain and spinal cord with one ascending and one descending pathway. Show each pathway crossing the midline and label the crossing decussation.

Then: Write beside the brain why a stroke on one side of the brain causes weakness on the opposite side.

B COMPARE TABLE

Draw a table with convergence, divergence, ascending pathway, descending pathway, and decussation down the side. Across the top: what it is and which way the signal travels. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a small body outline and shade the half the left side of the brain controls.

The Spinal Cord

184 Spinal cord external anatomy

MODULE 5 · LECTURE · LAB

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

A LABELED POSTERIOR VIEW

Draw the whole spinal cord from the back, running from the medulla down to its tapered end, with the vertebral column drawn faintly behind it. Label the cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, and posterior median sulcus. Mark the disc where the cord ends.

Then: Draw a small cross section beside it, label the anterior median fissure and posterior median sulcus, and show which groove is deeper.

B STRUCTURE AND LANDMARK TABLE

Draw a table with seven structures down the side: cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus. Across the top: what it looks like, where it sits on the cord, and what it does or serves. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the lower cord and mark where a lumbar puncture needle goes, then write why that spot does not risk the cord.

185 Roots, rootlets, and root ganglion

MODULE 5 · LECTURE · LAB

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

A LABELED CUTAWAY

Draw a cross section of the spinal cord with a row of rootlets leaving the back and the front on one side. Gather them into the posterior root and the anterior root, add the posterior root ganglion, and join the two roots into a spinal nerve. Label each part and the intervertebral foramen where the nerve exits.

Then: Write sensory or motor beside each root, and add arrows showing signals coming in on one root and going out on the other.

B FLOW CHAIN

Draw two chains of boxes with arrows. The sensory chain runs from a receptor in the skin through the spinal nerve, posterior root, and rootlets into the cord. The motor chain runs from the cord through the rootlets, anterior root, and spinal nerve out to a muscle. Add the posterior root ganglion and intervertebral foramen where they belong.

Then: Circle the box that holds the sensory cell bodies, and star the one nerve both chains travel in and write why it is called mixed.

186 Spinal cord internal anatomy

MODULE 5 · LAB

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

A LABELED CROSS SECTION

Draw a cross section of a thoracic segment of the cord as it looks on the slide or model, with the gray butterfly inside the white matter. Label the posterior, lateral, and anterior horns, the gray commissure, the central canal, the anterior white commissure, and the anterior, lateral, and posterior white columns.

Then: Write beside each horn, in a few words, what it holds, and write tracts inside one white column.

B LANDMARK TABLE

Draw a table with these down the side: the posterior, lateral, and anterior horns, the gray commissure, the anterior white commissure, the central canal, and the anterior, lateral, and posterior columns. Across the top: gray or white matter, how to find it on the section, and what it holds or connects. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a cervical and a thoracic section side by side and circle the lateral horn on the one that has it.

The Peripheral Nervous System

187 Nerve connective tissue wrappings

MODULE 5 · LECTURE · LAB

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

A LABELED NERVE CROSS SECTION

Draw a cross section of a peripheral nerve as it looks on the slide. Show the whole nerve with several fascicles inside it, then enlarge one fascicle so you can see a few axons. Label an axon, a fascicle, the endoneurium, the perineurium, and the epineurium.

Then: Color each wrapping a different shade and write beside each one the level it wraps: one axon, one fascicle, or the whole nerve.

B WRAPPING TABLE

Draw a table with the three wrappings down the side: endoneurium, perineurium, and epineurium. Across the top: inner, middle, or outer layer, the level it wraps (axon, fascicle, or whole nerve), and a tiny sketch of that level with the wrapping drawn around it. Use a word or a tiny sketch in each cell.

Then: Below the table, write how the perineurium and epineurium help a surgeon repair a cut nerve.

188 Spinal nerves and rami

MODULE 5 · LECTURE · LAB

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

A LABELED SPINAL NERVE

Draw a cross section of the spinal cord with one spinal nerve leaving through the intervertebral foramen. Show it splitting into the posterior ramus, the anterior ramus, and the meningeal branch, with the rami communicantes linking the anterior ramus to a sympathetic trunk ganglion. Label each branch and draw an arrow toward the territory it supplies.

Then: Beside the cord, write the number of spinal nerve pairs in each region, cervical through coccygeal, and the total.

B COUNT AND TERRITORY TABLE

Draw a table in two parts. In the top part, list the five cord regions down the side with the number of spinal nerve pairs in each, and the total. In the bottom part, list the posterior ramus, anterior ramus, meningeal branch, and rami communicantes, with the territory each supplies. Use a word or a tiny sketch in each cell.

Then: Star the ramus that forms the plexuses, and write the name and spinal levels of the thoracic nerves that do not form one.

189 Sensory receptor classification

MODULE 5 · LECTURE · LAB

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

A LABELED SKIN BLOCK

Draw a block of skin (epidermis, dermis, hypodermis) with one hair, and a skeletal muscle with its tendon underneath. Place and label free nerve endings, a tactile disc, a hair follicle receptor, a tactile corpuscle, a lamellar corpuscle, a bulbous corpuscle, a muscle spindle, and a tendon organ where the notes put them. Write what each one detects.

Then: Beside the drawing, write the three ways receptors are classified, and tag each receptor you drew as an exteroceptor or a proprioceptor.

B RECEPTOR TABLE

Draw a table with the nine named receptors down the side, from free nerve endings to joint kinesthetic receptors. Across the top: where it sits, what it detects, and its class by stimulus (such as mechanoreceptor or nociceptor). Use a word or a tiny sketch in each cell.

Then: Below the table, list the three classes by location and the three by structure, and write one receptor from your table beside each class where one fits.

The Cranial Nerves

190

Cranial nerves

MODULE 5 · LECTURE · LAB

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

A LABELED INFERIOR VIEW

Draw the inferior view of the brain and brainstem. Draw and label all twelve cranial nerves, I to XII, where each one attaches, along with the olfactory bulb and tract, optic chiasm, optic tract, and pontomedullary junction. Beside each nerve, write S, M, or B for sensory, motor, or both.

Then: Write the skull foramen at the end of each nerve, and circle the three openings that carry more than one nerve.

B CRANIAL NERVE TABLE

Draw a table with CN I to CN XII down the side. Across the top: name, class (sensory, motor, or mixed), primary function, and skull foramen. Write the names in order from memory first, using the memory sentence from the notes, then fill the other columns. Use a word or a tiny sketch in each cell.

Then: Color the rows that pass through the superior orbital fissure, the internal acoustic meatus, and the jugular foramen, using one color per opening.

MODULE 5

The Nerve Plexuses

191

Nerve plexuses

MODULE 5 · LECTURE · LAB

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

A LABELED PLEXUS MAP

Draw the spinal cord with the spinal nerves leaving one side, grouped and numbered by region. Bracket and label the cervical, brachial, lumbar, sacral, and coccygeal plexuses, and write the spinal nerves that form each one. Draw the intercostal nerves running straight out from the thoracic region without forming a plexus.

Then: On a small body outline beside it, shade in matching colors the region that loses movement and sensation when each plexus is damaged.

B DAMAGE TABLE

Draw a table with the five plexuses down the side. Across the top: the spinal nerves that form it, the region of the body it serves, and what a patient loses if it is damaged. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the phrenic nerve leaving its plexus to reach the diaphragm, and write what happens to breathing if it is damaged.

MODULE 5

The Autonomic Nervous System

192

Autonomic two-neuron pathway

MODULE 5 · LECTURE

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

A SIDE BY SIDE PATHWAYS

Draw a somatic pathway and an autonomic pathway side by side, each starting in the spinal cord. Show the somatic one as a single neuron running to skeletal muscle. Show the autonomic one as a preganglionic neuron, an autonomic ganglion, and a postganglionic neuron reaching an effector. Label the cell bodies, the myelinated and unmyelinated axons, and the synapse.

Then: Write the three autonomic effectors at the end of the autonomic pathway, and write voluntary or involuntary under each pathway.

B COMPARE TABLE

Draw a table with five features down the side: effector, control, number of motor neurons, effect on the effector, and awareness. Across the top: somatic nervous system and autonomic nervous system. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the four-step autonomic route as a chain of boxes with arrows, from the preganglionic neuron to the effector tissue.

193

Autonomic ganglia and divisions

MODULE 5 LECTURE

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

A LABELED GANGLIA MAP

Draw the brainstem and spinal cord down the middle with the sympathetic trunk beside the vertebral column. Bracket where each division leaves the CNS. Draw and label a sympathetic trunk ganglion, the prevertebral ganglia near the large abdominal arteries, and a terminal ganglion in an organ wall. Add the cardiac plexus on the heart, the celiac plexus on the abdominal organs, and the hypogastric plexus on the pelvic organs.

Then: Draw a small piece of gut wall and place the myenteric and submucosal plexuses between the layers where each one sits.

B COMPARE TABLE

Draw a table with sympathetic and parasympathetic across the top. Down the side: other name, origin in the CNS, location of its ganglia, the named ganglia that belong to it, and overall role. Use a word or a tiny sketch in each cell.

Then: Below the table, list the autonomic plexuses with what each supplies, and the two enteric plexuses with the gut layers each sits between.

PART B

The note sheets.

One sheet per competency, 61 sheets. The competency and both prompts sit on the left so nothing else has to be printed beside it. The box on the right is the whole assignment.

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133

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Kidney gross anatomy and layers

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

PICK ONE AND DO THAT ONE

☐ **A** LABELED LOCATION VIEW

Draw the back wall of the abdomen with the vertebrae from T12 to L3, and place both kidneys with the right one slightly lower. On one kidney, label the renal hilum with the ureter, renal artery, renal vein, and nerves. Around the other, draw and label the renal capsule, adipose capsule, and renal fascia.

Then: Write beside the right kidney why it sits lower, and write beside the left what retroperitoneal tells you about where it sits.

☐ **B** LAYERS TABLE

Draw a table with the renal capsule, adipose capsule, renal fascia, and renal hilum down the side. Across the top: where it sits (inner, middle, outer, or medial border), what it is made of or what passes through it, and its job. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch both kidneys between T12 and L3 with the right one lower, and mark the hilum on each.

DRAW IT

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134
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Internal kidney regions

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FRONTAL SECTION

Draw a kidney cut in a frontal section. Label the renal capsule, renal cortex, renal medulla, the renal pyramids with their bases toward the cortex, the renal columns between them, the renal papillae, minor calyces, major calyces, renal pelvis, and renal sinus. Show the ureter leaving at the hilum.

Then: Shade the parenchyma in one color and write beside it the two regions it includes and what it holds.

☐ **B** SORTING BOX

Draw two boxes, one labeled parenchyma and one labeled spaces that collect urine. Sort the cortex, renal columns, renal pyramids, renal papillae, minor calyces, major calyces, and renal pelvis into the two boxes. Beside the boxes, draw one pyramid with its papilla draining into a minor calyx, then a major calyx and the renal pelvis.

Then: Label a renal column beside your pyramid and write which region of the kidney it is an extension of.

DRAW IT

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135
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Nephron structure

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NEPHRON

Draw one nephron with its collecting duct. Label the renal corpuscle, the glomerulus, the glomerular capsule with its parietal and visceral layers, the podocytes, the capsular space, the proximal convoluted tubule, the descending and ascending limbs of the nephron loop, the distal convoluted tubule, and the collecting duct.

Then: Draw arrows for the filtrate from the capsular space to the collecting duct, and mark the segment lined with a brush border.

☐ **B** PARTS TABLE

Draw a table with the glomerulus, glomerular capsule, podocytes, capsular space, proximal convoluted tubule, nephron loop, distal convoluted tubule, and collecting duct down the side, with the renal corpuscle parts first and then the tubule parts in the order the filtrate meets them. Across the top: part of the renal corpuscle or the tubule, and one feature you would use to spot it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a podocyte wrapped around a glomerular capillary and label its foot processes and the filtration slits.

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136

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Cortical vs juxtamedullary nephrons

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

PICK ONE AND DO THAT ONE

☐ **A** SIDE BY SIDE NEPHRONS

Draw a strip of kidney with the cortex on top, the medulla below, and a line between them. Draw a cortical nephron and a juxtamedullary nephron side by side, showing how deep each nephron loop reaches. Wrap the cortical nephron in peritubular capillaries and the long juxtamedullary loop in the vasa recta.

Then: Write the percent of all nephrons each type makes up beside it, and star the type that concentrates the urine.

☐ **B** COMPARE TABLE

Draw a table with cortical nephrons and juxtamedullary nephrons down the side. Across the top: share of all nephrons, length of the loop, how far the loop reaches into the medulla, and the capillary bed that serves it. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch one renal pyramid with a short loop and a long loop drawn into it, and label each loop's nephron type.

DRAW IT

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137

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

JGA and filtration membrane

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CORPUSCLE CLOSE-UP

Draw a renal corpuscle with its afferent and efferent arterioles and the distal convoluted tubule touching them. Label the juxtaglomerular apparatus, the macula densa cells in the tubule, and the juxtaglomerular (granular) cells in the afferent arteriole. Add a zoomed box of the filtration membrane and label the fenestrated endothelium, basement membrane, and slit membranes.

Then: Draw a blood cell, a large protein, and a medium protein, each stopped at the layer that holds it back.

☐ **B** WHERE AND WHAT TABLE

Draw a table with the macula densa cells, the juxtaglomerular cells, and the three layers of the filtration membrane down the side, the layers in order from the blood to the capsular space. Across the top: where it sits, and what it senses, releases, or holds back. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the distal convoluted tubule touching the arterioles of its own corpuscle and circle where the juxtaglomerular apparatus forms.

DRAW IT

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138
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Kidney blood supply pathway

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VESSEL CUTAWAY

Draw a frontal section of a kidney showing two pyramids and the cortex. Draw and label the renal artery at the hilum, the segmental arteries branching in the renal sinus, the interlobar arteries running between the pyramids, the arcuate arteries along the pyramid bases, and the interlobular (cortical radiate) arteries up into the cortex. Add one nephron with its afferent arteriole, glomerulus, efferent arteriole, and peritubular capillaries.

Then: Draw the veins in a second shade beside the arteries and label the cortical radiate vein, arcuate vein, and renal vein.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows: renal artery, segmental and interlobar arteries, arcuate artery, interlobular (cortical radiate) artery, afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries and vasa recta, cortical radiate vein, arcuate vein, interlobar vein, renal vein. Under each artery box, write a word or two saying where it runs.

Then: Circle the two capillary beds in your chain and box the one vessel that sits between them.

DRAW IT

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139

COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE AND LAB

Ureters, bladder, urethra and urine path

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

PICK ONE AND DO THAT ONE

☐ **A** LABELED URINARY TRACT

Draw a kidney, ureter, and bladder, with the kidney and bladder cut open. Label the minor and major calyces, renal pelvis, ureter, detrusor muscle in the bladder wall, the trigone with the two ureteral openings and the internal urethral orifice, the internal urethral sphincter, and the external urethral sphincter. Add arrows for urine from the collecting duct to the outside.

Then: Beside the bladder, sketch a male and a female urethra, and label the prostatic, membranous, and spongy parts of the male one.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows for urine: collecting duct, renal papilla, minor calyx, major calyx, renal pelvis, ureter, urinary bladder, urethra. Under the bladder box, list the trigone, detrusor muscle, and both urethral sphincters. Split the urethra box into male (prostatic, membranous, spongy) and female, and write which one is longer.

Then: Write beside each sphincter whether it is smooth or skeletal muscle and whether you control it.

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140
COMP

MODULE 5 · THE URINARY SYSTEM · LECTURE

Clinical anatomy of urinary disorders

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw the urinary tract from the kidneys to the urethra, with the adipose capsule around one kidney. Write each disorder on the structure it affects: nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, renal failure, and polycystic kidney disease. Draw one stone in the renal pelvis and another lodged in the ureter.

Then: Draw a dropped kidney beside your map with a kinked ureter, and write what loss of fat has to do with it.

☐ **B** DISORDER TABLE

Draw a table with nephroptosis, kidney stones, urinary tract infection, glomerulonephritis, and polycystic kidney disease down the side. Across the top: the structure it affects and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a short female urethra beside a long male urethra and write why urinary tract infections are more common in women.

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141

COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Scrotum and testis structure

You should be able to: Identify the scrotum, dartos and cremaster muscles, tunica layers, seminiferous tubules, sustentacular and interstitial cells, rete testis, and efferent ductules.

PICK ONE AND DO THAT ONE

☐ **A** LABELED TESTIS SECTION

Draw the scrotum with one testis cut open inside it. Label the scrotal septum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, lobules, and seminiferous tubules. Add a zoomed circle of one tubule cut across, and label the sustentacular cells in its wall and the interstitial cells between the tubules.

Then: Write beside the dartos and the cremaster which kind of muscle each one is and what it moves.

☐ **B** STRUCTURE TABLE

Draw a table with the scrotum, dartos muscle, cremaster muscle, tunica vaginalis, tunica albuginea, seminiferous tubules, sustentacular cells, and interstitial cells down the side. Across the top: where it sits (scrotum, covering the testis, or inside the testis) and its job. Use a word or a tiny sketch in each cell.

Then: Circle the row whose cells produce testosterone, and star the two rows that set the temperature of the testes.

DRAW IT

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142

COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Sperm duct system and pathway

You should be able to: Trace the path of sperm from the seminiferous tubules to the outside, naming the epididymis, ductus deferens, ejaculatory duct, urethra, and spermatic cord contents.

PICK ONE AND DO THAT ONE

☐ **A** LABELED DUCT VIEW

Draw a side view of the male pelvis with the testis, the epididymis on it, and the ductus deferens running up out of the scrotum in the spermatic cord. Label the seminal vesicle, the ejaculatory duct, the prostate, and the prostatic, membranous, and spongy urethra. Add arrows for sperm from the seminiferous tubules to the outside.

Then: Draw the spermatic cord cut across as a circle, and label the ductus deferens, blood vessels, nerves, and cremaster muscle inside it.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows: seminiferous tubules, epididymis, ductus deferens, ejaculatory duct, prostatic urethra, membranous urethra, spongy urethra. Under each box, write in a word or two what happens to the sperm there. Box the one structure that produces sperm.

Then: Draw a bracket over the duct that runs inside the spermatic cord, and list the cord's other three contents beside it.

DRAW IT

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143 COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Sperm cell structure

You should be able to: Identify the head, acrosome, neck, middle piece, principal piece, and end piece of a sperm cell.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SPERM CELL

Draw one sperm cell large enough to fill the box. Label the head with its nucleus, the acrosome capping the head, the neck, the middle piece, the principal piece, and the end piece. Draw the middle piece packed with small mitochondria and the end piece tapering to a point.

Then: Write beside each part what it holds or does, in three words or fewer.

☐ **B** PARTS TABLE

Draw a table with the head, acrosome, neck, middle piece, principal piece, and end piece down the side, in order from front to back. Across the top: how you spot it and what it holds or does. Use a word or a tiny sketch in each cell.

Then: Draw a bracket beside the rows that make up the tail, and star the longest part of the tail.

DRAW IT

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144

COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Accessory glands and penis

You should be able to: Name the seminal vesicles, prostate, and bulbourethral glands and identify the root, body, glans, prepuce, and three erectile columns of the penis.

PICK ONE AND DO THAT ONE

☐ **A** LABELED PENIS VIEW

Draw a side view of the penis and a cross section of its body. On the side view, label the root, body, glans penis, and prepuce. On the cross section, label the two corpora cavernosa on the dorsal side and the corpus spongiosum on the ventral side surrounding the spongy urethra.

Then: Add the seminal vesicles, prostate, and bulbourethral glands as labeled boxes beside your drawing, and write what fluid each adds to semen.

☐ **B** TWO PART TABLE

Draw a table in two parts. First, put the seminal vesicles, prostate, and bulbourethral glands down the side, with what each adds to semen across the top. Second, put the root, body, glans penis, prepuce, corpora cavernosa, and corpus spongiosum down the side, with what each is. Use a word or a tiny sketch in each cell.

Then: Star the gland that adds the largest share of semen, and circle the erectile column that forms the glans.

DRAW IT

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145
COMP

MODULE 5 · THE MALE REPRODUCTIVE SYSTEM · LECTURE

Clinical anatomy of male disorders

You should be able to: Relate male reproductive disorders such as cryptorchidism and benign prostatic hyperplasia to the affected structure and its position.

PICK ONE AND DO THAT ONE

☐ **A** DISORDER MAP

Draw a side view of the male pelvis with the testis in the scrotum, the inguinal canal, and the urethra passing through the prostate. Write each disorder on the structure it affects: cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, prostate cancer, testicular cancer, and inguinal hernia.

Then: Draw an undescended testis with a dotted line to where it should sit, and write why staying at body temperature harms it.

☐ **B** DISORDER TABLE

Draw a table with cryptorchidism, hydrocele, varicocele, benign prostatic hyperplasia, and inguinal hernia down the side. Across the top: the structure it affects, where that structure sits, and what goes wrong there. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an enlarged prostate squeezing the urethra and write the urinary symptom it causes.

DRAW IT

split it up however the prompt needs

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146

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Ovary structure and follicle stages

You should be able to: Identify the ovary layers, ligaments, and follicle stages including the corpus luteum and corpus albicans.

PICK ONE AND DO THAT ONE

☐ **A** LABELED OVARY VIEW

Draw the uterus with one ovary beside it. Label the ovarian ligament, suspensory ligament, broad ligament, and mesovarium. Draw the ovary cut open and label the germinal epithelium, tunica albuginea, cortex, and medulla. In the cortex, draw ovarian follicles, a mature follicle, a corpus luteum, and a corpus albicans.

Then: Number the follicle structures in the order they appear, and mark where the oocyte leaves at ovulation.

☐ **B** SEQUENCE STRIP

Draw a row of circles showing a follicle growing into a mature follicle, releasing its oocyte, then becoming a corpus luteum and finally a corpus albicans. Beneath the row, list the ovary layers from the outside in, then the three ligaments with what each one attaches the ovary to.

Then: Circle the layer of the ovary where every follicle in your strip is found.

DRAW IT

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147

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Uterine tubes and ovum pathway

You should be able to: Trace the oocyte from the ovary to the uterus, naming the fimbriae, infundibulum, ampulla, and isthmus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED TUBE VIEW

Draw the uterus with one uterine tube and the ovary beside it. Label the fimbriae, infundibulum, ampulla, and isthmus, and show the infundibulum open to the pelvic cavity near the ovary. Draw an oocyte leaving a mature follicle, with arrows following it through the tube into the uterine cavity.

Then: Star where fertilization usually happens, and draw cilia on the tube lining with an arrow showing which way they move the oocyte.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows: ovary, fimbriae and infundibulum, ampulla, isthmus, uterus. Under each box, write what happens to the oocyte there and one feature of that region, such as its shape or width. Use a word or a tiny sketch under each box.

Then: Draw a dotted arrow from the open infundibulum into the pelvic cavity, and write one disorder this open route explains.

DRAW IT

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148

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Uterus regions, ligaments, and wall

You should be able to: Name the regions and supporting ligaments of the uterus and identify the perimetrium, myometrium, and endometrium.

PICK ONE AND DO THAT ONE

☐ **A** LABELED UTERUS VIEW

Draw the uterus cut open from the front, with the uterine tubes and the top of the vagina. Label the fundus, body, cervix, uterine cavity, and cervical canal with the internal and external os. Label the perimetrium, myometrium, and endometrium in the wall. Add the broad, round, uterosacral, and cardinal ligaments as lines running to what each attaches to.

Then: Zoom in on the endometrium and split it into the layer that sheds each month and the layer that rebuilds it.

☐ **B** REGION STACK AND TABLE

Draw a stack of three boxes from top to bottom for the fundus, body, and cervix, with the uterine cavity and cervical canal drawn inside. Beside it, draw a table with the four ligaments and the three wall layers down the side. Across the top: what it connects to or what it is made of. Use a word or a tiny sketch in each cell.

Then: Star the wall layer that contracts during childbirth, and circle the ligament that runs to the labia majora.

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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149

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Vagina and external genitalia

You should be able to: Identify the vagina, fornix, hymen, and the vulvar structures including the clitoris, vestibule, labia, and greater vestibular glands.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VULVA VIEW

Draw the vulva in an inferior view, with the mons pubis at the top. Label the labia majora, labia minora, clitoris, vestibule, urethral orifice, vaginal orifice, hymen, and the greater vestibular glands beside the vaginal opening. Outline the perineum around the genitalia and the anus.

Then: Sketch a side view of the vagina meeting the cervix, and label the fornix around the cervix.

☐ **B** STRUCTURE TABLE

Draw a table with the vagina, fornix, hymen, mons pubis, labia majora, labia minora, clitoris, vestibule, and greater vestibular glands down the side. Across the top: where it sits and what it is or does. Use a word or a tiny sketch in each cell.

Then: Mark the three structures that are homologous to male structures, and write the male match beside each.

DRAW IT

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150

COMP

MODULE 5 · THE FEMALE REPRODUCTIVE SYSTEM · LECTURE AND LAB

Mammary gland structure

You should be able to: Identify the lobes, lobules, alveoli, lactiferous ducts, nipple, areola, and suspensory ligaments of the mammary gland.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BREAST SECTION

Draw a side view of a breast cut open, sitting over the chest muscles. Label two or three lobes, the lobules inside one lobe, the alveoli inside one lobule, the lactiferous ducts running to the nipple, the areola, and the suspensory ligaments.

Then: Draw arrows for the path of milk from the alveoli to the nipple, and label each structure the arrows pass.

☐ **B** NESTING DIAGRAM

Draw nested shapes: alveoli inside a lobule, lobules inside a lobe, and the lobe draining by a lactiferous duct to the nipple. Beside it, draw a small table with the nipple, areola, and suspensory ligaments down the side and what each is or does across the top. Use a word or a tiny sketch in each cell.

Then: Write beside your lobe how many lobes one mammary gland has.

DRAW IT

split it up however the prompt needs

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151

COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Brain gross organization

You should be able to: Identify the four major brain divisions cerebrum, diencephalon, brainstem, and cerebellum, and distinguish the cortical gray matter from the deep white matter on a specimen.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the brain cut down the middle and seen from the side. Label the cerebrum, diencephalon, brainstem, and cerebellum, and outline each one so the four divisions are easy to tell apart. Along the edge of the cerebrum, draw a thin outer band of gray matter over a thick core of white matter, and label both.

Then: Write beside the gray matter band what it is made of, and beside the white matter what it is made of.

☐ **B** DIVISION TABLE

Draw a table with the cerebrum, diencephalon, brainstem, and cerebellum down the side. Across the top: where it sits and its main job. Use a word or a tiny sketch in each cell. Below the table, draw a small block of brain tissue with gray matter on the outside and white matter on the inside, and label both.

Then: Star the largest division and the second largest division, and write the cranial fossa where the cerebellum sits.

DRAW IT

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152
COMP

MODULE 5 · THE BRAIN · LECTURE

Neural tube brain development

You should be able to: Trace brain development from the neural tube through the primary and secondary vesicles and match each secondary vesicle to its adult structures and ventricle.

PICK ONE AND DO THAT ONE

☐ **A** BRANCHING FLOW CHAIN

Draw the neural tube at the top of the box. Branch it into the three primary vesicles, writing each one's formal name and common name (forebrain, midbrain, hindbrain). Branch those into the five secondary vesicles. Under each secondary vesicle, write the adult structures it becomes and its ventricle or fluid space.

Then: Write the germ layer the neural tube forms from, and circle the one primary vesicle that carries through without splitting.

☐ **B** VESICLE TABLE

Write neural tube at the top of the box. Below it, draw a table with the five secondary vesicles down the side. Across the top: the primary vesicle it came from, the adult structures it becomes, and its ventricle or fluid space. Use a word or a tiny sketch in each cell.

Then: Draw a simple side view of the adult brain and number each region 1 to 5 to match the secondary vesicle it grew from.

DRAW IT

split it up however the prompt needs

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153
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebral surface and lobes

You should be able to: Identify the cerebral hemispheres, five lobes, gyri, sulci, and the named central, lateral, and parieto-occipital sulci along with the longitudinal and transverse fissures on a specimen.

PICK ONE AND DO THAT ONE

☐ **A** SIDE AND TOP VIEWS

Draw the left side of the brain. Label the frontal, parietal, temporal, and occipital lobes, a gyrus, a sulcus, the precentral and postcentral gyri, and the central, lateral, and parieto-occipital sulci. Draw the insula as a dashed shape inside the lateral sulcus. Show the transverse fissure above the cerebellum. Beside it, draw a small top view and label both hemispheres and the longitudinal fissure.

Then: Write beside each of the four outer lobes the cranial bone that lies over it and gives it its name.

☐ **B** BOUNDARY TABLE

Sketch three folds of cortex and label a gyrus, a sulcus, and a fissure. Below, draw a table with the central sulcus, lateral sulcus, parieto-occipital sulcus, longitudinal fissure, and transverse fissure down the side. Across the top: the region on one side of it and the region on the other. Use a word or a tiny sketch in each cell.

Then: Draw the brain seen from above, label the longitudinal fissure, central sulcus, and each lobe you can see, then name the lobe no outside view shows.

DRAW IT

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154
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Functional cortical areas

You should be able to: Map the motor, sensory, and association areas of the cortex to their gyri and lobes, including primary motor, primary somatosensory, primary visual, auditory, gustatory, and olfactory cortices, Broca and Wernicke areas, and the prefrontal cortex.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CORTEX MAP

Draw the side view of the hemisphere where Broca's and Wernicke's areas are usually found. Draw the central and lateral sulci. Label the primary motor cortex, primary somatosensory cortex, Broca's area, Wernicke's area, prefrontal cortex, primary auditory cortex, and primary visual cortex. Draw the insula dashed for the gustatory cortex, and point an arrow to the medial temporal lobe for the olfactory cortex.

Then: Write M, S, or A on each labeled area to sort it as a motor, sensory, or association area.

☐ **B** AREA LOCATION TABLE

Draw a table with these areas down the side: primary motor, primary somatosensory, primary visual, primary auditory, primary gustatory, and primary olfactory cortex, Broca's area, Wernicke's area, and the prefrontal cortex. Across the top: lobe, gyrus or landmark, motor, sensory, or association, and job. Use a word or a tiny sketch in each cell.

Then: Add a row for each association area the notes place next to a primary area, and name the primary area it sits beside.

DRAW IT

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155

COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebral white matter tracts

You should be able to: Compare association, commissural, and projection fibers by what each connects and identify the corpus callosum and internal capsule on a specimen.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FRONTAL SECTION

Draw a frontal section through both hemispheres with gray cortex on the outside and white matter inside. Draw and label an association fiber arching between two gyri on the same side, the corpus callosum crossing between the hemispheres, and the projection fibers of the internal capsule running down beside the basal ganglia toward the brainstem.

Then: Shade the internal capsule and write beside it why a small stroke there can weaken a whole side of the body.

☐ **B** FIBER CLASS TABLE

Draw a table with association, commissural, and projection fibers down the side. Across the top: what it connects, a named example, and a tiny arrow sketch of its path. Use a word or a tiny sketch in each cell. Below, sketch a midline view with the corpus callosum and a frontal section with the internal capsule, and label both.

Then: Circle the largest commissural tract, and write which tract links Broca's and Wernicke's areas and which fiber class it belongs to.

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156
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Basal ganglia

You should be able to: Identify the caudate nucleus, putamen, and globus pallidus and locate the functionally grouped substantia nigra and subthalamic nucleus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FRONTAL SECTION

Draw a frontal section of one hemisphere with cortex on the outside, white matter inside, and deep masses of gray matter buried in the white matter. Label the caudate nucleus, the putamen, and the globus pallidus medial to the putamen. Below the hemisphere, sketch the diencephalon and midbrain, and mark the subthalamic nucleus and the substantia nigra.

Then: Draw dotted lines linking all five nuclei, and write the three roles the basal ganglia share.

☐ **B** LOCATION SORT

Draw three boxes stacked from top to bottom labeled cerebrum, diencephalon, and midbrain. Write each nucleus in the box where it sits: caudate nucleus, putamen, globus pallidus, subthalamic nucleus, and substantia nigra. Circle the three core nuclei and underline the two grouped with them only by function. Add an arrow showing the putamen lateral to the globus pallidus.

Then: Write beside the substantia nigra what it looks like and the disease its loss causes.

DRAW IT

split it up however the prompt needs

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157
COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Diencephalon

You should be able to: Identify the thalamus with the massa intermedia, the hypothalamus, the epithalamus with the pineal gland, and the third ventricle they enclose.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the thalamus with the massa intermedia, the hypothalamus below it, the epithalamus at the back with the pineal gland, and the third ventricle. Add the corpus callosum above, the infundibulum and optic chiasm below the hypothalamus, and the midbrain below as landmarks.

Then: Shade the third ventricle, and write the percent of the diencephalon the thalamus makes up.

☐ **B** THREE PART TABLE

Draw a table with the thalamus, hypothalamus, and epithalamus down the side. Across the top: where it sits, what it contains, and its main job. Use a word or a tiny sketch in each cell. Below the table, draw the third ventricle as a midline cavity and label it.

Then: Draw the right and left thalami on each side of your third ventricle, and join them across it with the massa intermedia.

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebellum

You should be able to: Identify the cerebellar hemispheres, vermis, folia, arbor vitae, deep nuclei, and the three cerebellar peduncles with their brainstem connections.

PICK ONE AND DO THAT ONE

☐ **A** BACK AND CUT VIEWS

Draw the cerebellum from behind and label the two hemispheres, the vermis, and the folia. Beside it, draw the cerebellum cut down the midline and attached to the brainstem. Label the cerebellar cortex, arbor vitae, a deep cerebellar nucleus, and the superior, middle, and inferior cerebellar peduncles as stalks running to the midbrain, pons, and medulla.

Then: Shade the gray matter on your cut view and leave the white matter blank.

☐ **B** PARTS TABLE

Draw a table with these down the side: cerebellar hemispheres, vermis, folia, arbor vitae, deep cerebellar nuclei, and the superior, middle, and inferior cerebellar peduncles. Across the top: what it is, gray or white matter, and where it sits or what it joins. Use a word or a tiny sketch in each cell.

Then: Draw an arrow on each peduncle row showing whether it mostly carries signals into the cerebellum or out of it.

DRAW IT

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Limbic system

You should be able to: Locate the cingulate gyrus, amygdala, and hippocampus as the limbic structures ringing the diencephalon.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the brain cut down the midline. Label the corpus callosum, the cingulate gyrus arching above it, and the diencephalon beneath it. Draw the hippocampus and amygdala as dashed shapes curving down and forward below the diencephalon, so that the three limbic structures form a ring around it.

Then: Write beside each limbic structure its main role from the notes.

☐ **B** LIMBIC COMPARE TABLE

Draw a table with the cingulate gyrus, amygdala, and hippocampus down the side. Across the top: where it sits, its role, and a tiny sketch. Use a word or a tiny sketch in each cell. Below the table, draw a small oval for the diencephalon and arrange the three structures around it as a ring.

Then: Write in the center of your ring the two jobs the limbic system does as a whole.

DRAW IT

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COMP

MODULE 5 · THE BRAIN · LECTURE AND LAB

Cerebral arterial supply

You should be able to: Trace the anterior and posterior circulations through the Circle of Willis and identify the anterior, middle, and posterior cerebral arteries, the communicating arteries, the internal carotid, vertebral, and basilar arteries, and the cerebellar branches.

PICK ONE AND DO THAT ONE

☐ **A** LABELED INFERIOR VIEW

Draw the base of the brain seen from below with the brainstem in the middle. Draw the vertebral arteries joining into the basilar artery, then the ring: posterior cerebral, posterior communicating, internal carotid, anterior cerebral, and anterior communicating arteries. Add the middle cerebral arteries and the cerebellar branches: posterior inferior, anterior inferior, and superior cerebellar arteries.

Then: Shade the anterior circulation one way and the posterior circulation another, and write what each one feeds.

☐ **B** TWO CIRCULATION CHAINS

Draw two flow chains side by side. Anterior: internal carotid artery to the anterior and middle cerebral arteries. Posterior: vertebral arteries to basilar artery to posterior cerebral arteries, with the posterior inferior cerebellar branch off each vertebral and the anterior inferior and superior cerebellar branches off the basilar. Join the two chains with the posterior communicating artery, and add the anterior communicating artery joining the right and left anterior cerebral arteries.

Then: Write beside each cerebral artery the part of the brain it supplies.

DRAW IT

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Brainstem regions

You should be able to: Locate the midbrain, pons, and medulla oblongata in order and describe how the brainstem connects the higher brain to the spinal cord.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SIDE VIEW

Draw the brainstem from the side with the diencephalon above it, the cerebellum behind it, and the spinal cord below. Label the midbrain, pons, and medulla oblongata from top to bottom, the basilar pons, and the pontomedullary junction. Mark the foramen magnum where the medulla becomes the spinal cord.

Then: Draw one arrow up and one arrow down through all three regions, and label them ascending sensory tracts and descending motor tracts.

☐ **B** REGION ORDER CHAIN

Draw a vertical chain of boxes from top to bottom: cerebrum and diencephalon, midbrain, pons, medulla oblongata, and spinal cord, with the cerebellum to the side. In each brainstem box, write its position (most superior, middle, most inferior) and one structure you can find on it on the model.

Then: Write beside the chain what passes through all three regions and how many cranial nerves have nuclei in the brainstem.

DRAW IT

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Midbrain structures

You should be able to: Identify the tectum with superior and inferior colliculi, the cerebral peduncles, the substantia nigra, and the cerebral aqueduct on a midbrain specimen.

PICK ONE AND DO THAT ONE

☐ **A** TWO LABELED VIEWS

Draw the midbrain from behind and in cross section. On the back view, label the tectum and its four colliculi, with the superior pair above the inferior pair. On the cross section, label the tectum, the cerebral aqueduct, the darkly pigmented substantia nigra, and the cerebral peduncles on the front.

Then: Draw an arrow into and out of the cerebral aqueduct, and name the ventricle at each end.

☐ **B** MIDBRAIN STRUCTURE TABLE

Draw a table with the tectum, superior colliculi, inferior colliculi, cerebral peduncles, substantia nigra, and cerebral aqueduct down the side. Across the top: how to spot it on the model, and its job. Use a word or a tiny sketch in each cell.

Then: Circle the two reflex centers, and write beside them whether you become conscious of the sight or sound there.

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Pons structures

You should be able to: Identify the pons, its pontine nuclei, the middle cerebellar peduncles, and the fourth ventricle behind it.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SIDE VIEW

Draw the brainstem from the left side with the cerebellum behind it. Label the midbrain, pons, and medulla, the basilar pons, and the thick middle cerebellar peduncle running back into the cerebellum. Dot the pontine nuclei inside the pons. Draw the fourth ventricle as a dashed space behind the pons and upper medulla.

Then: Draw an arrow from the cerebral cortex through the pontine nuclei and the middle cerebellar peduncle into the cerebellum.

☐ **B** PONS FLOW CHAIN

Draw a flow chain for motor information: cerebral cortex, pontine nuclei, middle cerebellar peduncles, cerebellum. Write what each step does in a few words. Under the chain, sketch the pons with the midbrain above, the medulla below, and the fourth ventricle behind it, and label all four.

Then: Write what the word pons means, and name the three regions it bridges.

DRAW IT

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Medulla oblongata structures

You should be able to: Identify the pyramids, olives, and the decussation of the pyramids on the medulla and locate the vital cardiac, respiratory, and vasomotor centers.

PICK ONE AND DO THAT ONE

☐ **A** LABELED ANTERIOR VIEW

Draw the brainstem from the front with the spinal cord below. Label the pons, the pontomedullary junction, the medulla oblongata, the two pyramids, the medullary olive on each side of the pyramids, and the decussation of the pyramids where the medulla meets the cord. Inside the medulla, draw and label boxes for the cardiac, respiratory, and vasomotor centers.

Then: Draw crossing arrows at the decussation, and write why a stroke in the right hemisphere weakens the left side of the body.

☐ **B** MEDULLA TABLE

Draw a table with the pyramids, medullary olive, decussation of the pyramids, cardiac center, respiratory center, and vasomotor center down the side. Across the top: seen on the surface or inside the medulla, and what it looks like or what it does. Use a word or a tiny sketch in each cell.

Then: Star the three vital centers, and write why injury to the medulla is immediately life threatening.

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COMP

MODULE 5 · THE BRAINSTEM · LECTURE AND LAB

Cranial nerve nuclei distribution

You should be able to: Match cranial nerve nuclei III through XII to the brainstem region that houses them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED BRAINSTEM VIEW

Draw the brainstem from the front, divided into the midbrain, pons, and medulla oblongata, and label the pontomedullary junction. Beside each region, write the Roman numerals and names of the cranial nerves whose nuclei it holds, from CN III down to CN XII.

Then: Write which two cranial nerves have no nuclei in the brainstem, and circle the one nerve you wrote beside two regions.

☐ **B** NERVE SORT TABLE

Draw a table with CN III through CN XII down the side. Across the top: the nerve's name, midbrain, pons, and medulla oblongata. Write each name, then put a check in the region that houses that nerve's nuclei. Use a word or a check in each cell.

Then: Count how many of the twelve cranial nerves have nuclei in the brainstem, and circle the nerve with checks in two regions.

DRAW IT

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166

COMP

MODULE 5 · THE BRAINSTEM · LECTURE

Reticular formation

You should be able to: Locate the reticular formation as a net of gray matter through the brainstem core and distinguish its ascending reticular activating system from its descending component.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CUTAWAY

Draw the brainstem from the side, cut open to show its core. Draw the reticular formation as a net of dots and lines woven through the midbrain, pons, and medulla. Draw arrows up from the net to the cerebral cortex labeled reticular activating system, and arrows down toward the spinal cord labeled descending component.

Then: Write beside each set of arrows two of its jobs from the notes.

☐ **B** COMPARE TABLE

Draw a table with the reticular activating system and the descending component down the side. Across the top: which way it sends signals, what it acts on, and its jobs. Use a word or a tiny sketch in each cell. Above the table, sketch the brainstem core with a gray matter net running through its white matter.

Then: Write under the table what happens when the reticular activating system shuts down and what severe damage to it can cause.

DRAW IT

split it up however the prompt needs

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167

COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Cranial meninges and dural reflections

You should be able to: Identify the dura, arachnoid, and pia mater and the dural reflections falx cerebri, falx cerebelli, and tentorium cerebelli, and trace dural venous sinus drainage from the superior sagittal sinus to the internal jugular vein.

PICK ONE AND DO THAT ONE

☐ **A** LABELED MIDSAGITTAL VIEW

Draw the inside of the skull in midsagittal view with the brain removed. Label the falx cerebri, falx cerebelli, and tentorium cerebelli. Draw the superior sagittal sinus on the falx cerebri, then the confluence of sinuses, a transverse sinus, a sigmoid sinus, and the internal jugular vein. In a corner, draw the dura, arachnoid, and pia from skull to brain.

Then: Add arrows along the sinuses for the direction blood drains, and label the periosteal and meningeal layers on your dura strip.

☐ **B** TABLE AND FLOW CHAIN

Draw a table with the dura mater, arachnoid mater, pia mater, falx cerebri, falx cerebelli, and tentorium cerebelli down the side. Across the top: where it sits and what it touches or separates. Use a word or a tiny sketch in each cell. Below the table, draw a chain of boxes from the superior sagittal sinus to the internal jugular vein.

Then: Mark the box in your chain where the arachnoid granulations return CSF to the blood.

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DRAW IT

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Dr. Sharilyn Rennie

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Spinal meninges and meningeal spaces

You should be able to: Identify the spinal meninges, the denticulate ligaments and filum terminale, and distinguish the epidural, subdural, and subarachnoid spaces by location and contents.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section of a vertebra with the spinal cord inside. Label the vertebral bone, the epidural space with fat and veins, the dura mater, subdural space, arachnoid mater, subarachnoid space with CSF, and pia mater. Draw a denticulate ligament from the pia to the dura. Beside it, sketch the lower cord with the filum terminale running to the coccyx.

Then: Star the space that holds CSF, and write potential space beside the space that is only a potential space.

☐ **B** SPACE COMPARE TABLE

Draw a table with the epidural, subdural, and subarachnoid spaces down the side. Across the top: the two layers it sits between, what fills it, and whether it is also found in the cranium. Use a word or a tiny sketch in each cell. Below the table, sketch the cord with two denticulate ligaments and the filum terminale.

Then: Write beside each anchor which meningeal layer it is made of and what it attaches the cord to.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Cord termination and lumbar puncture

You should be able to: Identify the conus medullaris, cauda equina, and lumbar cistern and list in order the layers a lumbar puncture needle crosses from skin to subarachnoid space.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SIDE VIEW

Draw a side view of the lower vertebral column with the spinal cord inside. Label the conus medullaris at the L1 to L2 disc, the cauda equina, the filum terminale, and the lumbar cistern below the cord. Draw a needle entering in the midline between the L3 and L4 spines, and label each layer it crosses, in order, from the skin to the subarachnoid space.

Then: Write beside the conus medullaris why the needle is placed between the L3 and L4 or L4 and L5 spines.

☐ **B** NEEDLE PATH SEQUENCE

Draw a vertical stack of boxes, one for each layer a lumbar puncture needle crosses, in order from the skin at the top to the subarachnoid space at the bottom. Name the layer and sketch what it looks like in each box. Beside the bottom box, sketch the lumbar cistern with the floating cauda equina, the conus medullaris above it, and the filum terminale.

Then: Mark the box where a clinician feels a firm ligament give way, and write the two spine levels where the needle can enter.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Ventricular system

You should be able to: Identify the lateral, third, and fourth ventricles and the central canal and describe how the interventricular foramina, cerebral aqueduct, and fourth ventricle apertures connect them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED VENTRICLE VIEW

Draw a side view of the brain outline with the ventricles inside it, as on the brain model. Label a lateral ventricle, an interventricular foramen of Monro, the third ventricle, the cerebral aqueduct, the fourth ventricle, and the central canal of the spinal cord. Draw the median aperture and the two lateral apertures opening from the fourth ventricle into the subarachnoid space.

Then: Write beside each ventricle the region of the brain it sits in, and write what lines every chamber.

☐ **B** CONNECTION TABLE

Draw a table with the lateral ventricles, third ventricle, fourth ventricle, and central canal down the side. Across the top: where it is, what it connects to next, and the opening or channel that makes the connection. Use a word or a tiny sketch in each cell.

Then: Circle the three narrow connection points in your table, and write beside the table what would happen if one were blocked.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

CSF production and circulation

You should be able to: Trace CSF from production at the choroid plexus through the ventricles and subarachnoid space to reabsorption at the arachnoid granulations.

PICK ONE AND DO THAT ONE

☐ **A** LABELED FLOW DRAWING

Draw a midsagittal view of the brain and upper spinal cord showing the ventricles, the subarachnoid space, and the superior sagittal sinus. Draw choroid plexus in the lateral, third, and fourth ventricles. Add numbered arrows for CSF from the lateral ventricles through the interventricular foramina, third ventricle, cerebral aqueduct, fourth ventricle, apertures, and subarachnoid space to the arachnoid granulations.

Then: Add a side arrow into the central canal, and write which cells of the choroid plexus produce the CSF.

☐ **B** FLOW CHAIN

Draw a chain of boxes with arrows for the one-way loop of CSF, from production at the choroid plexus of the lateral ventricles to reabsorption at the arachnoid granulations. Give each ventricle, opening, and space its own box. Add a tiny sketch in each box of what the structure looks like.

Then: Mark each box that holds choroid plexus, and add a final arrow into the venous blood of the superior sagittal sinus.

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COMP

MODULE 5 · MENINGES AND CEREBROSPINAL FLUID · LECTURE AND LAB

Blood–brain barrier structure

You should be able to: Name the three structural components of the blood-brain barrier the tight-junctioned capillary endothelium, the basement membrane, and the astrocyte end-feet.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CAPILLARY SECTION

Draw a cross section of a brain capillary with blood in its center. Label the capillary endothelial cells joined by tight junctions, the basement membrane wrapped around them, and the astrocyte end-feet covering the outside. Draw one whole star-shaped astrocyte with its end-feet reaching the capillary and a neuron nearby.

Then: Draw an arrow from the blood toward the neuron, and number each barrier component the arrow has to cross.

☐ **B** COMPARE TABLE

Draw a table with the three components of the blood-brain barrier down the side. Across the top: what it is, its position counting outward from the blood, and one feature that makes it part of the barrier. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch an ordinary capillary beside a brain capillary and label the pores on the ordinary capillary that the brain capillary lacks.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE

Nervous system divisions

You should be able to: Outline the organization of the nervous system into CNS and PNS and its somatic, autonomic, and enteric motor divisions.

PICK ONE AND DO THAT ONE

☐ **A** BRANCHING MAP

Draw a branching map that starts with the nervous system and splits into the CNS and PNS. Under the CNS, write the brain and spinal cord. Under the PNS, list nerves, ganglia, and sensory receptors, then branch into the somatic, autonomic, and enteric motor divisions. Split the autonomic division into sympathetic and parasympathetic.

Then: Write beside each motor division the effector it controls, and add the number of pairs of cranial and spinal nerves.

☐ **B** LABELED BODY OUTLINE

Draw a body outline with the brain and spinal cord shaded as the CNS. Draw a few nerves leaving the cord, one ganglion, and a sensory receptor in the skin, and label them as PNS. Draw a skeletal muscle, the heart, a gland, and a loop of gut wall with its own neurons.

Then: Label each effector with the motor division that controls it: somatic, autonomic, or enteric.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Neuron structure

You should be able to: Identify the cell body, dendrites, axon, axon hillock, Nissl bodies, and axon terminals of a neuron and state the direction a signal travels through them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NEURON

Draw a multipolar neuron as it looks on the neuron model. Label the cell body with its nucleus and nucleolus, the Nissl bodies, the dendrites, the axon hillock, the axon, and the axon terminals with their synaptic bulbs.

Then: Add arrows along the neuron for the direction a signal travels, from the dendrites through to the axon terminals.

☐ **B** SIGNAL FLOW CHAIN

Draw a chain of boxes in the order a signal travels through a neuron: dendrites, cell body, axon hillock, axon, and axon terminals. In each box, sketch the part and write its job in a word or two. Draw Nissl bodies inside the cell body box and write what they are made of and what they do.

Then: Add neurofibrils to the cell body box, and write why an axon dies when it is cut away from its cell body.

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COMP

MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Synapse structures

You should be able to: Identify the components of a chemical synapse including synaptic end bulbs, synaptic vesicles, and synaptic cleft, and name the three synapse types by the site they contact.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SYNAPSE CLOSE-UP

Draw the end of an axon meeting a second neuron. Label the axon terminal, a synaptic end bulb, the synaptic vesicles inside it, the synaptic cleft, and the next cell. Draw neurotransmitter molecules leaving the vesicles and crossing the cleft. Beside it, draw three tiny sketches of an axon ending on a dendrite, on a cell body, and on another axon.

Then: Name each of your three tiny sketches by the site it contacts: axodendritic, axosomatic, or axoaxonic.

☐ **B** SYNAPSE TYPE MAP

Draw one neuron with three axons from other neurons ending on it: one on a dendrite, one on the cell body, and one on its axon. Name each synapse by the site it contacts. Draw a zoom circle on one synapse showing the synaptic end bulb, synaptic vesicles, and synaptic cleft.

Then: Write beside the zoom circle what the vesicles store and what carries the signal across the cleft.

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MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Neuron classification

You should be able to: Classify neurons structurally as multipolar, bipolar, or unipolar and functionally as sensory, motor, or interneuron with an example of each.

PICK ONE AND DO THAT ONE

☐ **A** THREE SHAPE SKETCHES

Draw a multipolar, a bipolar, and a unipolar neuron side by side. Label the dendrites, cell body, and axon on each one, and write how many processes leave the cell body. Under the multipolar and bipolar shapes, write where each is found, and add the Purkinje cell and pyramidal cell under the shape they belong to.

Then: Label which shape most sensory neurons have, and which shape motor neurons and interneurons have.

☐ **B** FUNCTIONAL PATHWAY MAP

Draw a skin receptor, a cross section of the spinal cord, and a skeletal muscle. Connect them with a sensory neuron carrying a signal toward the CNS, an interneuron inside the cord, and a motor neuron carrying a command out to the muscle. Draw each neuron in its structural shape and label the shape.

Then: In the margin, add a bipolar neuron and name the three places it is found.

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MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Neuroglia identification

You should be able to: Name the four CNS glia and two PNS glia and describe the support role each one performs.

PICK ONE AND DO THAT ONE

☐ **A** LABELED GLIA SCENE

Draw a patch of CNS tissue with a neuron, a capillary, and the edge of a ventricle. Draw and label an astrocyte touching the capillary, an oligodendrocyte wrapping several axons, a microglial cell eating debris, and ependymal cells lining the ventricle. In a separate PNS box, draw Schwann cells on an axon and satellite cells around cell bodies in a ganglion.

Then: Write the support role of each glial cell beside its label in a few words.

☐ **B** SORTING BOX

Draw a box split into a CNS side and a PNS side. Sort the six glial cells into the correct side. For each cell, add a tiny sketch of its shape and a few words for the support role it performs.

Then: Circle the two cells that make myelin, and star the cell that helps form the blood-brain barrier.

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MODULE 5 · FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE · LECTURE AND LAB

Myelination and regeneration

You should be able to: Describe the myelin sheath, nodes of Ranvier, and neurolemma and contrast PNS and CNS myelination by the glial cell involved and by regeneration potential.

PICK ONE AND DO THAT ONE

☐ **A** LABELED AXON COMPARISON

Draw a PNS axon wrapped by a row of Schwann cells. Label the myelin sheath, the nodes of Ranvier, and the neurolemma. Beside it, draw one CNS oligodendrocyte with arms wrapping segments of several different axons, and label its myelin sheath and nodes of Ranvier. Show that the CNS axons have no neurolemma.

Then: Draw a cut across each axon, then write which one can regrow and the reason beside it.

☐ **B** COMPARE TABLE

Draw a table with PNS and CNS across the top. Down the side: the cell that makes the myelin, how many axons one cell wraps, whether a neurolemma is present, and whether a cut axon can regrow. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a myelinated axon, write what myelin is made of, and draw the impulse jumping from node to node.

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COMP

MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

CNS and PNS tissue terms

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

PICK ONE AND DO THAT ONE

☐ **A** TWO BY TWO GRID

Draw a grid with CNS and PNS across the top. Down the side: a cluster of cell bodies and a bundle of axons. Write nucleus, ganglion, tract, and nerve in the correct boxes. Add a tiny sketch in each box, with dots for cell bodies and lines for axons.

Then: Write a one-line definition under each term that uses the words cluster or bundle and CNS or PNS.

☐ **B** LABELED NERVOUS SYSTEM MAP

Draw the brain and spinal cord with one nerve leaving the cord and a ganglion along that nerve outside the CNS. Draw a nucleus as a cluster of dots inside the brain and a tract as a bundle of lines running up the cord. Label all four structures.

Then: Draw a dashed line where the CNS ends and the PNS begins, and label each structure cell bodies or axons.

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COMP

MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE AND LAB

Gray and white matter

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

PICK ONE AND DO THAT ONE

☐ **A** TWO LABELED SECTIONS

Draw a spinal cord cross section and a slice through the cerebrum side by side, as on the models. Shade gray matter one way and white matter another. On the brain, label the cortex and the white matter beneath it. On the cord, label the gray matter with its dorsal and ventral horns, the outer white matter, and the central canal.

Then: Add a small cerebellum with its gray cortex, and write what gray and white matter are made of in your shading key.

☐ **B** COMPARE TABLE

Draw a table with gray matter and white matter down the side. Across the top: what it is made of, where it sits in the brain, and where it sits in the spinal cord. Use a word or a tiny sketch in each cell.

Then: Below the table, draw two circles labeled brain and cord, and shade each to show which matter is on the outside.

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COMP

MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

Reflex arc components

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

PICK ONE AND DO THAT ONE

☐ **A** LABELED REFLEX ARC

Draw a receptor, a cross section of the spinal cord, and a muscle. Connect them with a sensory neuron running into the cord, a synapse in the cord, and a motor neuron running out to the muscle. Label the five components of the reflex arc and add arrows for the direction of the signal.

Then: Put an X at one spot on the arc and write what a missing reflex tells a clinician about that pathway.

☐ **B** FLOW CHAIN

Draw a chain of five boxes for the reflex arc in order, from receptor to effector. In each box, name the component, sketch it, and write its job in a few words. Draw an arrow between each pair of boxes for the direction of the signal.

Then: Circle the box where the synapses are, and write beside the chain why a reflex is fast.

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MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

Types of reflexes

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

PICK ONE AND DO THAT ONE

☐ **A** COMPARE TABLE

Draw a table with six reflex types down the side: somatic, autonomic, monosynaptic, polysynaptic, innate, and learned. Across the top: what the name is based on and one example. Use a word or a tiny sketch in each cell.

Then: Draw a bracket joining each pair of types, and write the one feature that splits that pair.

☐ **B** PAIRED SKETCHES

Draw three pairs of small sketches. For somatic and autonomic, show the effector each one uses. For monosynaptic and polysynaptic, show one synapse in the cord beside a chain with interneurons. For innate and learned, sketch one example of each. Label every sketch with its reflex type and its example.

Then: Write under each pair what the names are based on: the effector, the number of synapses, or whether it is inborn.

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MODULE 5 · GROSS ANATOMY AND NEURONAL INTEGRATION · LECTURE

Neuronal pools and pathways

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CIRCUIT DRAWING

Draw a convergence circuit with many neurons feeding into one, and a divergence circuit with one neuron spreading to many, with arrows. Below them, draw a front view of the brain and spinal cord with one ascending and one descending pathway. Show each pathway crossing the midline and label the crossing decussation.

Then: Write beside the brain why a stroke on one side of the brain causes weakness on the opposite side.

☐ **B** COMPARE TABLE

Draw a table with convergence, divergence, ascending pathway, descending pathway, and decussation down the side. Across the top: what it is and which way the signal travels. Use a word or a tiny sketch in each cell.

Then: Below the table, draw a small body outline and shade the half the left side of the brain controls.

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COMP

MODULE 5 · THE SPINAL CORD · LECTURE AND LAB

Spinal cord external anatomy

You should be able to: Identify the cervical and lumbar enlargements, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED POSTERIOR VIEW

Draw the whole spinal cord from the back, running from the medulla down to its tapered end, with the vertebral column drawn faintly behind it. Label the cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, and posterior median sulcus. Mark the disc where the cord ends.

Then: Draw a small cross section beside it, label the anterior median fissure and posterior median sulcus, and show which groove is deeper.

☐ **B** STRUCTURE AND LANDMARK TABLE

Draw a table with seven structures down the side: cervical enlargement, lumbar enlargement, conus medullaris, cauda equina, filum terminale, anterior median fissure, and posterior median sulcus. Across the top: what it looks like, where it sits on the cord, and what it does or serves. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch the lower cord and mark where a lumbar puncture needle goes, then write why that spot does not risk the cord.

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COMP

MODULE 5 · THE SPINAL CORD · LECTURE AND LAB

Roots, rootlets, and root ganglion

You should be able to: Identify the rootlets, posterior and anterior roots, posterior root ganglion, and intervertebral foramen and state which root carries sensory versus motor axons.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CUTAWAY

Draw a cross section of the spinal cord with a row of rootlets leaving the back and the front on one side. Gather them into the posterior root and the anterior root, add the posterior root ganglion, and join the two roots into a spinal nerve. Label each part and the intervertebral foramen where the nerve exits.

Then: Write sensory or motor beside each root, and add arrows showing signals coming in on one root and going out on the other.

☐ **B** FLOW CHAIN

Draw two chains of boxes with arrows. The sensory chain runs from a receptor in the skin through the spinal nerve, posterior root, and rootlets into the cord. The motor chain runs from the cord through the rootlets, anterior root, and spinal nerve out to a muscle. Add the posterior root ganglion and intervertebral foramen where they belong.

Then: Circle the box that holds the sensory cell bodies, and star the one nerve both chains travel in and write why it is called mixed.

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COMP

MODULE 5 · THE SPINAL CORD · LAB

Spinal cord internal anatomy

You should be able to: Identify the gray horns, central canal, commissures, and white columns on a cord cross-section and state the contents of the posterior, lateral, and anterior horns.

PICK ONE AND DO THAT ONE

☐ **A** LABELED CROSS SECTION

Draw a cross section of a thoracic segment of the cord as it looks on the slide or model, with the gray butterfly inside the white matter. Label the posterior, lateral, and anterior horns, the gray commissure, the central canal, the anterior white commissure, and the anterior, lateral, and posterior white columns.

Then: Write beside each horn, in a few words, what it holds, and write tracts inside one white column.

☐ **B** LANDMARK TABLE

Draw a table with these down the side: the posterior, lateral, and anterior horns, the gray commissure, the anterior white commissure, the central canal, and the anterior, lateral, and posterior columns. Across the top: gray or white matter, how to find it on the section, and what it holds or connects. Use a word or a tiny sketch in each cell.

Then: Below the table, sketch a cervical and a thoracic section side by side and circle the lateral horn on the one that has it.

DRAW IT

split it up however the prompt needs

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Pass 1 from the notes, before the video. Pass 2 in the second color, after it, in the same box.

Dr. Sharilyn Rennie

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The box is **drawn**, not written. Pick one prompt and divide the box however it needs. Words go inside the drawing, in labels and little boxes with arrows between them. Sentences running across the page do not count.

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COMP

MODULE 5 · THE PERIPHERAL NERVOUS SYSTEM · LECTURE AND LAB

Nerve connective tissue wrappings

You should be able to: Identify the axon, fascicle, endoneurium, perineurium, and epineurium of a peripheral nerve and state the level each layer wraps.

PICK ONE AND DO THAT ONE

☐ **A** LABELED NERVE CROSS SECTION

Draw a cross section of a peripheral nerve as it looks on the slide. Show the whole nerve with several fascicles inside it, then enlarge one fascicle so you can see a few axons. Label an axon, a fascicle, the endoneurium, the perineurium, and the epineurium.

Then: Color each wrapping a different shade and write beside each one the level it wraps: one axon, one fascicle, or the whole nerve.

☐ **B** WRAPPING TABLE

Draw a table with the three wrappings down the side: endoneurium, perineurium, and epineurium. Across the top: inner, middle, or outer layer, the level it wraps (axon, fascicle, or whole nerve), and a tiny sketch of that level with the wrapping drawn around it. Use a word or a tiny sketch in each cell.

Then: Below the table, write how the perineurium and epineurium help a surgeon repair a cut nerve.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

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COMP

MODULE 5 · THE PERIPHERAL NERVOUS SYSTEM · LECTURE AND LAB

Spinal nerves and rami

You should be able to: State the count of the 31 spinal nerve pairs by region and identify the posterior ramus, anterior ramus, meningeal branch, and rami communicantes by the territory each supplies.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SPINAL NERVE

Draw a cross section of the spinal cord with one spinal nerve leaving through the intervertebral foramen. Show it splitting into the posterior ramus, the anterior ramus, and the meningeal branch, with the rami communicantes linking the anterior ramus to a sympathetic trunk ganglion. Label each branch and draw an arrow toward the territory it supplies.

Then: Beside the cord, write the number of spinal nerve pairs in each region, cervical through coccygeal, and the total.

☐ **B** COUNT AND TERRITORY TABLE

Draw a table in two parts. In the top part, list the five cord regions down the side with the number of spinal nerve pairs in each, and the total. In the bottom part, list the posterior ramus, anterior ramus, meningeal branch, and rami communicantes, with the territory each supplies. Use a word or a tiny sketch in each cell.

Then: Star the ramus that forms the plexuses, and write the name and spinal levels of the thoracic nerves that do not form one.

DRAW IT

split it up however the prompt needs

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COMP

MODULE 5 · THE PERIPHERAL NERVOUS SYSTEM · LECTURE AND LAB

Sensory receptor classification

You should be able to: Classify sensory receptors by stimulus, location, and structure and match named receptors such as tactile corpuscles, lamellar corpuscles, and muscle spindles to their location and stimulus.

PICK ONE AND DO THAT ONE

☐ **A** LABELED SKIN BLOCK

Draw a block of skin (epidermis, dermis, hypodermis) with one hair, and a skeletal muscle with its tendon underneath. Place and label free nerve endings, a tactile disc, a hair follicle receptor, a tactile corpuscle, a lamellar corpuscle, a bulbous corpuscle, a muscle spindle, and a tendon organ where the notes put them. Write what each one detects.

Then: Beside the drawing, write the three ways receptors are classified, and tag each receptor you drew as an exteroceptor or a proprioceptor.

☐ **B** RECEPTOR TABLE

Draw a table with the nine named receptors down the side, from free nerve endings to joint kinesthetic receptors. Across the top: where it sits, what it detects, and its class by stimulus (such as mechanoreceptor or nociceptor). Use a word or a tiny sketch in each cell.

Then: Below the table, list the three classes by location and the three by structure, and write one receptor from your table beside each class where one fits.

DRAW IT

split it up however the prompt needs

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190

COMP

MODULE 5 · THE CRANIAL NERVES · LECTURE AND LAB

Cranial nerves

You should be able to: Name the twelve cranial nerves in order, classify each as sensory, motor, or mixed, and match each to its primary function and skull foramen.

PICK ONE AND DO THAT ONE

☐ **A** LABELED INFERIOR VIEW

Draw the inferior view of the brain and brainstem. Draw and label all twelve cranial nerves, I to XII, where each one attaches, along with the olfactory bulb and tract, optic chiasm, optic tract, and pontomedullary junction. Beside each nerve, write S, M, or B for sensory, motor, or both.

Then: Write the skull foramen at the end of each nerve, and circle the three openings that carry more than one nerve.

☐ **B** CRANIAL NERVE TABLE

Draw a table with CN I to CN XII down the side. Across the top: name, class (sensory, motor, or mixed), primary function, and skull foramen. Write the names in order from memory first, using the memory sentence from the notes, then fill the other columns. Use a word or a tiny sketch in each cell.

Then: Color the rows that pass through the superior orbital fissure, the internal acoustic meatus, and the jugular foramen, using one color per opening.

DRAW IT

split it up however the prompt needs

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191

COMP

MODULE 5 · THE NERVE PLEXUSES · LECTURE AND LAB

Nerve plexuses

You should be able to: Name the cervical, brachial, lumbar, sacral, and coccygeal plexuses with the spinal nerves that form each and predict the body region affected if a given plexus is damaged.

PICK ONE AND DO THAT ONE

☐ **A** LABELED PLEXUS MAP

Draw the spinal cord with the spinal nerves leaving one side, grouped and numbered by region. Bracket and label the cervical, brachial, lumbar, sacral, and coccygeal plexuses, and write the spinal nerves that form each one. Draw the intercostal nerves running straight out from the thoracic region without forming a plexus.

Then: On a small body outline beside it, shade in matching colors the region that loses movement and sensation when each plexus is damaged.

☐ **B** DAMAGE TABLE

Draw a table with the five plexuses down the side. Across the top: the spinal nerves that form it, the region of the body it serves, and what a patient loses if it is damaged. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the phrenic nerve leaving its plexus to reach the diaphragm, and write what happens to breathing if it is damaged.

DRAW IT

split it up however the prompt needs

PASS 1 COLOR _____

PASS 2 COLOR _____

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 NAME _____ DATE _____

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192

COMP

MODULE 5 · THE AUTONOMIC NERVOUS SYSTEM · LECTURE

Autonomic two-neuron pathway

You should be able to: Contrast the somatic and autonomic motor systems and trace the two-neuron pathway through a preganglionic neuron, autonomic ganglion, and postganglionic neuron to an effector.

PICK ONE AND DO THAT ONE

☐ **A** SIDE BY SIDE PATHWAYS

Draw a somatic pathway and an autonomic pathway side by side, each starting in the spinal cord. Show the somatic one as a single neuron running to skeletal muscle. Show the autonomic one as a preganglionic neuron, an autonomic ganglion, and a postganglionic neuron reaching an effector. Label the cell bodies, the myelinated and unmyelinated axons, and the synapse.

Then: Write the three autonomic effectors at the end of the autonomic pathway, and write voluntary or involuntary under each pathway.

☐ **B** COMPARE TABLE

Draw a table with five features down the side: effector, control, number of motor neurons, effect on the effector, and awareness. Across the top: somatic nervous system and autonomic nervous system. Use a word or a tiny sketch in each cell.

Then: Below the table, draw the four-step autonomic route as a chain of boxes with arrows, from the preganglionic neuron to the effector tissue.

PASS 1 COLOR _____

PASS 2 COLOR _____

DRAW IT

split it up however the prompt needs

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Dr. Sharilyn Rennie

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193

COMP

MODULE 5 · THE AUTONOMIC NERVOUS SYSTEM · LECTURE

Autonomic ganglia and divisions

You should be able to: Compare the sympathetic and parasympathetic divisions by CNS origin and ganglion location and identify the sympathetic trunk, prevertebral ganglia, terminal ganglia, and autonomic and enteric plexuses.

PICK ONE AND DO THAT ONE

☐ **A** LABELED GANGLIA MAP

Draw the brainstem and spinal cord down the middle with the sympathetic trunk beside the vertebral column. Bracket where each division leaves the CNS. Draw and label a sympathetic trunk ganglion, the prevertebral ganglia near the large abdominal arteries, and a terminal ganglion in an organ wall. Add the cardiac plexus on the heart, the celiac plexus on the abdominal organs, and the hypogastric plexus on the pelvic organs.

Then: Draw a small piece of gut wall and place the myenteric and submucosal plexuses between the layers where each one sits.

☐ **B** COMPARE TABLE

Draw a table with sympathetic and parasympathetic across the top. Down the side: other name, origin in the CNS, location of its ganglia, the named ganglia that belong to it, and overall role. Use a word or a tiny sketch in each cell.

Then: Below the table, list the autonomic plexuses with what each supplies, and the two enteric plexuses with the gut layers each sits between.

DRAW IT

split it up however the prompt needs

PASS 1 COLOR _____

PASS 2 COLOR _____

Pass 1 from the notes, before the video. Pass 2 in the second color, after it, in the same box.

Dr. Sharilyn Rennie

PART C

Check your drawing.

Do this after both passes. Find your competency and the prompt you picked, and tick off everything on the list that is in your box. Anything missing goes in with your second color. Then look for the common mistake named underneath.

133 Kidney gross anatomy and layers

PROMPT A · LABELED LOCATION VIEW

- [ ] Both kidneys sit against the back wall of the abdomen, between T12 and L3
- [ ] Right kidney drawn slightly lower than the left
- [ ] Renal hilum is the notch on the medial border, facing the spine
- [ ] Ureter, renal artery, renal vein and nerves all enter or leave at the hilum
- [ ] Renal capsule is the innermost layer, the smooth fibrous covering right on the kidney
- [ ] Adipose capsule is the fat layer around the renal capsule
- [ ] Renal fascia is the outermost layer, anchoring the kidney to the abdominal wall
- [ ] Right kidney note: it sits lower because the liver sits above it
- [ ] Left kidney note: retroperitoneal means it lies behind the peritoneum, against the back wall of the abdomen

Check: From the inside out the layers go capsule, fat, fascia. Make sure the renal fascia is your outermost layer, not the capsule.

PROMPT B · LAYERS TABLE

- [ ] Renal capsule: inner layer, smooth fibrous tissue, keeps the kidney's shape
- [ ] Adipose capsule: middle layer, fat, cushions the kidney and holds it in place
- [ ] Renal fascia: outer layer, connective tissue, anchors the kidney to the abdominal wall
- [ ] Renal hilum: medial border, where the ureter, vessels and nerves pass in and out
- [ ] Sketch shows both kidneys between T12 and L3
- [ ] Right kidney drawn slightly lower than the left
- [ ] Hilum marked on the medial side of each kidney, facing the spine

Check: The hilum is on the medial border, facing the spine. Check that both of your hilums point inward, not out to the sides.

134 Internal kidney regions

PROMPT A · LABELED FRONTAL SECTION

- [ ] Renal capsule drawn as the thin covering around the outside
- [ ] Renal cortex is the outer band of tissue just under the capsule
- [ ] Renal medulla is the inner region, made of the pyramids
- [ ] Renal pyramids are cones with their wide bases toward the cortex
- [ ] Renal columns are strips of cortex dipping down between the pyramids
- [ ] Renal papilla is the tip of each pyramid, pointing in toward a minor calyx
- [ ] Minor calyces cup the papillae and join into major calyces
- [ ] Major calyces empty into the renal pelvis, which narrows into the ureter at the hilum
- [ ] Renal sinus is the space just inside the hilum that holds the calyces, renal pelvis, vessels and fat
- [ ] Parenchyma shaded: the cortex (with the renal columns) and the medulla (the pyramids), which together hold the nephrons

Check: The pyramids point their bases out toward the cortex and their tips in toward the calyces. An upside down pyramid is the usual slip.

PROMPT B · SORTING BOX

- [ ] Parenchyma box: cortex, renal columns, renal pyramids, renal papillae
- [ ] Urine spaces box: minor calyces, major calyces, renal pelvis
- [ ] Pyramid drawn with its papilla at the tip, pointing inward
- [ ] Papilla drains into a minor calyx
- [ ] Minor calyx drains into a major calyx, then into the renal pelvis
- [ ] Renal column drawn beside the pyramid
- [ ] The renal column is an extension of the renal cortex

Check: The renal papilla is the tip of a pyramid, so it goes in the parenchyma box, not with the urine spaces.

135 Nephron structure

PROMPT A · LABELED NEPHRON

- [ ] Renal corpuscle is the glomerulus plus the glomerular capsule around it
- [ ] Glomerulus drawn as a tuft of capillaries inside the capsule
- [ ] Parietal layer is the outer wall of the capsule; visceral layer hugs the glomerulus
- [ ] Podocytes form the visceral layer, wrapped around the glomerular capillaries
- [ ] Capsular space is the gap between the parietal and visceral layers
- [ ] Proximal convoluted tubule leaves the capsule and coils
- [ ] Nephron loop drawn as a U, with a descending limb going down and an ascending limb coming back up
- [ ] Distal convoluted tubule is the second coiled segment, emptying into the collecting duct
- [ ] Filtrate arrows run capsular space, proximal tubule, descending limb, ascending limb, distal tubule, collecting duct
- [ ] Brush border marked on the proximal convoluted tubule

Check: The brush border belongs on the proximal convoluted tubule, the first coiled segment, not the distal one.

PROMPT B · PARTS TABLE

- [ ] Corpuscle rows come first: glomerulus, glomerular capsule, podocytes, capsular space
- [ ] Glomerulus: renal corpuscle; a tuft of leaky, fenestrated capillaries
- [ ] Glomerular capsule: renal corpuscle; the cup around the glomerulus that collects filtrate
- [ ] Podocytes: renal corpuscle; cells of the inner (visceral) layer with foot processes
- [ ] Capsular space: renal corpuscle; the space inside the capsule where filtrate collects
- [ ] Proximal convoluted tubule: tubule; first coiled segment with a brush border
- [ ] Nephron loop: tubule; U shape with a descending and an ascending limb
- [ ] Distal convoluted tubule: tubule; second coiled segment
- [ ] Collecting duct: comes after the tubule and is shared by many nephrons; carries fluid down the pyramid toward the papilla
- [ ] Sketch shows a podocyte's foot processes wrapped on a capillary, with the filtration slits as the gaps between the processes

Check: Filtration slits are the gaps between the podocyte foot processes, not holes in the capillary wall.

136 Cortical vs juxtamedullary nephrons

PROMPT A · SIDE BY SIDE NEPHRONS

- [ ] Strip shows the cortex on top, the medulla below, and a line between them
- [ ] Both renal corpuscles sit in the cortex: the cortical one in the outer cortex, the juxtamedullary one deep in the cortex next to the medulla
- [ ] Cortical nephron has a short loop that barely dips into the medulla
- [ ] Juxtamedullary nephron has a long loop that reaches deep into the medulla
- [ ] Peritubular capillaries wrap the cortical nephron
- [ ] Vasa recta run alongside the long juxtamedullary loop
- [ ] Cortical nephrons labeled about 85 percent of all nephrons
- [ ] Juxtamedullary nephrons labeled about 15 percent
- [ ] Star on the juxtamedullary nephron, the type that concentrates the urine

Check: The juxtamedullary nephrons are the rare ones, about 15 percent, but they are the ones that concentrate urine. Check you did not swap the percents.

PROMPT B · COMPARE TABLE

- [ ] Cortical: about 85 percent of nephrons
- [ ] Cortical: short loop that barely enters the medulla
- [ ] Cortical: served by the peritubular capillaries
- [ ] Juxtamedullary: about 15 percent of nephrons
- [ ] Juxtamedullary: long loop that reaches deep into the medulla
- [ ] Juxtamedullary: served by the vasa recta
- [ ] Pyramid sketch shows a short loop staying near the base of the pyramid, labeled cortical
- [ ] Long loop reaching down toward the tip of the pyramid, labeled juxtamedullary

Check: The vasa recta go with the long juxtamedullary loops, and the peritubular capillaries go with the cortical nephrons. Make sure you did not swap the capillary beds.

137 JGA and filtration membrane

PROMPT A · LABELED CORPUSCLE CLOSE-UP

- [ ] Afferent arteriole drawn going into the glomerulus and efferent arteriole coming out
- [ ] Distal convoluted tubule drawn touching the arterioles of its own corpuscle
- [ ] Juxtaglomerular apparatus labeled where the tubule meets the arterioles
- [ ] Macula densa cells are in the wall of the distal tubule
- [ ] Juxtaglomerular (granular) cells are in the wall of the afferent arteriole
- [ ] Zoom box layers in order from the blood: fenestrated endothelium, basement membrane, slit membranes
- [ ] Slit membranes drawn between the podocyte processes
- [ ] Blood cell stopped at the fenestrated endothelium
- [ ] Large protein stopped at the basement membrane
- [ ] Medium protein stopped at the slit membranes

Check: The juxtaglomerular cells sit in the afferent arteriole, the vessel going in, and the macula densa sits in the distal tubule. These two get swapped a lot.

PROMPT B · WHERE AND WHAT TABLE

- [ ] Macula densa: in the wall of the distal convoluted tubule; senses the sodium and chloride in the fluid
- [ ] Juxtaglomerular cells: in the wall of the afferent arteriole; release renin when blood pressure is low
- [ ] Layers listed in order from the blood: fenestrated endothelium, basement membrane, slit membranes
- [ ] Fenestrated endothelium: the pored capillary wall; lets plasma through but holds back blood cells
- [ ] Basement membrane: between the capillary and the podocytes; holds back large plasma proteins
- [ ] Slit membranes: between the podocyte processes; hold back medium sized proteins
- [ ] Sketch shows the distal tubule touching the arterioles of its own corpuscle
- [ ] Circle drawn where the tubule touches the arterioles, marking the juxtaglomerular apparatus

Check: Each layer holds back something smaller than the one before: cells, then large proteins, then medium proteins. Check that your rows run from the blood side out.

138 Kidney blood supply pathway

PROMPT A · LABELED VESSEL CUTAWAY

- [ ] Renal artery enters at the hilum
- [ ] Segmental arteries branch in the renal sinus, before reaching the tissue
- [ ] Interlobar arteries run between the pyramids, through the renal columns
- [ ] Arcuate arteries arch along the bases of the pyramids, where cortex meets medulla
- [ ] Interlobular (cortical radiate) arteries run up into the cortex
- [ ] Afferent arteriole carries blood into the glomerulus
- [ ] Efferent arteriole carries blood out of the glomerulus to the peritubular capillaries
- [ ] Peritubular capillaries wrap the tubule of the nephron
- [ ] Veins drawn in a second shade: cortical radiate vein in the cortex, arcuate vein along the pyramid base, renal vein leaving at the hilum

Check: Only the interlobar arteries run between the pyramids. The segmental arteries branch earlier, in the renal sinus.

PROMPT B · FLOW CHAIN

- [ ] Boxes in order: renal artery, segmental and interlobar arteries, arcuate artery, interlobular (cortical radiate) artery, afferent arteriole, glomerulus, efferent arteriole, peritubular capillaries and vasa recta, cortical radiate vein, arcuate vein, interlobar vein, renal vein
- [ ] Renal artery: from the aorta into the hilum
- [ ] Segmental and interlobar arteries: renal sinus, then between the pyramids
- [ ] Arcuate artery: arches along the bases of the pyramids
- [ ] Interlobular (cortical radiate) artery: up into the cortex
- [ ] Afferent arteriole: into the glomerulus
- [ ] Circles on the two capillary beds: the glomerulus and the peritubular capillaries with the vasa recta
- [ ] Box on the efferent arteriole, the one vessel between the two capillary beds

Check: The vessel between the two capillary beds is the efferent arteriole. If you boxed the afferent arteriole, look again.

139 Ureters, bladder, urethra and urine path

PROMPT A · LABELED URINARY TRACT

- [ ] Minor calyces cup the papillae and feed into major calyces
- [ ] Major calyces empty into the renal pelvis, which narrows into the ureter
- [ ] Ureter runs down to the back of the bladder
- [ ] Detrusor muscle labeled in the bladder wall
- [ ] Trigone is the triangle on the bladder floor, with the two ureteral openings at the upper corners and the internal urethral orifice at the lowest point
- [ ] Internal urethral sphincter at the bladder outlet
- [ ] External urethral sphincter below it, around the urethra
- [ ] Urine arrows run collecting duct, minor calyx, major calyx, renal pelvis, ureter, bladder, urethra, outside
- [ ] Male urethra drawn longer, with prostatic, membranous and spongy parts from the bladder down
- [ ] Female urethra drawn short

Check: The trigone's three corners are the two ureteral openings and the one urethral opening. The urethral opening should be the lowest corner.

PROMPT B · FLOW CHAIN

- [ ] Boxes in order: collecting duct, renal papilla, minor calyx, major calyx, renal pelvis, ureter, urinary bladder, urethra
- [ ] Under the bladder: trigone, detrusor muscle, internal urethral sphincter, external urethral sphincter
- [ ] Male urethra split into prostatic, membranous and spongy parts, in that order
- [ ] Female urethra in its own part of the box
- [ ] The male urethra is the longer one
- [ ] Internal urethral sphincter: smooth muscle, not under your control
- [ ] External urethral sphincter: skeletal muscle, under your control

Check: The internal sphincter is smooth muscle you cannot control; the external one is skeletal muscle you can. Check you did not swap them.

140 Clinical anatomy of urinary disorders

PROMPT A · DISORDER MAP

- [ ] Nephroptosis written on the kidney and its adipose capsule
- [ ] Kidney stones: one drawn in the renal pelvis and one lodged in the ureter
- [ ] Urinary tract infection written on the urethra, bladder and kidney
- [ ] Glomerulonephritis written on the glomeruli inside the kidney
- [ ] Renal failure written on both kidneys, since it affects the nephrons of both
- [ ] Polycystic kidney disease written on a kidney drawn with many fluid filled cysts
- [ ] Dropped kidney drawn lower than normal with its ureter kinked
- [ ] Note: when the fat of the adipose capsule is lost, the kidney loses its support and drops, kinking the ureter and blocking urine

Check: A urinary tract infection can sit in the urethra, the bladder or the kidney. Make sure you did not tie it to only one spot.

PROMPT B · DISORDER TABLE

- [ ] Nephroptosis: kidney and its adipose capsule; the fat support is lost and the kidney drops
- [ ] Kidney stones: kidney and ureter; hardened mineral deposits form in the kidney and can lodge in the ureter
- [ ] Urinary tract infection: urethra, bladder or kidneys; usually a bacterial infection
- [ ] Glomerulonephritis: glomeruli; inflammation damages the filtration membrane, so protein and blood leak into the urine
- [ ] Polycystic kidney disease: kidneys; many fluid filled cysts enlarge them (inherited)
- [ ] Sketch shows a short female urethra beside a long male urethra
- [ ] Note: the short female urethra gives bacteria a shorter climb to the bladder

Check: Nephroptosis is about the fat that holds the kidney in place. If your row only says the kidney is damaged, add the adipose capsule.

141 Scrotum and testis structure

PROMPT A · LABELED TESTIS SECTION

- [ ] Scrotal septum divides the scrotum down the middle
- [ ] Dartos muscle drawn in the scrotal wall, under the skin
- [ ] Cremaster muscle drawn coming down the spermatic cord and looping around the testis
- [ ] Tunica vaginalis partly covers the outside of the testis
- [ ] Tunica albuginea is the white fibrous capsule right on the testis, dividing it into lobules
- [ ] Lobules each hold coiled seminiferous tubules
- [ ] Zoom circle shows sustentacular cells in the wall of the tubule
- [ ] Interstitial cells drawn in the spaces between the tubules
- [ ] Dartos note: smooth muscle that wrinkles the scrotal skin
- [ ] Cremaster note: skeletal muscle that pulls the testis up toward the warmer body

Check: Interstitial cells sit between the tubules, not in the tubule wall. The sustentacular cells are the ones in the wall.

PROMPT B · STRUCTURE TABLE

- [ ] Scrotum: pouch that supports the testes and keeps them cooler than the body core
- [ ] Dartos muscle: scrotum; wrinkles the skin to reduce heat loss
- [ ] Cremaster muscle: scrotum, in the spermatic cord around the testis; pulls the testis up toward the warmer body
- [ ] Tunica vaginalis: covering the testis; serous membrane that partly covers it
- [ ] Tunica albuginea: covering the testis; white fibrous capsule that divides it into lobules
- [ ] Seminiferous tubules: inside the testis; where sperm are made
- [ ] Sustentacular cells: inside the testis; nourish developing sperm and form the blood testis barrier
- [ ] Interstitial cells: inside the testis; make testosterone
- [ ] Circle on the interstitial cells row
- [ ] Stars on the dartos and cremaster rows

Check: The two temperature rows are the two muscles, the dartos and the cremaster, not the tunics.

142 Sperm duct system and pathway

PROMPT A · LABELED DUCT VIEW

- [ ] Epididymis sits on the testis
- [ ] Ductus deferens runs up out of the scrotum inside the spermatic cord
- [ ] Seminal vesicle duct joins the ductus deferens to form the ejaculatory duct
- [ ] Ejaculatory duct empties into the prostatic urethra
- [ ] Prostatic urethra runs through the prostate
- [ ] Membranous urethra runs through the perineum
- [ ] Spongy urethra runs through the penis
- [ ] Sperm arrows run seminiferous tubules, epididymis, ductus deferens, ejaculatory duct, prostatic, membranous and spongy urethra, outside
- [ ] Spermatic cord circle holds the ductus deferens, blood vessels, nerves and cremaster muscle

Check: The ejaculatory duct starts where the ductus deferens meets the seminal vesicle duct and empties into the prostatic urethra. Check it does not end in the bladder.

PROMPT B · FLOW CHAIN

- [ ] Seminiferous tubules: sperm are made
- [ ] Epididymis: sperm are stored and gain the ability to move
- [ ] Ductus deferens: sperm are carried up out of the scrotum
- [ ] Ejaculatory duct: the ductus deferens joins the seminal vesicle duct
- [ ] Prostatic urethra: sperm pass through the prostate
- [ ] Membranous urethra: sperm pass through the perineum
- [ ] Spongy urethra: sperm pass through the penis to the outside
- [ ] Box on the seminiferous tubules, the only structure that makes sperm
- [ ] Bracket over the ductus deferens
- [ ] Other spermatic cord contents: blood vessels, nerves, cremaster muscle

Check: Only the seminiferous tubules make sperm. The epididymis stores and matures them, so it should not be boxed.

143 Sperm cell structure

PROMPT A · LABELED SPERM CELL

- [ ] Head drawn with its nucleus inside
- [ ] Acrosome drawn as a cap over the front of the head
- [ ] Neck is the short part just behind the head
- [ ] Middle piece packed with small mitochondria
- [ ] Principal piece drawn as the longest part of the tail
- [ ] End piece tapers to a point
- [ ] Head: holds 23 chromosomes
- [ ] Acrosome: enzymes to enter the oocyte
- [ ] Neck: holds the centrioles
- [ ] Middle piece: mitochondria power movement
- [ ] Principal piece: drives the swimming
- [ ] End piece: short thin tip of the tail

Check: The mitochondria go in the middle piece, not in the head or the end piece.

PROMPT B · PARTS TABLE

- [ ] Rows run front to back: head, acrosome, neck, middle piece, principal piece, end piece
- [ ] Head: holds the nucleus with 23 chromosomes
- [ ] Acrosome: cap over the head; its enzymes help the sperm enter the oocyte
- [ ] Neck: just behind the head; holds the centrioles
- [ ] Middle piece: packed with mitochondria that power movement
- [ ] Principal piece: the longest part of the tail
- [ ] End piece: the tapering tip of the tail
- [ ] Tail bracket covers the neck, middle piece, principal piece and end piece
- [ ] Star on the principal piece

Check: The notes count the neck as part of the tail, so the tail bracket covers four rows, starting at the neck.

144 Accessory glands and penis

PROMPT A · LABELED PENIS VIEW

- [ ] Root is the attached part of the penis
- [ ] Body is the free, movable part
- [ ] Glans penis is the enlarged tip
- [ ] Prepuce is the fold of skin over the glans
- [ ] Cross section shows two corpora cavernosa on the dorsal side
- [ ] Corpus spongiosum is the single column on the ventral side
- [ ] Spongy urethra runs through the corpus spongiosum
- [ ] Seminal vesicles: alkaline fluid rich in fructose that nourishes sperm
- [ ] Prostate: milky fluid that helps sperm stay active
- [ ] Bulbourethral glands: alkaline mucus that neutralizes urine in the urethra and lubricates

Check: The urethra runs through the single corpus spongiosum on the ventral side, not between the two corpora cavernosa.

PROMPT B · TWO PART TABLE

- [ ] Seminal vesicles: alkaline, fructose rich fluid that nourishes the sperm
- [ ] Prostate: milky fluid that helps the sperm stay active
- [ ] Bulbourethral glands: alkaline mucus that neutralizes urine in the urethra and lubricates
- [ ] Root: the attached part of the penis
- [ ] Body: the free, movable part
- [ ] Glans penis: the enlarged tip
- [ ] Prepuce: the foreskin over the glans
- [ ] Corpora cavernosa: the two dorsal columns of erectile tissue
- [ ] Corpus spongiosum: the single ventral column around the spongy urethra
- [ ] Star on the seminal vesicles, and circle on the corpus spongiosum

Check: The glans is an extension of the corpus spongiosum, not the corpora cavernosa. Check which column you circled.

145 Clinical anatomy of male disorders

PROMPT A · DISORDER MAP

- [ ] Cryptorchidism written on the testis, which failed to drop into the scrotum
- [ ] Hydrocele written on the tunica vaginalis around the testis, where fluid collects
- [ ] Varicocele written on the pampiniform plexus, the veins in the spermatic cord above the testis
- [ ] Benign prostatic hyperplasia written on the prostate, around the urethra
- [ ] Prostate cancer written on the prostate
- [ ] Testicular cancer written on the testis
- [ ] Inguinal hernia written at the inguinal canal, with a loop of intestine pushing through
- [ ] Undescended testis drawn above the scrotum, with a dotted line down to where it belongs
- [ ] Note: sperm production needs a temperature a few degrees below the body core, so at body temperature the warmth damages the sperm forming cells

Check: Hydrocele is fluid around the testis; varicocele is swollen veins. Check you did not swap them.

PROMPT B · DISORDER TABLE

- [ ] Cryptorchidism: testis; belongs in the scrotum, outside the body; fails to descend
- [ ] Hydrocele: tunica vaginalis; covering the testis in the scrotum; serous fluid collects there
- [ ] Varicocele: pampiniform plexus, the veins of the testis; in the spermatic cord above the testis; the veins swell and twist, more often on the left
- [ ] Benign prostatic hyperplasia: prostate; around the urethra below the bladder; grows larger, not cancer
- [ ] Inguinal hernia: abdominal wall; at the inguinal canal in the groin; a loop of intestine pushes through
- [ ] Sketch shows an enlarged prostate pressing on the urethra that runs through it
- [ ] Symptom written: urination becomes difficult

Check: The prostate wraps around the urethra, so benign prostatic hyperplasia squeezes the urethra, not the bladder.

146 Ovary structure and follicle stages

PROMPT A · LABELED OVARY VIEW

- [ ] Ovarian ligament anchors the ovary to the uterus
- [ ] Suspensory ligament attaches the ovary to the pelvic wall
- [ ] Broad ligament drapes over the uterus and tube
- [ ] Mesovarium is the part of the broad ligament that holds the ovary
- [ ] Ovary layers from outside in: germinal epithelium, tunica albuginea, cortex, medulla
- [ ] Follicles, mature follicle, corpus luteum and corpus albicans all drawn in the cortex
- [ ] Numbered in order: 1 follicle, 2 mature follicle, 3 corpus luteum, 4 corpus albicans
- [ ] Ovulation marked where the mature follicle releases its oocyte at the ovary surface

Check: The ovarian ligament ties the ovary to the uterus; the suspensory ligament ties it to the pelvic wall. These two get swapped a lot.

PROMPT B · SEQUENCE STRIP

- [ ] Strip order: growing follicle, mature follicle, oocyte released, corpus luteum, corpus albicans
- [ ] Mature follicle drawn large and fluid filled
- [ ] Corpus luteum forms from what is left after ovulation
- [ ] Corpus albicans is the scar left when the corpus luteum breaks down
- [ ] Layers from the outside in: germinal epithelium, tunica albuginea, cortex, medulla
- [ ] Ovarian ligament: ovary to the uterus
- [ ] Suspensory ligament: ovary to the pelvic wall
- [ ] Broad ligament: holds the ovary by the mesovarium, as part of the fold over the uterus and tubes
- [ ] Circle on the cortex

Check: The corpus luteum forms after ovulation, from the follicle left behind. If it comes before the oocyte release in your strip, reorder.

147 Uterine tubes and ovum pathway

PROMPT A · LABELED TUBE VIEW

- [ ] Infundibulum is the funnel end of the tube, open to the pelvic cavity near the ovary
- [ ] Fimbriae are finger-like fringes on the infundibulum, reaching toward the ovary
- [ ] Ampulla is the widest and longest part of the tube
- [ ] Isthmus is the narrow part joining the tube to the uterus
- [ ] Oocyte drawn leaving a mature follicle
- [ ] Arrows run ovary, fimbriae and infundibulum, ampulla, isthmus, uterine cavity
- [ ] Star in the ampulla
- [ ] Cilia drawn on the tube lining
- [ ] Cilia arrow points toward the uterus

Check: Fertilization usually happens in the ampulla, the wide part of the tube, not in the uterus. Check your star.

PROMPT B · FLOW CHAIN

- [ ] Ovary: the mature follicle releases the oocyte at ovulation
- [ ] Fimbriae and infundibulum: the fimbriae sweep the oocyte into the funnel of the tube
- [ ] Infundibulum feature: funnel shaped, open to the pelvic cavity
- [ ] Ampulla: sperm may fertilize the oocyte here; widest and longest part
- [ ] Isthmus: the oocyte or early embryo passes through; narrow and thick walled
- [ ] Uterus: the oocyte or embryo reaches the uterine cavity
- [ ] Dotted arrow drawn from the open infundibulum into the pelvic cavity
- [ ] Disorder named: pelvic inflammatory disease, an infection that climbs through the uterus and tubes into the pelvic cavity (a rare ectopic pregnancy in the abdominal cavity also counts)

Check: The ampulla is the wide part and the isthmus is the narrow part next to the uterus. Check you did not flip them.

148 Uterus regions, ligaments, and wall

PROMPT A · LABELED UTERUS VIEW

- [ ] Fundus is the dome above where the tubes enter
- [ ] Body is the tapered middle part
- [ ] Cervix is the narrow lower part opening into the vagina
- [ ] Uterine cavity is inside the body; cervical canal is inside the cervix
- [ ] Internal os opens into the uterine cavity; external os opens into the vagina
- [ ] Wall layers from outside in: perimetrium, myometrium, endometrium
- [ ] Broad ligament runs from the uterus to the pelvic walls; round ligament from the uterus to the labia majora
- [ ] Uterosacral ligament runs from the cervix to the sacrum; cardinal ligament from the cervix and upper vagina to the side walls of the pelvis
- [ ] Endometrium zoom: stratum functionalis sheds each month
- [ ] Stratum basalis is the deeper layer that rebuilds it

Check: The internal os opens into the uterine cavity and the external os opens into the vagina. Check you did not flip them.

PROMPT B · REGION STACK AND TABLE

- [ ] Stack runs fundus on top, body in the middle, cervix at the bottom
- [ ] Uterine cavity drawn inside the body, cervical canal inside the cervix
- [ ] Broad ligaments: uterus to the pelvic walls
- [ ] Round ligaments: uterus to the labia majora
- [ ] Uterosacral ligaments: cervix to the sacrum
- [ ] Cardinal ligaments: pelvic wall to the cervix and vagina
- [ ] Perimetrium: outer serous layer, part of the visceral peritoneum
- [ ] Myometrium: thick middle layer of smooth muscle
- [ ] Endometrium: inner mucous lining; its functional layer sheds and its basal layer rebuilds it
- [ ] Star on the myometrium, and circle on the round ligament

Check: The layer that contracts during childbirth is the myometrium, the muscle layer, not the endometrium.

149 Vagina and external genitalia

PROMPT A · LABELED VULVA VIEW

- [ ] Mons pubis at the top, over the pubic symphysis
- [ ] Labia majora are the larger outer folds
- [ ] Labia minora are the smaller folds inside the labia majora
- [ ] Clitoris sits where the labia minora meet at the front
- [ ] Vestibule is the area between the labia minora
- [ ] Urethral orifice sits between the clitoris and the vaginal orifice
- [ ] Hymen drawn as a thin fold partly closing the vaginal orifice
- [ ] Greater vestibular glands sit beside the vaginal opening
- [ ] Perineum outlined as a diamond around the genitalia and anus
- [ ] Side view shows the fornix as the recess of the vagina around the cervix

Check: Front to back the openings go clitoris, urethral opening, vaginal opening, anus. A urethral opening drawn behind the vaginal opening is the usual slip.

PROMPT B · STRUCTURE TABLE

- [ ] Vagina: runs from the cervix to the outside; the birth canal
- [ ] Fornix: the recess of the vagina around the cervix
- [ ] Hymen: thin fold at the vaginal opening that partly closes it
- [ ] Mons pubis: fatty cushion over the pubic symphysis
- [ ] Labia majora: the larger, hair covered outer folds
- [ ] Labia minora: the smaller, hairless folds inside the labia majora
- [ ] Clitoris: erectile tissue where the labia minora meet at the front
- [ ] Vestibule: the area between the labia minora that holds the vaginal and urethral openings
- [ ] Greater vestibular glands: beside the vaginal opening; secrete lubricating mucus
- [ ] Marked homologs, all three: labia majora match the scrotum, clitoris matches the penis, greater vestibular glands match the bulbourethral glands

Check: The vestibule is the area between the labia minora, not between the labia majora.

150 Mammary gland structure

PROMPT A · LABELED BREAST SECTION

- [ ] Breast drawn sitting over the chest muscles
- [ ] Two or three lobes labeled
- [ ] Lobules drawn inside one lobe
- [ ] Alveoli drawn as small sacs inside one lobule
- [ ] Lactiferous ducts run from the lobes to the nipple
- [ ] Areola is the pigmented ring of skin around the nipple
- [ ] Suspensory ligaments drawn as supporting bands through the breast
- [ ] Milk arrows run alveoli, small ducts of the lobule, lactiferous duct, lactiferous sinus, nipple

Check: The alveoli are the smallest sacs inside the lobules, and the lobe is the biggest unit. Check the nesting goes small inside big.

PROMPT B · NESTING DIAGRAM

- [ ] Alveoli drawn inside a lobule
- [ ] Lobules drawn inside a lobe
- [ ] Lobe drains through a lactiferous duct to the nipple
- [ ] Nipple: the pigmented projection where the lactiferous ducts open
- [ ] Areola: the circular pigmented skin around the nipple
- [ ] Suspensory ligaments: the Cooper's ligaments that support the breast
- [ ] Lobe note: one mammary gland has 15 to 20 lobes

Check: The lactiferous ducts open at the nipple, not on the areola.

151 Brain gross organization

PROMPT A · LABELED MIDSAGITTAL VIEW

- [ ] Cerebrum is the largest part, filling the top of the drawing
- [ ] Diencephalon sits deep in the middle, between the cerebrum and the brainstem
- [ ] Brainstem is the stalk running down toward the spinal cord
- [ ] Cerebellum sits behind the brainstem, under the back of the cerebrum
- [ ] Each of the four divisions is outlined
- [ ] Thin outer band of gray matter drawn along the edge of the cerebrum
- [ ] Thick core of white matter drawn beneath it
- [ ] Gray matter note: neuron cell bodies, dendrites and unmyelinated fibers
- [ ] White matter note: bundles of myelinated axons

Check: Gray matter is the thin outer layer of the cerebrum and white matter is the core. Check you did not reverse them.

PROMPT B · DIVISION TABLE

- [ ] Cerebrum: top of the brain; conscious thought, sensation and voluntary movement
- [ ] Diencephalon: deep, between the cerebrum and the brainstem; relays sensory signals and controls homeostasis
- [ ] Brainstem: the stalk below the diencephalon, in front of the cerebellum; connects the brain to the spinal cord
- [ ] Cerebellum: behind the brainstem; coordinates movement
- [ ] Tissue block shows gray matter on the outside
- [ ] White matter on the inside
- [ ] Star on the cerebrum as the largest division
- [ ] Star on the cerebellum as the second largest
- [ ] Cerebellum sits in the posterior cranial fossa

Check: The second largest division is the cerebellum, not the brainstem or the diencephalon.

152 Neural tube brain development

PROMPT A · BRANCHING FLOW CHAIN

- [ ] Neural tube at the top
- [ ] Primary vesicles: prosencephalon (forebrain), mesencephalon (midbrain), rhombencephalon (hindbrain)
- [ ] Prosencephalon splits into the telencephalon and diencephalon
- [ ] Rhombencephalon splits into the metencephalon and myelencephalon
- [ ] Telencephalon: cerebrum and lateral ventricles
- [ ] Diencephalon: thalamus, hypothalamus, epithalamus and third ventricle
- [ ] Mesencephalon: midbrain and cerebral aqueduct
- [ ] Metencephalon: pons, cerebellum and upper fourth ventricle
- [ ] Myelencephalon: medulla oblongata and lower fourth ventricle
- [ ] Germ layer: ectoderm; circle on the mesencephalon

Check: There are five secondary vesicles because the mesencephalon carries through without splitting. If you have only four, you dropped it.

PROMPT B · VESICLE TABLE

- [ ] Telencephalon: from the prosencephalon; cerebrum; lateral ventricles
- [ ] Diencephalon: from the prosencephalon; thalamus, hypothalamus, epithalamus; third ventricle
- [ ] Mesencephalon: from the mesencephalon; midbrain; cerebral aqueduct
- [ ] Metencephalon: from the rhombencephalon; pons and cerebellum; upper fourth ventricle
- [ ] Myelencephalon: from the rhombencephalon; medulla oblongata; lower fourth ventricle
- [ ] Brain sketch numbered 1 cerebrum, 2 diencephalon, 3 midbrain
- [ ] 4 on both the pons and the cerebellum, 5 on the medulla

Check: The cerebellum grows from the metencephalon along with the pons, not from the myelencephalon.

153 Cerebral surface and lobes

PROMPT A · SIDE AND TOP VIEWS

- [ ] Frontal lobe at the front, parietal lobe behind the central sulcus, temporal lobe below the lateral sulcus, occipital lobe at the back
- [ ] A gyrus labeled as a raised ridge and a sulcus as a groove
- [ ] Central sulcus runs between the frontal and parietal lobes
- [ ] Precentral gyrus is just in front of the central sulcus; postcentral gyrus just behind it
- [ ] Lateral sulcus separates the temporal lobe from the frontal and parietal lobes
- [ ] Parieto-occipital sulcus separates the parietal and occipital lobes
- [ ] Insula drawn dashed, deep inside the lateral sulcus
- [ ] Transverse fissure drawn between the cerebrum and the cerebellum
- [ ] Top view shows the right and left hemispheres split by the longitudinal fissure
- [ ] Bones: frontal bone, parietal bone, temporal bone, occipital bone, each beside its lobe

Check: The precentral gyrus is in front of the central sulcus and the postcentral gyrus is behind it. These two get swapped a lot.

PROMPT B · BOUNDARY TABLE

- [ ] Gyrus labeled as a raised ridge, sulcus as a shallow groove, fissure as the deepest groove
- [ ] Central sulcus: frontal lobe and parietal lobe
- [ ] Lateral sulcus: frontal and parietal lobes above, temporal lobe below
- [ ] Parieto-occipital sulcus: parietal lobe and occipital lobe
- [ ] Longitudinal fissure: left hemisphere and right hemisphere
- [ ] Transverse fissure: cerebrum and cerebellum
- [ ] Top view shows the longitudinal fissure down the middle and the central sulcus across each side
- [ ] Frontal, parietal and occipital lobes labeled on the top view
- [ ] The lobe no outside view shows: the insula

Check: The lateral sulcus sits between the temporal lobe and the lobes above it. The line between the frontal and parietal lobes is the central sulcus.

154 Functional cortical areas

PROMPT A · LABELED CORTEX MAP

- [ ] Left hemisphere drawn, the side where Broca's and Wernicke's areas usually sit
- [ ] Primary motor cortex on the precentral gyrus, in front of the central sulcus
- [ ] Primary somatosensory cortex on the postcentral gyrus, behind the central sulcus
- [ ] Broca's area on the inferior frontal gyrus
- [ ] Wernicke's area where the temporal and parietal lobes meet
- [ ] Prefrontal cortex at the front of the frontal lobe
- [ ] Primary auditory cortex on the superior temporal gyrus, just below the lateral sulcus
- [ ] Primary visual cortex in the occipital lobe
- [ ] Gustatory cortex on the dashed insula, and the olfactory arrow pointing to the medial temporal lobe
- [ ] M on the primary motor cortex and Broca's area; S on the somatosensory, auditory, visual, gustatory and olfactory areas; A on Wernicke's area and the prefrontal cortex

Check: Broca's area is in the frontal lobe and Wernicke's area is back where the temporal and parietal lobes meet. Speaking is in front, understanding is behind.

PROMPT B · AREA LOCATION TABLE

- [ ] Primary motor: frontal lobe; precentral gyrus; M; starts voluntary movement
- [ ] Primary somatosensory: parietal lobe; postcentral gyrus; S; conscious touch, pressure and body position
- [ ] Primary visual: occipital lobe; calcarine sulcus; S; raw visual input
- [ ] Primary auditory: temporal lobe; superior temporal gyrus; S; hearing input
- [ ] Primary gustatory: insula and the lower postcentral gyrus; S; taste
- [ ] Primary olfactory: medial temporal lobe at the uncus; S; smell
- [ ] Broca's area: frontal lobe; inferior frontal gyrus, usually left; M; motor production of speech
- [ ] Wernicke's area: temporal and parietal junction, usually left; A; understanding spoken and written language
- [ ] Prefrontal cortex: front of the frontal lobe; A; judgment, planning and personality
- [ ] Added rows: somatosensory association area beside the primary somatosensory cortex, visual association area beside the primary visual cortex, auditory association area beside the primary auditory cortex

Check: The primary motor cortex is on the precentral gyrus (frontal) and the primary somatosensory cortex is on the postcentral gyrus (parietal). Check you did not swap them.

155 Cerebral white matter tracts

PROMPT A · LABELED FRONTAL SECTION

- [ ] Gray cortex drawn on the outside, white matter inside
- [ ] Association fiber arches between two gyri on the same side
- [ ] Corpus callosum crosses the midline between the two hemispheres
- [ ] Internal capsule runs down beside the basal ganglia toward the brainstem
- [ ] Internal capsule labeled as projection fibers
- [ ] Internal capsule shaded
- [ ] Note: nearly all traffic between the cortex and the body funnels through this one narrow band
- [ ] So a small stroke there cuts a great deal of wiring at once and can weaken the whole opposite side of the body

Check: The corpus callosum crosses side to side between the hemispheres; projection fibers run up and down. Check that each fiber goes the right direction.

PROMPT B · FIBER CLASS TABLE

- [ ] Association fibers: connect gyri within the same hemisphere; example: arcuate fasciculus
- [ ] Commissural fibers: connect matching gyri between the two hemispheres; example: corpus callosum (also the anterior and posterior commissures)
- [ ] Projection fibers: connect the cortex with lower brain regions and the spinal cord; example: internal capsule
- [ ] Arrow sketches: association stays on one side, commissural crosses the midline, projection runs up and down
- [ ] Midline sketch shows the corpus callosum
- [ ] Frontal section sketch shows the internal capsule beside the basal ganglia
- [ ] Circle on the corpus callosum
- [ ] Arcuate fasciculus links Broca's and Wernicke's areas
- [ ] Arcuate fasciculus is an association fiber

Check: The arcuate fasciculus stays in one hemisphere, so it is an association tract, not a commissural one.

156 Basal ganglia

PROMPT A · LABELED FRONTAL SECTION

- [ ] Gray cortex outside and white matter inside, with deep gray masses buried in the white matter
- [ ] Caudate nucleus labeled as one of the deep masses
- [ ] Putamen drawn lateral, toward the outside
- [ ] Globus pallidus drawn medial to the putamen
- [ ] Subthalamic nucleus marked in the diencephalon
- [ ] Substantia nigra marked in the midbrain
- [ ] Dotted lines link all five nuclei
- [ ] Roles: starting and regulating movement, suppressing unwanted movement, setting muscle tone

Check: The globus pallidus is medial, closer to the midline, and the putamen is lateral. Check your section.

PROMPT B · LOCATION SORT

- [ ] Cerebrum box: caudate nucleus, putamen, globus pallidus
- [ ] Diencephalon box: subthalamic nucleus
- [ ] Midbrain box: substantia nigra
- [ ] Circles on the caudate nucleus, putamen and globus pallidus
- [ ] Underlines on the subthalamic nucleus and substantia nigra
- [ ] Arrow shows the putamen lateral to the globus pallidus
- [ ] Substantia nigra looks dark (darkly pigmented)
- [ ] Its loss causes Parkinson disease

Check: The substantia nigra goes in the midbrain box and the subthalamic nucleus goes in the diencephalon box, even though both work with the basal ganglia.

157 Diencephalon

PROMPT A · LABELED MIDSAGITTAL VIEW

- [ ] Thalamus in the center, with the massa intermedia drawn as a round patch in its middle
- [ ] Hypothalamus below the thalamus
- [ ] Epithalamus at the back, with the pineal gland
- [ ] Third ventricle labeled as the midline space around the massa intermedia
- [ ] Corpus callosum above
- [ ] Optic chiasm and infundibulum below the hypothalamus
- [ ] Midbrain below
- [ ] Third ventricle shaded
- [ ] Thalamus note: about 80 percent of the diencephalon

Check: The pineal gland belongs to the epithalamus at the back. The infundibulum hangs from the hypothalamus at the front.

PROMPT B · THREE PART TABLE

- [ ] Thalamus: paired oval masses on each side of the third ventricle; holds the thalamic nuclei; main sensory relay to the cortex
- [ ] Hypothalamus: below the thalamus; connects to the pituitary through the infundibulum; master control for homeostasis
- [ ] Epithalamus: small part at the back; holds the pineal gland and habenular nuclei
- [ ] Epithalamus job: the pineal gland secretes melatonin
- [ ] Third ventricle drawn as a midline cavity
- [ ] Right and left thalami drawn on each side of the third ventricle
- [ ] Massa intermedia drawn as a bridge across the ventricle joining the two thalami

Check: The massa intermedia is a bridge that crosses the third ventricle to join the two thalami. It is not part of the hypothalamus.

158 Cerebellum

PROMPT A · BACK AND CUT VIEWS

- [ ] Back view shows the two cerebellar hemispheres
- [ ] Vermis is the narrow band in the midline between them
- [ ] Folia drawn as fine leaf-like folds on the surface
- [ ] Cut view shows the cerebellar cortex as the outer layer
- [ ] Arbor vitae is the branching, tree-like white matter inside
- [ ] Deep cerebellar nucleus drawn as gray matter buried in the white matter
- [ ] Superior cerebellar peduncle runs to the midbrain
- [ ] Middle cerebellar peduncle runs to the pons
- [ ] Inferior cerebellar peduncle runs to the medulla
- [ ] Cortex and deep nucleus shaded; arbor vitae left blank

Check: Shade the deep cerebellar nuclei too. They are gray matter even though they sit inside the white arbor vitae.

PROMPT B · PARTS TABLE

- [ ] Cerebellar hemispheres: the two side halves; gray cortex over white matter
- [ ] Vermis: the narrow midline band joining the hemispheres
- [ ] Folia: fine leaf-like folds of the surface; gray matter
- [ ] Arbor vitae: branching, tree-like pattern inside; white matter
- [ ] Deep cerebellar nuclei: buried in the white matter; gray matter
- [ ] Peduncles: stalks of white matter joining the cerebellum to the brainstem
- [ ] Superior peduncle joins the midbrain; arrow out of the cerebellum
- [ ] Middle peduncle joins the pons; arrow into the cerebellum
- [ ] Inferior peduncle joins the medulla; arrow into the cerebellum

Check: The superior peduncle mostly carries signals out. The middle and inferior peduncles carry signals in.

159 Limbic system

PROMPT A · LABELED MIDSAGITTAL VIEW

- [ ] Corpus callosum drawn across the middle
- [ ] Cingulate gyrus arches just above the corpus callosum
- [ ] Diencephalon labeled beneath the corpus callosum
- [ ] Hippocampus and amygdala drawn dashed, curving down and forward below the diencephalon into the temporal lobe, with the amygdala at the front end
- [ ] The three limbic structures together form a ring around the diencephalon
- [ ] Cingulate gyrus role: connects emotion with behavior
- [ ] Amygdala role: fear and other emotional responses, linking emotion to memory
- [ ] Hippocampus role: forming new long-term memories

Check: The hippocampus forms new memories and the amygdala handles fear. Check you did not swap their roles.

PROMPT B · LIMBIC COMPARE TABLE

- [ ] Cingulate gyrus: arches above the corpus callosum; connects emotion with behavior
- [ ] Amygdala: front of the medial temporal lobe, just in front of the hippocampus; fear and other emotional responses, linking emotion to memory
- [ ] Hippocampus: medial temporal lobe, in the floor of the inferior horn of the lateral ventricle; forming new long-term memories
- [ ] A small sketch in each row
- [ ] Oval drawn for the diencephalon with the three structures arranged around it
- [ ] Center of the ring, the two jobs: emotion and memory

Check: The cingulate gyrus is the part above the diencephalon, arching over the corpus callosum. Check it is not drawn below.

160 Cerebral arterial supply

PROMPT A · LABELED INFERIOR VIEW

- [ ] Two vertebral arteries join into one basilar artery on the front of the brainstem
- [ ] Basilar artery splits into the right and left posterior cerebral arteries
- [ ] Posterior communicating arteries join each posterior cerebral artery to the internal carotid
- [ ] Each internal carotid divides into the anterior and middle cerebral arteries
- [ ] Anterior communicating artery joins the right and left anterior cerebral arteries, closing the front of the ring
- [ ] Middle cerebral arteries run out to the sides, off the ring
- [ ] Posterior inferior cerebellar artery branches from each vertebral artery
- [ ] Anterior inferior and superior cerebellar arteries branch from the basilar artery
- [ ] Anterior circulation (internal carotids and their branches) shaded: feeds most of the cerebrum
- [ ] Posterior circulation (vertebral and basilar arteries and their branches) shaded: feeds the brainstem, cerebellum and occipital lobe

Check: The posterior inferior cerebellar artery comes off the vertebral artery. The other two cerebellar arteries come off the basilar. Check where your three start.

PROMPT B · TWO CIRCULATION CHAINS

- [ ] Anterior chain: internal carotid artery splits into the anterior and middle cerebral arteries
- [ ] Posterior chain: vertebral arteries join into the basilar artery, which splits into the posterior cerebral arteries
- [ ] Posterior inferior cerebellar artery branches off each vertebral artery
- [ ] Anterior inferior and superior cerebellar arteries branch off the basilar artery
- [ ] Posterior communicating artery joins the internal carotid to the posterior cerebral artery
- [ ] Anterior communicating artery joins the right and left anterior cerebral arteries
- [ ] Anterior cerebral artery: medial surface of the frontal and parietal lobes, including the leg and foot areas of the motor and sensory cortex
- [ ] Middle cerebral artery: most of the lateral surface of the cerebrum, including the face and arm areas of the motor and sensory cortex
- [ ] Posterior cerebral artery: occipital lobe and the underside of the temporal lobe

Check: The anterior communicating artery joins the two anterior cerebral arteries to each other. The posterior communicating artery is the one that links the front and back circulations.

161 Brainstem regions

PROMPT A · LABELED SIDE VIEW

- [ ] Diencephalon above the brainstem, cerebellum behind it, spinal cord below
- [ ] Midbrain on top
- [ ] Pons in the middle
- [ ] Medulla oblongata at the bottom
- [ ] Basilar pons drawn as the rounded bulge on the front of the pons
- [ ] Pontomedullary junction labeled where the pons meets the medulla
- [ ] Foramen magnum marked where the medulla becomes the spinal cord
- [ ] Arrow up through all three regions labeled ascending sensory tracts
- [ ] Arrow down through all three regions labeled descending motor tracts

Check: The medulla, not the pons, becomes the spinal cord at the foramen magnum.

PROMPT B · REGION ORDER CHAIN

- [ ] Chain from the top: cerebrum and diencephalon, midbrain, pons, medulla oblongata, spinal cord
- [ ] Cerebellum drawn off to the side, behind the brainstem
- [ ] Midbrain: most superior; a structure such as the cerebral peduncles or cerebral aqueduct
- [ ] Pons: middle; a structure such as the basilar pons or middle cerebellar peduncle
- [ ] Medulla oblongata: most inferior; a structure such as the pyramids or medullary olive
- [ ] Every ascending sensory tract and descending motor tract passes through all three
- [ ] Ten cranial nerves have nuclei in the brainstem

Check: Ten of the twelve cranial nerves have brainstem nuclei, CN III through CN XII. Twelve is the usual wrong answer.

162 Midbrain structures

PROMPT A · TWO LABELED VIEWS

- [ ] Back view shows the tectum as four rounded bumps
- [ ] Superior colliculi are the upper pair
- [ ] Inferior colliculi are the lower pair
- [ ] Cross section shows the tectum at the back as the roof
- [ ] Cerebral aqueduct drawn as a small channel near the back, just in front of the tectum
- [ ] Substantia nigra drawn as a dark band just behind the crus cerebri, the front part of each peduncle
- [ ] Cerebral peduncles on the front
- [ ] Arrow into the aqueduct from the third ventricle above
- [ ] Arrow out of the aqueduct into the fourth ventricle below

Check: Cerebrospinal fluid runs from the third ventricle above down into the fourth ventricle below. Check that your arrows point down.

PROMPT B · MIDBRAIN STRUCTURE TABLE

- [ ] Tectum: the roof on the back of the midbrain, made of four bumps; holds the reflex centers
- [ ] Superior colliculi: the upper pair of bumps; visual reflexes, like turning the eyes and head toward a sight
- [ ] Inferior colliculi: the lower pair of bumps; auditory reflexes, like turning toward a sound
- [ ] Cerebral peduncles: paired stalks on the front; carry descending motor tracts
- [ ] Substantia nigra: a dark band; helps regulate movement with the basal ganglia
- [ ] Cerebral aqueduct: a narrow channel through the midbrain; carries cerebrospinal fluid from the third to the fourth ventricle
- [ ] Circles on the superior and inferior colliculi
- [ ] Note: you do not become conscious of the sight or sound there; they run reflexes only

Check: Upper colliculi serve the eyes and lower colliculi serve the ears. Check you did not swap sight and sound.

163 Pons structures

PROMPT A · LABELED SIDE VIEW

- [ ] Midbrain on top, pons in the middle, medulla at the bottom
- [ ] Cerebellum drawn behind the brainstem
- [ ] Basilar pons is the rounded bulge on the front
- [ ] Middle cerebellar peduncle drawn as a thick stalk running back into the cerebellum
- [ ] Pontine nuclei dotted inside the pons
- [ ] Fourth ventricle drawn dashed behind the pons and upper medulla, in front of the cerebellum
- [ ] Arrow runs cerebral cortex, pontine nuclei, middle cerebellar peduncle, cerebellum

Check: The fourth ventricle sits behind the pons and upper medulla, between the brainstem and the cerebellum. It is not inside the pons.

PROMPT B · PONS FLOW CHAIN

- [ ] Cerebral cortex: sends motor information
- [ ] Pontine nuclei: relay the motor information on to the cerebellum
- [ ] Middle cerebellar peduncles: carry the signals from the pontine nuclei into the cerebellum
- [ ] Cerebellum: uses the information to coordinate movement
- [ ] Sketch shows the midbrain above the pons and the medulla below
- [ ] Fourth ventricle drawn behind the pons
- [ ] Pons means bridge
- [ ] It bridges the medulla, the midbrain and the cerebellum

Check: Signals from the pons enter the cerebellum through the middle cerebellar peduncle, not the superior or inferior one.

164 Medulla oblongata structures

PROMPT A · LABELED ANTERIOR VIEW

- [ ] Pons above, medulla oblongata below, spinal cord at the bottom
- [ ] Pontomedullary junction labeled at the groove where the pons meets the medulla
- [ ] Two pyramids drawn as ridges down the front of the medulla, one on each side of the midline
- [ ] Medullary olive drawn as an oval bump to the outside of each pyramid
- [ ] Decussation of the pyramids drawn where the medulla meets the cord
- [ ] Boxes inside the medulla for the cardiac, respiratory and vasomotor centers
- [ ] Crossing arrows drawn at the decussation
- [ ] Note: most motor fibers cross to the opposite side here, so the right hemisphere controls the left side of the body

Check: The olives sit to the outside of the pyramids, not between them.

PROMPT B · MEDULLA TABLE

- [ ] Pyramids: on the surface; paired ridges on the front made by the descending motor tracts
- [ ] Medullary olive: on the surface; an oval bump beside each pyramid
- [ ] Decussation of the pyramids: on the surface at the lower medulla; where most motor fibers cross to the opposite side
- [ ] Cardiac center: inside; adjusts the rate and force of the heartbeat
- [ ] Respiratory center: inside; sets the basic rhythm of breathing
- [ ] Vasomotor center: inside; adjusts blood vessel diameter, and so blood pressure
- [ ] Stars on the cardiac, respiratory and vasomotor centers
- [ ] Note: these centers run the heartbeat and breathing, so medulla injury is immediately life threatening

Check: The three vital centers are inside the medulla. Only the pyramids, olives and decussation show on the surface.

165 Cranial nerve nuclei distribution

PROMPT A · LABELED BRAINSTEM VIEW

- [ ] Midbrain on top, pons in the middle, medulla at the bottom
- [ ] Pontomedullary junction labeled between the pons and the medulla
- [ ] Midbrain: CN III oculomotor and CN IV trochlear
- [ ] Pons: CN V trigeminal, CN VI abducens, CN VII facial, CN VIII vestibulocochlear
- [ ] Medulla: CN VIII vestibulocochlear, CN IX glossopharyngeal, CN X vagus, CN XII hypoglossal, and CN XI accessory (grouped here, though its nucleus sits in the upper cervical cord)
- [ ] No nuclei in the brainstem: CN I olfactory and CN II optic
- [ ] Circle on CN VIII, written beside both the pons and the medulla

Check: CN VIII is the one nerve listed beside both the pons and the medulla. If any other nerve shows up twice, recheck.

PROMPT B · NERVE SORT TABLE

- [ ] CN III oculomotor and CN IV trochlear: checked under midbrain
- [ ] CN V trigeminal: pons
- [ ] CN VI abducens: pons
- [ ] CN VII facial: pons
- [ ] CN VIII vestibulocochlear: checked under both pons and medulla
- [ ] CN IX glossopharyngeal and CN X vagus: medulla
- [ ] CN XI accessory and CN XII hypoglossal: medulla (CN XI is grouped here, though its nucleus sits in the upper cervical cord)
- [ ] Count: 10 of the 12 cranial nerves have brainstem nuclei
- [ ] Circle on CN VIII

Check: CN VI abducens and CN VII facial belong to the pons, not the medulla. Check where you put your checks for them.

166 Reticular formation

PROMPT A · LABELED CUTAWAY

- [ ] Brainstem drawn from the side, cut open to show the core
- [ ] Net of dots and lines runs through the midbrain, pons and medulla
- [ ] Arrows up from the net to the cerebral cortex, labeled reticular activating system
- [ ] Arrows down toward the spinal cord, labeled descending component
- [ ] Reticular activating system jobs, any two: keeps you conscious and alert, holds attention, filters out unimportant stimuli
- [ ] Descending component jobs, any two: sets resting muscle tone, helps keep posture, holds the medulla's centers for heart rate, breathing and blood vessel diameter

Check: The reticular formation runs through all three brainstem regions, not only the medulla.

PROMPT B · COMPARE TABLE

- [ ] Sketch above the table shows a net of gray matter running through the white matter core of the brainstem
- [ ] Reticular activating system: sends signals up
- [ ] Reticular activating system: acts on the cerebral cortex
- [ ] Reticular activating system jobs: consciousness and arousal, attention, filtering out unimportant stimuli
- [ ] Descending component: sends signals down toward the spinal cord
- [ ] Descending component: acts on the spinal cord, and through it the skeletal muscles
- [ ] Descending component jobs: resting muscle tone, posture, and through the medullary centers heart rate, breathing and blood vessel diameter
- [ ] When the reticular activating system shuts down, the result is sleep
- [ ] Severe damage to it can cause coma

Check: Muscle tone belongs to the descending component. The reticular activating system is the upward part that keeps the cortex awake.

167 Cranial meninges and dural reflections

PROMPT A · LABELED MIDSAGITTAL VIEW

- [ ] Falx cerebri hangs down in the midline between the right and left cerebral hemispheres
- [ ] Tentorium cerebelli is the shelf between the cerebrum above and the cerebellum below
- [ ] Falx cerebelli sits below the tentorium, between the right and left cerebellar hemispheres
- [ ] Superior sagittal sinus runs along the top edge of the falx cerebri
- [ ] Confluence of sinuses at the back of the skull, at the internal occipital protuberance
- [ ] Transverse sinus runs out to the side from the confluence, then turns down as the S-shaped sigmoid sinus
- [ ] Sigmoid sinus ends at the jugular foramen, where the internal jugular vein leaves the skull
- [ ] Arrows run front to back along the superior sagittal sinus, then out, down and into the internal jugular vein
- [ ] Corner strip in order from skull to brain: dura, arachnoid, pia, with the pia clinging to the brain surface
- [ ] Dura drawn as two layers: periosteal layer against the skull, meningeal layer beneath it

Check: The tentorium cerebelli separates the cerebrum from the cerebellum; it does not split the cerebellum. That job belongs to the falx cerebelli.

PROMPT B · TABLE AND FLOW CHAIN

- [ ] Dura mater: outermost, against the skull, tough and two-layered
- [ ] Arachnoid mater: middle layer, lace-like, between the dura and the pia
- [ ] Pia mater: innermost, thin and full of blood vessels, clinging to the brain surface
- [ ] Falx cerebri: in the longitudinal fissure, separating the right and left cerebral hemispheres
- [ ] Falx cerebelli: in the midline below the tentorium, separating the right and left cerebellar hemispheres
- [ ] Tentorium cerebelli: in the transverse fissure, roofing the cerebellum, separating the cerebrum above from the cerebellum below
- [ ] Chain in order: superior sagittal sinus, confluence of sinuses, transverse sinus, sigmoid sinus, internal jugular vein
- [ ] Arachnoid granulations marked on the superior sagittal sinus box

Check: The arachnoid granulations empty CSF into the superior sagittal sinus, not into the internal jugular vein. Check which box you marked.

168 Spinal meninges and meningeal spaces

PROMPT A · LABELED CROSS SECTION

- [ ] Vertebral bone on the outside of the drawing
- [ ] Epidural space between the bone and the dura, filled with fat and veins
- [ ] Dura mater as the tough outer membrane around the cord
- [ ] Subdural space between the dura and the arachnoid
- [ ] Arachnoid mater as the middle membrane
- [ ] Subarachnoid space with CSF between the arachnoid and the pia
- [ ] Pia mater clinging right on the surface of the cord
- [ ] Denticulate ligament running from the pia out to the dura
- [ ] Filum terminale running from the tapered end of the cord down to the coccyx
- [ ] Star on the subarachnoid space, and potential space written beside the subdural space

Check: CSF is in the subarachnoid space, under the arachnoid. If your star is on the epidural or subdural space, move it.

PROMPT B · SPACE COMPARE TABLE

- [ ] Epidural space: between the dura and the vertebral bone
- [ ] Epidural space holds fat and veins in the spine; in the cranium it is only a potential space, because the dura is attached to the skull
- [ ] Subdural space: between the dura and the arachnoid
- [ ] Subdural space is only a potential space with a thin film of fluid, and it is found in the cranium too
- [ ] Subarachnoid space: between the arachnoid and the pia
- [ ] Subarachnoid space holds CSF and is found in the cranium too
- [ ] Denticulate ligaments are made of pia and anchor the cord to the dura along its length
- [ ] Filum terminale is made of pia and anchors the end of the cord to the coccyx

Check: The epidural space is a real space only in the spine. In the cranium the dura is attached to the skull, so the epidural space there is only a potential space.

169 Cord termination and lumbar puncture

PROMPT A · LABELED SIDE VIEW

- ☐ Conus medullaris, the tapered end of the cord, at the L1 to L2 disc
- ☐ Cauda equina hanging down below the conus as a bundle of nerve roots
- ☐ Filum terminale running from the conus down to the coccyx
- ☐ Lumbar cistern below the conus, filled with CSF
- ☐ Needle entering in the midline between the L3 and L4 spines
- ☐ Needle layers in order: skin, subcutaneous fat, supraspinous ligament, interspinous ligament
- ☐ Then ligamentum flavum, epidural space, dura mater, arachnoid mater
- ☐ Needle tip ends in the subarachnoid space of the lumbar cistern
- ☐ Note by the conus: the cord ends at L1 to L2, so a needle below that meets only CSF and the floating cauda equina, which drifts aside

Check: The cord ends at the L1 to L2 disc, not at the bottom of the spine. If your cord runs down to the sacrum, shorten it.

PROMPT B · NEEDLE PATH SEQUENCE

- ☐ Box 1 skin, box 2 subcutaneous fat
- ☐ Box 3 supraspinous ligament along the tips of the spinous processes
- ☐ Box 4 interspinous ligament between neighboring spinous processes
- ☐ Box 5 ligamentum flavum joining the laminae
- ☐ Box 6 epidural space with fat and veins
- ☐ Box 7 dura mater, box 8 arachnoid mater, box 9 subarachnoid space where CSF is drawn
- ☐ Lumbar cistern sketched below the conus medullaris, with the cauda equina floating in CSF and the filum terminale running down the middle
- ☐ Mark on the ligamentum flavum box as the firm ligament that gives way
- ☐ Entry levels written as between the L3 and L4 spines or the L4 and L5 spines

Check: There are three ligaments in a row, and the ligamentum flavum is the last and deepest of them. Make sure it sits just above the epidural space.

170 Ventricular system

PROMPT A · LABELED VENTRICLE VIEW

- ☐ A lateral ventricle drawn as the large curved chamber in the cerebral hemisphere
- ☐ Interventricular foramen of Monro joining the lateral ventricle to the third ventricle
- ☐ Third ventricle as a thin midline chamber in the diencephalon
- ☐ Cerebral aqueduct as a narrow channel through the midbrain to the fourth ventricle
- ☐ Fourth ventricle between the brainstem and the cerebellum
- ☐ Central canal running down the spinal cord, continuous with the fourth ventricle
- ☐ Median aperture and two lateral apertures opening from the fourth ventricle into the subarachnoid space
- ☐ Regions written: lateral ventricle in the cerebrum, third in the diencephalon, fourth between the brainstem and the cerebellum
- ☐ Ependymal cells written as the lining of every chamber

Check: The cerebral aqueduct joins the third ventricle to the fourth. The lateral ventricles reach the third ventricle through the interventricular foramina instead.

PROMPT B · CONNECTION TABLE

- ☐ Lateral ventricles: one in each cerebral hemisphere; connect to the third ventricle through the interventricular foramina
- ☐ Third ventricle: in the diencephalon; connects to the fourth ventricle through the cerebral aqueduct
- ☐ Fourth ventricle: between the brainstem and the cerebellum; opens into the subarachnoid space through the median aperture and two lateral apertures
- ☐ Fourth ventricle also continues straight down into the central canal
- ☐ Central canal: runs the length of the spinal cord; continuous above with the fourth ventricle
- ☐ Circled: the interventricular foramina, the cerebral aqueduct, and the apertures
- ☐ If one is blocked, CSF backs up behind it, the ventricles swell, and pressure in the skull rises (hydrocephalus)

Check: The apertures open into the subarachnoid space, not into the central canal. The central canal joins the fourth ventricle directly, with no aperture.

171 CSF production and circulation

PROMPT A · LABELED FLOW DRAWING

- ☐ Choroid plexus drawn in the lateral, third, and fourth ventricles
- ☐ Arrow 1 starts in a lateral ventricle
- ☐ Arrow 2 through an interventricular foramen into the third ventricle
- ☐ Arrow 3 down the cerebral aqueduct to the fourth ventricle
- ☐ Arrow 4 out through the median and lateral apertures into the subarachnoid space
- ☐ Arrows then run up around the brain in the subarachnoid space to the arachnoid granulations
- ☐ Arachnoid granulations drawn poking into the superior sagittal sinus
- ☐ Side arrow from the fourth ventricle down into the central canal
- ☐ Ependymal cells of the choroid plexus written as the cells that make CSF

Check: The side arrow into the central canal leaves from the fourth ventricle, not from the subarachnoid space. Check where yours starts.

PROMPT B · FLOW CHAIN

- ☐ Box 1 lateral ventricles, where the choroid plexus makes CSF
- ☐ Box 2 interventricular foramina
- ☐ Box 3 third ventricle
- ☐ Box 4 cerebral aqueduct
- ☐ Box 5 fourth ventricle
- ☐ Box 6 the apertures, one median and two lateral
- ☐ Box 7 subarachnoid space around the brain and cord
- ☐ Box 8 arachnoid granulations, where CSF is reabsorbed
- ☐ Choroid plexus marks on the lateral, third, and fourth ventricle boxes only
- ☐ Final arrow into the venous blood of the superior sagittal sinus

Check: The foramina, the aqueduct, and the apertures are passageways with no choroid plexus. Only the three ventricle boxes get the mark.

172 Blood-brain barrier structure

PROMPT A · LABELED CAPILLARY SECTION

- [ ] Blood in the center of the capillary
- [ ] Capillary wall made of endothelial cells joined by tight junctions, with no pores
- [ ] Basement membrane wrapped around the outside of the endothelial cells
- [ ] Astrocyte end-feet covering the outside of the basement membrane
- [ ] One star-shaped astrocyte with several end-feet reaching the capillary
- [ ] A neuron drawn near the astrocyte, outside the capillary
- [ ] Arrow from the blood toward the neuron
- [ ] Numbered in order: 1 endothelial cells with tight junctions, 2 basement membrane, 3 astrocyte end-feet

Check: The astrocyte end-feet are the outermost layer, outside the basement membrane. If your arrow meets the end-feet first, the order is flipped.

PROMPT B · COMPARE TABLE

- [ ] Capillary endothelial cells: the cells lining the brain capillary
- [ ] Endothelial cells are first, the layer in contact with the blood
- [ ] Endothelial feature: tight junctions between the cells and no pores
- [ ] Basement membrane: the supporting layer that wraps the endothelium
- [ ] Basement membrane is second counting out from the blood
- [ ] Astrocyte end-feet: foot-like projections of astrocytes
- [ ] Astrocyte end-feet are third, the outermost layer, enveloping the capillary and helping control the barrier
- [ ] Ordinary capillary sketched with pores labeled in its wall, beside a brain capillary with none

Check: The end-feet come from astrocytes, not from oligodendrocytes or neurons. Check your label.

173 Nervous system divisions

PROMPT A · BRANCHING MAP

- [ ] Nervous system splits into the CNS and the PNS
- [ ] CNS: brain and spinal cord
- [ ] PNS: nerves, ganglia, and sensory receptors
- [ ] PNS motor side branches into somatic, autonomic, and enteric
- [ ] Autonomic splits into sympathetic and parasympathetic
- [ ] Somatic: skeletal muscle
- [ ] Autonomic: smooth muscle, cardiac muscle, and glands
- [ ] Enteric: smooth muscle and glands in the wall of the GI tract
- [ ] 12 pairs of cranial nerves and 31 pairs of spinal nerves

Check: Sympathetic and parasympathetic belong under autonomic, not under somatic. Trace your branch lines.

PROMPT B · LABELED BODY OUTLINE

- [ ] Brain and spinal cord shaded and labeled CNS
- [ ] Nerves leaving the cord labeled PNS
- [ ] A ganglion outside the cord labeled PNS
- [ ] A sensory receptor in the skin labeled PNS
- [ ] Skeletal muscle labeled somatic
- [ ] Heart labeled autonomic
- [ ] Gland labeled autonomic
- [ ] Loop of gut wall with its own neurons labeled enteric

Check: The heart is muscle, but it is cardiac muscle, so it is autonomic, not somatic.

174 Neuron structure

PROMPT A · LABELED NEURON

- [ ] Multipolar shape: many branching dendrites and a single axon
- [ ] Cell body with a nucleus and a dark nucleolus inside
- [ ] Nissl bodies drawn as dark specks in the cell body
- [ ] Dendrites branching off the cell body like a tree
- [ ] Axon hillock as the cone where the cell body narrows into the axon
- [ ] One long axon leaving the hillock
- [ ] Axon terminals branching at the far end, each with a swollen synaptic bulb
- [ ] Arrows run one way: dendrites, cell body, axon hillock, axon, axon terminals

Check: Nissl bodies are in the cell body and dendrites, never in the axon hillock or the axon. If you drew them in the axon, take them out.

PROMPT B · SIGNAL FLOW CHAIN

- [ ] Box 1 dendrites: receive incoming signals
- [ ] Box 2 cell body: holds the nucleus, the metabolic center
- [ ] Box 3 axon hillock: where the impulse starts (the trigger zone)
- [ ] Box 4 axon: carries the impulse away
- [ ] Box 5 axon terminals: contact the target cell
- [ ] Nissl bodies drawn in the cell body box, made of rough ER and ribosomes
- [ ] Nissl bodies make proteins
- [ ] Neurofibrils drawn as fibers in the cell body box
- [ ] The axon depends on the cell body for proteins, so the part cut away from the cell body dies

Check: The axon hillock comes between the cell body and the axon. If it is after the axon in your chain, the order is off.

175 Synapse structures

PROMPT A · LABELED SYNAPSE CLOSE-UP

- [ ] Axon terminal of the first neuron ending in a swollen synaptic end bulb
- [ ] Synaptic vesicles drawn inside the end bulb
- [ ] Synaptic cleft as a gap between the end bulb and the next cell
- [ ] The next cell labeled on the far side of the cleft
- [ ] Neurotransmitter molecules leaving the vesicles and crossing the cleft
- [ ] Sketch of an axon ending on a dendrite labeled axodendritic
- [ ] Sketch of an axon ending on a cell body labeled axosomatic
- [ ] Sketch of an axon ending on another axon labeled axoaxonic

Check: There is a gap between the two cells. If your end bulb touches the next cell, open up the cleft.

PROMPT B · SYNAPSE TYPE MAP

- [ ] One neuron with an axon from another neuron ending on its dendrite, labeled axodendritic
- [ ] A second axon ending on its cell body, labeled axosomatic
- [ ] A third axon ending on its axon, labeled axoaxonic
- [ ] Zoom circle showing the synaptic end bulb
- [ ] Synaptic vesicles inside the end bulb
- [ ] Synaptic cleft as the gap between the bulb and the next cell
- [ ] Vesicles store neurotransmitter
- [ ] Neurotransmitter molecules carry the signal across the cleft

Check: Vesicles belong in the sending end bulb, not in the receiving cell. Check which side of the cleft they are on.

176 Neuron classification

PROMPT A · THREE SHAPE SKETCHES

- [ ] Multipolar: many dendrites and one axon leaving the cell body
- [ ] Bipolar: one dendrite and one axon, on opposite ends of the cell body
- [ ] Unipolar: one short process leaves the cell body, then splits like a T into a branch from the receptor and a branch that runs into the CNS
- [ ] Process counts written: multipolar many, bipolar two, unipolar one
- [ ] Under multipolar: brain and spinal cord, and motor neurons
- [ ] Under bipolar: retina, inner ear, and olfactory area
- [ ] Purkinje cell and pyramidal cell both written under multipolar
- [ ] Most sensory neurons labeled unipolar
- [ ] Motor neurons and interneurons labeled multipolar

Check: Bipolar has two processes and unipolar has one. Count the lines leaving each cell body before you label it.

PROMPT B · FUNCTIONAL PATHWAY MAP

- [ ] Sensory neuron runs from the skin receptor into the spinal cord
- [ ] Sensory neuron drawn and labeled unipolar
- [ ] Interneuron drawn inside the cord, linking the sensory neuron to the motor neuron
- [ ] Interneuron drawn and labeled multipolar
- [ ] Motor neuron runs from the cord out to the skeletal muscle
- [ ] Motor neuron drawn and labeled multipolar
- [ ] Arrows run receptor to cord to muscle
- [ ] Bipolar neuron in the margin with retina, inner ear, and olfactory area

Check: The sensory neuron is the unipolar one. If you drew all three as multipolar, redraw the sensory neuron.

177 Neuroglia identification

PROMPT A · LABELED GLIA SCENE

- [ ] Astrocyte drawn star-shaped, touching the capillary
- [ ] Oligodendrocyte with arms wrapping several axons
- [ ] Microglial cell drawn small, eating debris
- [ ] Ependymal cells lining the edge of the ventricle
- [ ] PNS box: Schwann cells wrapped along an axon
- [ ] PNS box: satellite cells around the cell bodies in a ganglion
- [ ] Astrocyte role: support, helps form the blood-brain barrier, guides neuron growth, keeps the chemical environment steady
- [ ] Oligodendrocyte role: makes CNS myelin; microglia role: remove debris and microbes
- [ ] Ependymal role: line the ventricles and central canal and help make and move CSF
- [ ] Schwann cell role: makes PNS myelin and helps axons regrow; satellite cell role: support the cell bodies in ganglia

Check: Oligodendrocytes make myelin in the CNS, Schwann cells in the PNS. Make sure each one is on the correct side of your drawing.

PROMPT B · SORTING BOX

- [ ] CNS side: astrocytes, oligodendrocytes, microglia, ependymal cells
- [ ] PNS side: Schwann cells and satellite cells
- [ ] Astrocyte: star shape; supports neurons, helps form the blood-brain barrier, keeps the chemical environment steady
- [ ] Oligodendrocyte: makes the myelin sheaths of the CNS
- [ ] Microglia: small phagocytes that remove debris and microbes
- [ ] Ependymal cells: line the ventricles and central canal and help make and move CSF
- [ ] Schwann cell: makes PNS myelin and helps axons regrow
- [ ] Satellite cell: surrounds and supports cell bodies in ganglia
- [ ] Circled: oligodendrocyte and Schwann cell
- [ ] Starred: astrocyte

Check: The CNS side should have four cells and the PNS side two. If the count is off, one cell is on the wrong side.

178 Myelination and regeneration

PROMPT A · LABELED AXON COMPARISON

- [ ] Row of Schwann cells along the PNS axon, one cell per segment
- [ ] Myelin sheath as the wrapped layers around the axon
- [ ] Nodes of Ranvier as the gaps between neighboring Schwann cells
- [ ] Neurolemma as the outer layer of each Schwann cell
- [ ] One oligodendrocyte with several arms, each wrapping a segment of a different axon
- [ ] Myelin sheath and nodes of Ranvier labeled on the CNS axons
- [ ] No neurolemma on the CNS axons
- [ ] PNS axon can regrow because the neurolemma guides it
- [ ] CNS axon has very limited regrowth because there is no neurolemma

Check: One Schwann cell wraps one segment of one axon. If your Schwann cell reaches several axons, you drew an oligodendrocyte.

PROMPT B · COMPARE TABLE

- [ ] Cell that makes myelin: Schwann cell in the PNS, oligodendrocyte in the CNS
- [ ] Axons per cell: a Schwann cell wraps one segment of one axon
- [ ] Axons per cell: one oligodendrocyte wraps segments of several axons
- [ ] Neurolemma: present in the PNS, absent in the CNS
- [ ] Cut axon: can regrow in the PNS, very limited in the CNS
- [ ] Myelinated axon sketched with gaps (nodes) between the myelin segments
- [ ] Myelin is made of lipid and protein in many layers
- [ ] Impulse drawn jumping from node to node

Check: Regrowth depends on the neurolemma. If you wrote that the CNS regrows well, check that row again.

179 CNS and PNS tissue terms

PROMPT A · TWO BY TWO GRID

- [ ] CNS and cluster of cell bodies: nucleus
- [ ] PNS and cluster of cell bodies: ganglion
- [ ] CNS and bundle of axons: tract
- [ ] PNS and bundle of axons: nerve
- [ ] Dots sketched in the nucleus and ganglion boxes
- [ ] Lines sketched in the tract and nerve boxes
- [ ] Each term has a one-line definition using cluster or bundle and CNS or PNS

Check: Nucleus and ganglion are both clusters of cell bodies. The only difference is CNS or PNS, so check which column each one is in.

PROMPT B · LABELED NERVOUS SYSTEM MAP

- [ ] Nucleus drawn as dots inside the brain
- [ ] Tract drawn as lines running up inside the cord
- [ ] Nerve leaving the cord
- [ ] Ganglion drawn as a swelling along the nerve, outside the CNS
- [ ] Dashed line at the surface of the cord where the nerve leaves
- [ ] Nucleus and ganglion labeled cell bodies
- [ ] Tract and nerve labeled axons

Check: A bundle of axons inside the cord is a tract, not a nerve. Nerves are only outside the CNS.

180 Gray and white matter

PROMPT A · TWO LABELED SECTIONS

- [ ] Cord section: gray butterfly in the center, white matter around it
- [ ] Gray matter labeled with dorsal (posterior) horns and ventral (anterior) horns
- [ ] Central canal in the middle of the gray matter
- [ ] Cerebrum slice: gray cortex on the outside, white matter beneath it
- [ ] Small cerebellum with gray cortex on the outside
- [ ] Key: gray matter is neuron cell bodies, dendrites, and unmyelinated fibers
- [ ] Key: white matter is myelinated axons

Check: Gray is outside in the brain but inside in the cord. Check that your cord shading is the reverse of your brain shading.

PROMPT B · COMPARE TABLE

- [ ] Gray matter is made of neuron cell bodies, dendrites, and unmyelinated fibers
- [ ] White matter is made of myelinated axons running in bundles
- [ ] Brain: gray matter in the outer cortex of the cerebrum and cerebellum, plus deep nuclei inside
- [ ] Brain: white matter beneath the cortex
- [ ] Cord: gray matter in the inner core
- [ ] Cord: white matter in the outer part
- [ ] Brain circle shaded gray on the outside
- [ ] Cord circle shaded white on the outside

Check: White matter is white because of myelin. If you wrote cell bodies for white matter, swap the rows.

181 Reflex arc components

PROMPT A · LABELED REFLEX ARC

- [ ] Receptor that detects the stimulus
- [ ] Sensory neuron carrying the signal from the receptor into the cord
- [ ] Integration center: the synapse in the cord
- [ ] Motor neuron carrying the command out of the cord
- [ ] Effector: the muscle that responds
- [ ] All five components labeled
- [ ] Arrows run receptor, sensory neuron, cord, motor neuron, muscle
- [ ] X placed on one part of the arc
- [ ] A missing reflex points to a break somewhere along that pathway

Check: The integration center counts as one of the five components. If you have only four labels, you left out the synapse in the cord.

PROMPT B · FLOW CHAIN

- [ ] Box 1 receptor: detects the stimulus
- [ ] Box 2 sensory neuron: carries the signal to the CNS
- [ ] Box 3 integration center: one or more synapses in the CNS where the signal is processed
- [ ] Box 4 motor neuron: carries the command to the effector
- [ ] Box 5 effector: the muscle or gland that responds
- [ ] Arrows between every pair of boxes, pointing toward the effector
- [ ] Integration center circled
- [ ] A reflex is fast because it runs a short, fixed pathway

Check: The sensory neuron comes before the integration center and the motor neuron comes after it. Check your box order.

182 Types of reflexes

PROMPT A · COMPARE TABLE

- [ ] Somatic: based on the effector, skeletal muscle; example the patellar (knee-jerk) reflex
- [ ] Autonomic: based on the effector, smooth muscle, cardiac muscle, or a gland; example the pupillary light reflex
- [ ] Monosynaptic: based on one synapse in the integration center; example the stretch reflex
- [ ] Polysynaptic: based on two or more synapses with interneurons; example the withdrawal reflex
- [ ] Innate: present from birth; example the blink reflex
- [ ] Learned: built by practice and repetition; example catching a dropped object
- [ ] Bracket on somatic and autonomic: the effector
- [ ] Bracket on monosynaptic and polysynaptic: the number of synapses
- [ ] Bracket on innate and learned: inborn or learned

Check: The withdrawal reflex runs through interneurons, so it is polysynaptic, not monosynaptic.

PROMPT B · PAIRED SKETCHES

- [ ] Somatic sketch shows skeletal muscle as the effector, labeled with the patellar reflex
- [ ] Autonomic sketch shows smooth muscle, cardiac muscle, or a gland, labeled with the pupillary light reflex
- [ ] Monosynaptic sketch: sensory neuron synapses directly on the motor neuron in the cord, labeled stretch reflex
- [ ] Polysynaptic sketch: interneurons between the sensory and motor neurons, labeled withdrawal reflex
- [ ] Innate sketch labeled blink reflex
- [ ] Learned sketch labeled catching a dropped object
- [ ] Under the first pair: the effector
- [ ] Under the second pair: the number of synapses
- [ ] Under the third pair: whether it is inborn or learned

Check: The pupillary light reflex is autonomic because its effector is smooth muscle in the iris. Make sure it is not under somatic.

183 Neuronal pools and pathways

PROMPT A · LABELED CIRCUIT DRAWING

- [ ] Convergence: many neurons with arrows feeding onto one neuron
- [ ] Divergence: one neuron with arrows spreading to many neurons
- [ ] Ascending pathway with arrows running up the cord to the brain
- [ ] Descending pathway with arrows running down from the brain
- [ ] Each pathway crosses the midline
- [ ] Each crossing labeled decussation
- [ ] Stroke note: pathways cross, so each side of the brain controls the opposite side of the body

Check: Convergence arrows point in toward one neuron and divergence arrows spread out. Check you did not draw both the same way.

PROMPT B · COMPARE TABLE

- [ ] Convergence: many neurons feed their signals onto one; signal travels in toward one neuron
- [ ] Divergence: one neuron spreads its signal to many; signal travels out to many neurons
- [ ] Ascending pathway: tracts carrying sensory information; signal travels up the cord to the brain
- [ ] Descending pathway: tracts carrying motor commands; signal travels down from the brain to the body
- [ ] Decussation: a pathway crossing to the opposite side; signal travels across the midline
- [ ] Body outline with the right half shaded

Check: The left side of the brain controls the right side of the body. If you shaded the left half, switch it.

184 Spinal cord external anatomy

PROMPT A · LABELED POSTERIOR VIEW

- [ ] Cord drawn from the medulla down to a tapered end, with the vertebral column faint behind it
- [ ] Cervical enlargement in the neck region, where the nerves to the upper limbs arise
- [ ] Lumbar enlargement lower down, where the nerves to the lower limbs arise
- [ ] Conus medullaris as the tapered cone at the end of the cord
- [ ] Cord end marked at the L1 to L2 disc
- [ ] Cauda equina hanging below the conus as a bundle of nerve roots
- [ ] Filum terminale running from the conus to the coccyx
- [ ] Posterior median sulcus down the middle of the back of the cord
- [ ] Cross section: anterior median fissure on the front drawn deep, posterior median sulcus on the back drawn shallow

Check: The anterior median fissure is the deep groove and the posterior median sulcus is the shallow one. Check which side of your cross section has the deep notch.

PROMPT B · STRUCTURE AND LANDMARK TABLE

- [ ] Cervical enlargement: a wider part of the cord from C4 to T1; serves the upper limbs
- [ ] Lumbar enlargement: a wider part of the lower cord, behind the T9 to T12 vertebrae; serves the lower limbs
- [ ] Conus medullaris: tapered cone at the end of the cord, at the L1 to L2 disc
- [ ] Cauda equina: hair-like bundle of nerve roots below the conus, inside the vertebral canal
- [ ] Filum terminale: thread of pia from the conus; anchors the cord to the coccyx
- [ ] Anterior median fissure: deep groove down the front
- [ ] Posterior median sulcus: shallow groove down the back; with the fissure it divides the cord into right and left halves
- [ ] Needle marked between the L3 and L4 or L4 and L5 spines, below the conus
- [ ] Below the conus there is no cord, only CSF and the cauda equina, which drifts aside

Check: The cord ends at L1 to L2, well above the end of the vertebral column. The cauda equina below it is nerve roots, not cord.

185 Roots, rootlets, and root ganglion

PROMPT A · LABELED CUTAWAY

- [ ] Row of rootlets leaving the back of the cord and a row leaving the front
- [ ] Back rootlets gather into the posterior root
- [ ] Front rootlets gather into the anterior root
- [ ] Posterior root ganglion as a swelling on the posterior root only
- [ ] The two roots join to form the spinal nerve
- [ ] Spinal nerve leaving through the intervertebral foramen
- [ ] Posterior root labeled sensory, anterior root labeled motor
- [ ] Arrows point into the cord on the posterior root and out of the cord on the anterior root

Check: Only the posterior root has a ganglion. If you put a swelling on the anterior root, take it off.

PROMPT B · FLOW CHAIN

- [ ] Sensory chain: receptor in the skin, spinal nerve, posterior root, rootlets, cord
- [ ] Arrows on the sensory chain point toward the cord
- [ ] Motor chain: cord, rootlets, anterior root, spinal nerve, muscle
- [ ] Arrows on the motor chain point away from the cord
- [ ] Posterior root ganglion placed on the posterior root, in the sensory chain only
- [ ] Intervertebral foramen placed where the spinal nerve passes out of the vertebral canal, in both chains
- [ ] Posterior root ganglion circled as the home of the sensory cell bodies
- [ ] Spinal nerve starred
- [ ] It is mixed because it carries both sensory and motor axons

Check: The sensory cell bodies are in the posterior root ganglion, outside the cord. Do not circle the cord or the rootlets.

186 Spinal cord internal anatomy

PROMPT A · LABELED CROSS SECTION

- [ ] Gray butterfly in the center, white matter around it
- [ ] Posterior horns pointing toward the back
- [ ] Small lateral horns pointing out to the sides
- [ ] Anterior horns pointing toward the front
- [ ] Gray commissure as the gray bridge across the middle, around the central canal
- [ ] Central canal as a small hole at the very center
- [ ] Anterior white commissure as a white bridge just in front of the gray commissure
- [ ] Anterior, lateral, and posterior white columns labeled
- [ ] Posterior horn: interneurons and incoming sensory axons; lateral horn: autonomic motor nuclei; anterior horn: somatic motor nuclei for skeletal muscle
- [ ] Tracts written inside one white column

Check: The anterior white commissure sits in front of the gray commissure, toward the anterior median fissure. Check that yours is not behind it.

PROMPT B · LANDMARK TABLE

- [ ] Posterior horn: gray; the back points of the butterfly; holds interneurons and incoming sensory axons
- [ ] Lateral horn: gray; small side point between the other two horns; holds autonomic motor nuclei
- [ ] Anterior horn: gray; the front points of the butterfly; holds somatic motor nuclei for skeletal muscle
- [ ] Gray commissure: gray; the bar across the middle; connects the two sides
- [ ] Anterior white commissure: white; just in front of the gray commissure; connects the two sides
- [ ] Central canal: in the center of the gray commissure; holds CSF
- [ ] Anterior, lateral, and posterior columns: white; front, side, and back of the white matter; hold tracts
- [ ] Cervical section drawn large and oval with lots of white matter and no lateral horn
- [ ] Thoracic section drawn smaller, with the lateral horn circled

Check: The lateral horn is found only from T1 to L2. It belongs on your thoracic sketch, not your cervical one.

187 Nerve connective tissue wrappings

PROMPT A · LABELED NERVE CROSS SECTION

- [ ] Whole nerve with several fascicles inside it
- [ ] One fascicle enlarged to show a few axons
- [ ] An axon labeled inside the fascicle
- [ ] Endoneurium around each single axon
- [ ] Perineurium as a ring around each fascicle
- [ ] Epineurium around the whole nerve, on the outside
- [ ] Levels written: endoneurium one axon, perineurium one fascicle, epineurium the whole nerve

Check: Endo is inside and epi is outside. If your endoneurium is around the whole nerve, the labels are flipped.

PROMPT B · WRAPPING TABLE

- [ ] Endoneurium: inner layer; wraps one axon
- [ ] Perineurium: middle layer; wraps one fascicle
- [ ] Epineurium: outer layer; wraps the whole nerve
- [ ] Each sketch shows the wrapping drawn around its own level
- [ ] The perineurium and epineurium hold a cut nerve's shape
- [ ] That lets a surgeon line up the fascicles and sew the nerve back together

Check: The perineurium goes around a fascicle, a group of axons, not around one axon. Check your middle row.

188 Spinal nerves and rami

PROMPT A · LABELED SPINAL NERVE

- [ ] Spinal nerve leaving through the intervertebral foramen
- [ ] Posterior ramus with an arrow to the deep back muscles and the skin of the back of the neck and trunk
- [ ] Anterior ramus with an arrow to the limbs and the side and front of the trunk
- [ ] Meningeal branch turning back into the vertebral canal, toward the vertebrae, ligaments, cord blood vessels, and meninges
- [ ] Rami communicantes linking the anterior ramus to a sympathetic trunk ganglion
- [ ] Counts: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal
- [ ] Total of 31 pairs

Check: There are 8 pairs of cervical nerves even though there are only 7 cervical vertebrae. Check your cervical number.

PROMPT B · COUNT AND TERRITORY TABLE

- [ ] Cervical 8 pairs, thoracic 12 pairs
- [ ] Lumbar 5 pairs, sacral 5 pairs, coccygeal 1 pair
- [ ] Total 31 pairs
- [ ] Posterior ramus: deep back muscles and the skin of the back of the neck and trunk
- [ ] Anterior ramus: muscles and skin of the limbs and the side and front of the trunk
- [ ] Meningeal branch: vertebrae, ligaments, blood vessels of the cord, and meninges
- [ ] Rami communicantes: carry sympathetic axons to and from the sympathetic trunk
- [ ] Anterior ramus starred
- [ ] Intercostal nerves, T2 to T12

Check: The anterior ramus forms the plexuses, not the posterior ramus. Check which one has your star.

189 Sensory receptor classification

PROMPT A · LABELED SKIN BLOCK

- [ ] Skin block with epidermis, dermis, and hypodermis, one hair, and muscle with its tendon below
- [ ] Free nerve endings as bare tips in the tissue; detect pain, temperature, and crude touch
- [ ] Tactile disc in the deepest layer of the epidermis; detects light touch
- [ ] Hair follicle receptor wrapped around the hair follicle; detects hair movement
- [ ] Tactile corpuscle in a dermal papilla; detects fine touch
- [ ] Lamellar corpuscle in the deep dermis or hypodermis; detects deep pressure and vibration
- [ ] Bulbous corpuscle in the deep dermis; detects steady pressure
- [ ] Muscle spindle inside the muscle (detects stretch) and tendon organ inside the tendon (detects tension)
- [ ] Three ways to classify: by stimulus, by location, by structure
- [ ] Skin receptors tagged exteroceptor; muscle spindle and tendon organ tagged proprioceptor

Check: The tactile corpuscle sits high, in the dermal papillae, and the lamellar corpuscle sits deep. If they are at the same depth, move one.

PROMPT B · RECEPTOR TABLE

- [ ] Free nerve endings: most tissues; pain, temperature, and some touch; nociceptor, thermoreceptor, or mechanoreceptor
- [ ] Tactile discs: deepest layer of the epidermis; light touch; mechanoreceptor
- [ ] Hair follicle receptors: around hair follicles; hair movement; mechanoreceptor
- [ ] Tactile corpuscles: dermal papillae of hairless skin; fine touch; mechanoreceptor
- [ ] Lamellar corpuscles: deep dermis and hypodermis; deep pressure and vibration; mechanoreceptor
- [ ] Bulbous corpuscles: deep dermis, hypodermis, and joint capsules; steady pressure and skin stretch; mechanoreceptor
- [ ] Muscle spindles in skeletal muscle (stretch), tendon organs in tendons (tension), joint kinesthetic receptors in joint capsules (position and movement); all mechanoreceptors
- [ ] By location: exteroceptors (such as tactile corpuscles), interoceptors (free nerve endings in the organs), proprioceptors (such as muscle spindles)
- [ ] By structure: free nerve endings (such as hair follicle receptors) and encapsulated nerve endings (such as lamellar corpuscles); specialized receptor cells (such as hair cells of the inner ear) also count

Check: Free nerve endings are the only receptors in the table that sense pain. Everything else in the table is a mechanoreceptor.

190 Cranial nerves

PROMPT A · LABELED INFERIOR VIEW

- [ ] CN I olfactory bulb and tract under the frontal lobe
- [ ] CN II optic nerves meeting at the optic chiasm, with the optic tract behind it
- [ ] CN III and CN IV at the midbrain, with CN IV the thin nerve curving around the side
- [ ] CN V large, on the side of the pons
- [ ] CN VI, VII, and VIII at the pontomedullary junction, with VI nearest the midline
- [ ] CN IX, X, XI, and XII on the medulla, with XI also running up from the upper spinal cord
- [ ] S beside I, II, and VIII; B beside V, VII, IX, and X; M beside III, IV, VI, XI, and XII
- [ ] Foramina: I cribriform plate, II optic canal, III, IV, and VI superior orbital fissure, V superior orbital fissure, foramen rotundum, and foramen ovale
- [ ] Foramina: VII and VIII internal acoustic meatus, IX, X, and XI jugular foramen, XII hypoglossal canal
- [ ] Circled: superior orbital fissure, internal acoustic meatus, and jugular foramen

Check: Only four nerves are both: V, VII, IX, and X. Count your Bs.

PROMPT B · CRANIAL NERVE TABLE

- [ ] Names in order: olfactory, optic, oculomotor, trochlear, trigeminal, abducens
- [ ] Then: facial, vestibulocochlear, glossopharyngeal, vagus, accessory, hypoglossal
- [ ] Sensory: I, II, VIII. Motor: III, IV, VI, XI, XII. Mixed: V, VII, IX, X
- [ ] Functions: I smell, II vision, III most eye movements, eyelid, pupil, IV superior oblique, V face sensation and chewing, VI lateral rectus
- [ ] Functions: VII facial expression, front tongue taste, tears and saliva, VIII hearing and balance, IX back tongue taste, swallowing, parotid saliva, X parasympathetic control of the chest and abdominal organs, voice, swallowing, XI sternocleidomastoid and trapezius, XII tongue
- [ ] Foramina: I cribriform plate, II optic canal, III, IV, VI superior orbital fissure, VII and VIII internal acoustic meatus, IX, X, XI jugular foramen, XII hypoglossal canal
- [ ] V foramina: V1 superior orbital fissure, V2 foramen rotundum, V3 foramen ovale
- [ ] One color for the superior orbital fissure rows: III, IV, V, VI
- [ ] One color for the internal acoustic meatus rows: VII and VIII
- [ ] One color for the jugular foramen rows: IX, X, XI

Check: Only V1 of the trigeminal passes through the superior orbital fissure. V2 uses the foramen rotundum and V3 the foramen ovale.

191 Nerve plexuses

PROMPT A · LABELED PLEXUS MAP

- [ ] Spinal nerves numbered: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal
- [ ] Cervical plexus: C1 to C4, with a small part from C5
- [ ] Brachial plexus: C5 to T1
- [ ] Lumbar plexus: L1 to L4
- [ ] Sacral plexus: L4 to S4
- [ ] Coccygeal plexus: S4, S5, and the coccygeal nerve
- [ ] Intercostal nerves (T2 to T12) running straight out with no plexus
- [ ] Shaded: neck and upper shoulder for cervical, shoulder and whole upper limb for brachial
- [ ] Shaded: lower abdominal wall and front and inner thigh for lumbar, buttock, back of thigh, leg and foot for sacral
- [ ] Shaded: small patch of skin over the coccyx for coccygeal

Check: The brachial plexus runs from C5 to T1, so it takes in the first thoracic nerve. Check that your bracket reaches T1.

PROMPT B · DAMAGE TABLE

- [ ] Cervical: C1 to C4, with a small part from C5; neck and upper shoulder; loss of neck and shoulder movement and feeling
- [ ] Brachial: C5 to T1; shoulder and whole upper limb; weakness and numbness of the whole arm on that side
- [ ] Lumbar: L1 to L4; lower abdominal wall, external genitals, part of the lower limb; weakness and numbness of the lower abdominal wall and front and inner thigh
- [ ] Sacral: L4 to S4; buttock, perineum, most of the lower limb; weakness and numbness of the buttock, back of the thigh, leg, and foot
- [ ] Coccygeal: S4, S5, and the coccygeal nerve; skin over the coccyx; numbness of that small patch
- [ ] Phrenic nerve drawn leaving the cervical plexus and running down to the diaphragm
- [ ] Phrenic damage weakens the diaphragm on that side and breathing suffers; a high neck injury can stop breathing

Check: The phrenic nerve comes from the cervical plexus, not the brachial plexus. Check where your nerve starts.

192 Autonomic two-neuron pathway

PROMPT A · SIDE BY SIDE PATHWAYS

- [ ] Somatic: one neuron with its cell body in the spinal cord
- [ ] Somatic axon myelinated, running straight to skeletal muscle
- [ ] Autonomic: preganglionic cell body in the spinal cord
- [ ] Preganglionic axon labeled lightly myelinated, running to the autonomic ganglion
- [ ] Synapse labeled in the autonomic ganglion
- [ ] Postganglionic cell body in the ganglion
- [ ] Postganglionic axon labeled unmyelinated, running to the effector
- [ ] Autonomic effectors: cardiac muscle, smooth muscle, and glands
- [ ] Somatic labeled voluntary, autonomic labeled involuntary

Check: The preganglionic axon is the lightly myelinated one and the postganglionic axon is unmyelinated. Check that you did not swap them.

PROMPT B · COMPARE TABLE

- [ ] Effector: skeletal muscle for somatic; cardiac muscle, smooth muscle, and glands for autonomic
- [ ] Control: voluntary for somatic, involuntary for autonomic
- [ ] Motor neurons: one for somatic; two with a synapse in a ganglion for autonomic
- [ ] Effect: always excitation for somatic; excitation or inhibition for autonomic
- [ ] Awareness: normally conscious for somatic, normally not conscious for autonomic
- [ ] Chain in order: preganglionic neuron, autonomic ganglion, postganglionic neuron, effector tissue

Check: The somatic system only ever excites its muscle. Inhibition belongs in the autonomic column.

193 Autonomic ganglia and divisions

PROMPT A · LABELED GANGLIA MAP

- [ ] Sympathetic trunk drawn as a chain of ganglia beside the vertebral column
- [ ] Sympathetic bracket on the T1 to L2 segments
- [ ] Parasympathetic brackets on the brainstem and on the S2 to S4 segments
- [ ] A sympathetic trunk ganglion labeled in the chain
- [ ] Prevertebral ganglia in front of the vertebral column, near the large abdominal arteries
- [ ] A terminal ganglion in the wall of an organ
- [ ] Cardiac plexus on the heart, celiac plexus on the abdominal organs, hypogastric plexus on the pelvic organs
- [ ] Gut wall: myenteric plexus between the longitudinal and circular muscle layers
- [ ] Gut wall: submucosal plexus in the submucosa, between the muscularis mucosae and the circular muscle layer

Check: Sympathetic leaves the middle of the cord, T1 to L2. Parasympathetic leaves the two ends, the brainstem and S2 to S4.

PROMPT B · COMPARE TABLE

- [ ] Other name: thoracolumbar for sympathetic, craniosacral for parasympathetic
- [ ] Origin: T1 to L2 for sympathetic; brainstem nuclei of CN III, VII, IX, and X and S2 to S4 for parasympathetic
- [ ] Ganglia: close to the cord for sympathetic; close to or inside the organ for parasympathetic
- [ ] Sympathetic ganglia: sympathetic trunk ganglia and prevertebral ganglia (celiac, superior mesenteric, inferior mesenteric, aorticorenal, renal)
- [ ] Parasympathetic ganglia: terminal ganglia, including the ciliary, pterygopalatine, submandibular, and otic
- [ ] Role: fight or flight for sympathetic, rest and digest for parasympathetic
- [ ] Plexuses: cardiac to the heart, pulmonary to the bronchial tree, celiac to most of the upper abdominal organs
- [ ] Plexuses: superior and inferior mesenteric to the intestines, hypogastric to the pelvic organs, renal to the kidneys and ureters
- [ ] Myenteric plexus between the longitudinal and circular muscle layers
- [ ] Submucosal plexus in the submucosa, between the muscularis mucosae and the circular muscle layer

Check: Thoracolumbar is sympathetic and craniosacral is parasympathetic. Check you did not swap the names.